

Trade Credit and Sectoral Comovement during Recessions^{*}

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July 27, 2026

Abstract

We examine sectoral output comovement during economic downturns, focusing on the role of trade credit in amplifying or dampening shock transmission. Analyzing nine postwar U.S. recessions, we find that the Great Recession is the only episode with a pronounced and robust increase in sectoral comovement after filtering out aggregate fluctuations; during this episode, sector pairs with measured trade-credit contractions exhibited 40% higher comovement. A multisector model with endogenous trade credit highlights an asymmetric mechanism: relatively unconstrained suppliers expand trade credit to cushion financially distressed customers, whereas sufficiently constrained suppliers reduce trade credit, amplifying financial distress. Quantitatively, relative to a fixed-trade-credit counterfactual, endogenous trade credit increases average pairwise output-growth correlations during the Great Recession and deepens the decline in aggregate consumption growth at the 2008Q4 trough. Recalibrating the economy to the early-1980s and COVID-19 recessions yields the opposite result: endogenous trade credit moderates the increase in sectoral comovement and the decline in aggregate consumption.

Keywords: Trade Credit, Sectoral Comovement, Financial Frictions

JEL Classifications: C67, E23, E32, E44, E51, F40, G30

^{*}We thank the editor, Francesco Lippi, and three anonymous referees for their careful reading and constructive comments, which substantially improved the paper. We thank Eric van Wincoop and Eric Young, for their enormous support. We also thank David Baqaee, Huberto Ennis, Andrew Foerster, Zhen Huo, Yang Jiao, Pete Klenow, Shaowen Luo, Toshihiko Mukoyama, Daniel Murphy, Jose Mustre-del-Rio, Alessandro Rebucci, Christiern Rose, Pierre Sarte, Nicholas Sly, Kwok Ping Tsang, Kieran Walsh, Tao Zha, Hulai Zhang, and seminar participants at FRB Kansas City, FRB Cleveland, HKU, SUFE, Virginia Tech, Jinan, SED meetings, and Catholic University of Chile for helpful discussions and comments. We also thank Alvaro Castillo for outstanding research assistance. Zhang gratefully acknowledges financial support from the Bankard Fund for Political Economy and the ASEAN Business Research Initiative. We thank Egon Zakrajšek for providing us with the spread data.

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1 Introduction

Aggregate volatility can be strongly correlated with the comovement of output growth across sectors or firms (Carvalho, 2014), which may arise from two key factors: first, aggregate shocks driving overall fluctuations, and second, the propagation and amplification of sectoral shocks through production and credit linkages.¹ While prior research has been crucial in explaining average sectoral comovement (e.g., Raddatz, 2010; Li and Martin, 2019), less attention has been paid to the dynamics of sectoral comovement across different types of recessions (e.g., financial vs. economic recessions), which influence the magnitude of macroeconomic fluctuations. In this paper, we study endogenous trade-credit adjustment as one mechanism through which sectoral shocks may propagate more strongly during financial crises, potentially increasing comovement and amplifying the downturn, as seen during the Great Recession in the U.S.

Using annual sectoral output growth data from the Bureau of Economic Analysis (BEA), we find that sectoral output-growth comovement increased significantly during both the Great Recession and the COVID-19 recession, but not during the early-1980s recessions, despite its comparable magnitude. After controlling for recession severity, the increase in sectoral comovement during the Great Recession remains pronounced, whereas that observed during the COVID-19 recession largely disappears. We then zoom into the Great Recession with quarterly data and find that sectoral comovement increased more for more-interconnected sectors. The average correlation rose from 0.02 to 0.25 between sectors trading intermediates one-way (only one sector providing intermediates to the other) and from 0.1 to 0.52 between sectors trading two-way (both sectors are intermediate input providers and users of each other). Interestingly, this pattern is not observed between sectors in the no-trading group or during other recessions, indicating that a change in the nature of sectoral connections contributed to the rise in comovement during the Great Recession.

Next, we examine whether contractions in trade credit are associated with changes in sectoral comovement during the Great Recession. We classify a sector pair as experiencing a trade-credit contraction if, between 2008Q3 and 2009Q1, both the supplier’s trade credit provided (measured by the accounts-receivable-to-sales ratio) and the customer’s trade credit received (measured by the accounts-payable-to-costs ratio) decline by more than the median decline among public firms. We find that sector pairs experiencing a

¹For the second factor, see, for example, Lucas (1981), Long and Plosser (1983), Hornstein and Praschnik (1997), Horvath (1998), Shea (2002), Foerster et al. (2011), Acemoglu et al. (2012), Acemoglu et al. (2016), Barrot and Sauvagnat (2016), Di Giovanni et al. (2018), Baqaee and Farhi (2019), vom Lehn and Winberry (2020), Cun et al. (2023), among others.

trade-credit contraction exhibit substantially stronger comovement. Specifically, among two-way trading pairs, the average output-growth correlation during the Great Recession is 0.18 (40%) higher for pairs with trade-credit contractions than for those without.²

To reconcile our sectoral observations and quantify the role of trade credit during recessions, we develop a multisector production-network model, building on [Bigio and La'O \(2020\)](#), with endogenous trade-credit provision. Trade credit emerges for two reasons. First, motivated by [Smith \(1987\)](#) and [Antras and Foley \(2015\)](#), delivered intermediate inputs are subject to quality uncertainty. Deferred payment allows repayment to depend on realized input quality, creating a role for trade credit as part of the payment contract. Providing trade credit is therefore costly for suppliers because it delays cash inflows, increases their dependence on external finance, and exposes them to repayment risk. Second, in the spirit of [Cuñat \(2007\)](#) and [Brugués \(2026\)](#), trade credit also helps sustain supplier-customer relationships. By extending trade credit and exerting relationship-maintenance effort, suppliers preserve valuable trading relationships and generate relationship surplus.

Firms operate both as input customers and input suppliers. As customers, they choose labor and intermediate inputs, taking suppliers' trade credit as given. As suppliers, they choose how much trade credit to extend to customers and the effort devoted to maintaining those relationships. Firms are subject to working-capital constraints: labor costs, upfront payments for intermediate inputs, and relationship-maintenance expenditures must be financed before production by pledging a fraction of future output as collateral. Therefore, trade credit creates a financial linkage within the production network. Receiving trade credit relaxes customers' working-capital requirements, whereas extending trade credit delays suppliers' cash inflows and tightens their own financing constraints.

The supplier's trade-off between supporting customer input demand and preserving valuable trading relationships, on the one hand, and conserving its own liquidity and limiting the costs of extending trade credit, on the other, gives rise to the model's central mechanism. When a customer becomes more financially constrained, trade credit becomes more valuable because it lowers the customer's effective cost of purchasing inputs and sustains input demand. The response of trade credit to customer financial distress depends on the supplier's own financial condition. If the supplier is relatively uncon-

²These results document an empirical association rather than a causal effect. In Online Appendix III, we exploit the collapse of Lehman Brothers (LB) as a quasi-natural experiment. Combining pre-crisis firm-to-firm supply-chain networks with firm-to-bank relationships, we provide suggestive evidence that customers exposed to LB exhibit stronger comovement with their suppliers than otherwise similar unexposed customers, particularly when the exposed customers experience larger declines in trade credit following the LB collapse.

strained, it is willing to expand trade credit to preserve the relationship and maintain input sales. However, when the supplier itself becomes sufficiently financially constrained, the shadow value of upfront payment and the shadow cost of relationship maintenance increase. At that point, the supplier reduces trade credit precisely when the customer values it most. Trade credit therefore switches from a cushion that absorbs financial distress to an amplifier, tightening trade credit along production links and strengthening sectoral comovement.

We bring the model to the data by first calibrating the stationary economy to match key sectoral moments from input-output tables and firm balance sheets, including production shares, consumption shares, input-expenditure shares, and trade-credit usage. The calibrated model closely matches these targeted moments and also performs well on untargeted trade-credit moments. We then recover time-varying sectoral productivity and financial conditions by requiring the model to match observed sectoral output and bond spreads. We show that although productivity and financial shocks become more correlated during the Great Recession, this increase alone cannot explain which sector pairs comove more: sector pairs with stronger output-growth comovement are not systematically those experiencing more highly correlated productivity or financial shocks.

We then use the calibrated model to study the role of trade credit in shaping sectoral comovement and aggregate growth through several quantitative exercises. First, after showing that the model matches sector-level trade-credit moments during the Great Recession, we use the model-implied bilateral trade-credit responses to examine the asymmetric role of trade credit. We find that sector pairs in which trade credit switches from cushioning to amplifying customer financial distress during the Great Recession exhibit substantially stronger comovement: their average output-growth correlation increases from 0.06 before the recession to 0.66 during the crisis, compared with an increase from 0.03 to 0.52 for other sector pairs. Consistent with this result, we also show that the sector pairs with the highest output comovement during the Great Recession experience the sharpest contraction in model-implied bilateral trade credit.

Second, we quantify the role of endogenous trade credit adjustment by fixing bilateral trade credit at its pre-recession level. With endogenous trade credit, the average pairwise output-growth correlation rises from 0.04 before the Great Recession to 0.56 during the crisis.³ When trade credit is fixed, the corresponding increase is much smaller, from

³A possible concern is that the rise in sectoral comovement during the Great Recession simply reflects more highly correlated underlying shocks. To examine this possibility, in Online Appendix IV.6, we compare simulations using the estimated crisis-period covariance matrices of recovered productivity and financial conditions with counterfactual simulations that remove cross-sector shock correlations. We find that, although the stronger cross-sector correlations of recovered productivity and financial conditions during

0.12 to 0.26. The aggregate effect is also economically meaningful: at the trough of the recession, aggregate consumption growth falls by 7.4% in the benchmark economy, compared with 6.1% in the fixed-trade-credit counterfactual. One-shock counterfactuals suggest that this amplification is mainly associated with financial shocks, while productivity shocks affect comovement primarily through the production network itself.

Third, we recover productivity and financial conditions under a fixed-trade-credit economy to examine what fundamentals are needed when the trade-credit propagation margin is shut down. To match the same sectoral output and bond-spread data, the fixed-trade-credit model requires a larger increase in the comovement of recovered productivity and financial conditions. This comparison shows that endogenous trade-credit adjustment changes the model’s attribution of the Great Recession: part of the rise in sectoral comovement is accounted for by propagation through endogenous trade credit adjustment, rather than by more correlated fundamentals alone.

Finally, we apply the same framework to the early-1980s and COVID-19 recessions. These episodes provide useful contrasts because the U.S. economy experienced similarly severe, or even deeper, aggregate downturns, but the underlying shocks and financial environment differed from the Great Recession. For each episode, we recalibrate the stationary economy using period-specific input-output and trade credit moments. In both cases, the fixed-trade-credit counterfactual generates a larger increase in sectoral comovement and a deeper aggregate contraction than the benchmark economy with endogenous trade credit. Thus, unlike during the Great Recession, endogenous trade credit does not appear to amplify shock transmission in these episodes. Instead, it helps moderate sectoral comovement and dampen the aggregate downturn.

Taken together, our findings point to endogenous trade-credit adjustment as an empirically relevant and quantitatively important mechanism for propagating shocks through production networks. We do not claim that trade credit is the only mechanism behind the rise in sectoral comovement during the Great Recession. Rather, our objective is to quantify the contribution of this mechanism within a model disciplined by sectoral output, bond spreads, and trade-credit moments.

Contribution to the literature: Our paper builds on and contributes to three strands of the existing literature. The first incorporates financial frictions into multisector real business cycle models to study how financial frictions propagate and amplify sector-specific shocks in macroeconomic aggregates (see, for example, [Bigio and La’O, 2020](#);

the Great Recession contribute to higher sectoral comovement, removing these correlations still leaves substantial comovement, indicating that endogenous propagation through production linkages and trade credit also plays an important role.

Altinoglu, 2021; Luo, 2020; Reischer, 2020; Miranda-Pinto and Young, 2022; Shao, 2020; Cun et al., 2023). We contribute to this literature by identifying an asymmetric trade-credit channel through which financial disruptions can either amplify or dampen sectoral comovement and aggregate downturns, complementing the evidence in Kaminsky and Reinhart (1999) and Reinhart and Rogoff (2011). We also provide empirical and quantitative evidence that the role of trade credit differs systematically across recessions because suppliers' endogenous responses depend on prevailing financial conditions.

Within this literature, Altinoglu (2021) is particularly related because the author introduces supplier credit into a production-network economy and shows that credit obligations between firms transmit liquidity shocks through production networks. The mechanism operates through a given credit network: shocks to customers impair their ability to pay suppliers and thereby propagate upstream. Our paper instead focuses on the endogenous adjustment of trade credit itself. Suppliers choose whether to expand or contract trade credit in response to customers' and their own financial conditions, so trade credit can either cushion or amplify financial distress.

Our model is also closely related to Luo (2020) and Reischer (2020), both of which generate asymmetric effects of trade credit. In Luo (2020), trade credit attenuates relatively small shocks through liquidity sharing but amplifies large shocks through illiquidity contagion. In Reischer (2020), amplification arises when increases in bank borrowing costs translate into even larger increases in the cost of trade-credit finance. Our mechanism is different. The asymmetry does not depend only on shock size or on an exogenous rise in the cost of trade-credit finance. Instead, it arises from suppliers' endogenous trade-credit choices: when suppliers are relatively unconstrained, they extend trade credit to preserve customer relationships and support demand; when suppliers become sufficiently constrained, they reduce trade credit to protect their own liquidity. This state-dependent supplier response is central to our explanation of why trade credit amplifies sectoral comovement during the Great Recession but plays a different role in other downturns.

The second strand studies the role of trade credit in liquidity provision and in the propagation of financial distress (see, for example, Smith, 1987; Kiyotaki and Moore, 1997a; Petersen and Rajan, 1997; Love et al., 2007; Giannetti et al., 2011; Garcia-Appendini and Montoriol-Garriga, 2013; Jacobson and von Schedvin, 2015; Garcia-Marin et al., 2019; Costello, 2020; Giannetti et al., 2021; Grigoris et al., 2023; Adelino et al., 2023; Hardy et al., 2026). This literature shows that suppliers can provide liquidity to constrained customers, but also that trade-credit claims can transmit distress through default and balance-sheet losses. Our paper studies how this margin shapes sectoral comovement over U.S. recessions. On the theoretical side, we build on information- and relationship-

based motives for trade credit, in the spirit of [Smith \(1987\)](#), [Cuñat \(2007\)](#), [Brugués \(2026\)](#), and related work, and embed them in a multisector production-network model. The model generates an asymmetric role for trade credit: financially healthy suppliers can use trade credit to cushion customer distress, whereas financially constrained suppliers may instead reduce trade credit and propagate shocks. The mechanism is closely related to the “deep-pocket” view in [Kiyotaki and Moore \(1997a\)](#): whether trade credit cushions or amplifies a customer liquidity shock depends critically on the supplier’s own financial capacity. Using the model, we quantify the potential contribution of this state-dependent trade-credit channel to sectoral comovement across recessions.

The third strand of related research examines sectoral comovement over the business cycle and the forces that generate it (see, for example, [Long and Plosser, 1983](#); [Hornstein and Praschnik, 1997](#); [Christiano and Fitzgerald, 1998](#); [Raddatz, 2010](#); [Carvalho, 2014](#); [Li and Martin, 2019](#); [Bonadio et al., 2025](#); [Huo et al., 2025](#)). This literature studies how aggregate shocks, sectoral shocks, and input-output linkages shape the joint movement of sectoral activity. Our paper contributes to this literature by focusing on nine postwar U.S. Recessions. In particular, we contrast patterns of sectoral comovement across three severe crisis: the Great Recession, the early-80s recessions, and the COVID-19 recession. We study how financial linkages embedded in production relationships affect sectoral comovement and aggregate output. The paper most closely related to ours in this strand is [Li and Martin \(2019\)](#), who document a large increase in sectoral comovement during the Great Recession and attribute much of it to a Great-Recession-specific common factor. We complement their reduced-form decomposition with a structural production-network model featuring state-dependent trade-credit linkages. This framework helps explain why the rise in comovement is concentrated among sectors connected through input-output relationships and is especially strong when trade credit contracts. It also provides a mechanism through which heterogeneous productivity and financial conditions can generate a sharp increase in comovement without requiring a uniform common shock across all sectors.

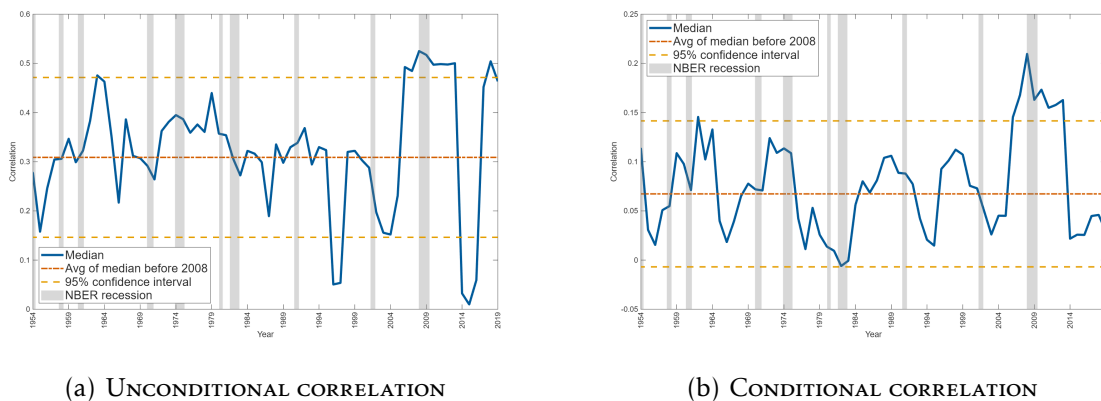
2 Empirical Evidence

In this section, we document our main observations regarding the behavior of sectoral comovement during recessions. First, we find that sectoral output growth comoved strongly during the Great Recession, even after controlling for the size of recessions. In other postwar recessions, sectors either did not comove, or the comovement was not robust after filtering out the aggregate component. Second, the rise in sectoral comovement during the

Great Recession was more pronounced in pairs of sectors that provide intermediate inputs to each other than in those in which only one provides inputs to the other, and much more than among pairs without input-trading relationships. Third, among sector pairs mutually connected through intermediate-input trade, those experiencing larger-than-median contractions in trade credit comoved more strongly, by more than 40 percent.

2.1 Observation I: rise in pairwise correlation

We begin by analyzing the annual growth rate of sectoral outputs from the BEA to investigate the dynamics of sectoral comovement following World War II (see Online Appendices I.1 and I.2 for details on the sectoral data the construction of the sectoral comovement measure). Panel (a) of Figure 1 shows the evolution of the median value of pairwise correlations over an eight-year rolling window, with the red dotted line representing the 1950-2005 average, and the yellow dashed lines indicating the 95% confidence interval.⁴ Our analysis reveals that the median generally fluctuated within the 95% confidence interval, except during the Great Recession and the COVID-19 recession, where significant increases are observed. This pattern is not evident in other recessions.



Note: The figure uses BEA real annual gross output by industry for 57 private non-farm, non-FIRE sectors. Pairwise correlations are calculated using an eight-year rolling window. The year on the horizontal axis corresponds to the fourth year of each time window. Panel (a) reports unconditional correlations of sectoral output growth. Panel (b) reports conditional correlations, computed from the first difference of residuals after regressing log sectoral output on log U.S. GDP. The dashed horizontal line marks the pre-2008 average median correlation, the outer dashed lines mark the 95% confidence interval around the pre-recession average of the rolling medians, and shaded areas denote NBER recessions.

Figure 1
EVOLUTION OF MEDIAN PAIRWISE CORRELATIONS AFTER WWII

⁴We report the median because the cross-sectional distribution of pairwise correlations can be skewed, making the median a more robust measure of the typical sector pair than the average. Our empirical conclusions, however, are based on the full distribution of pairwise correlations, which is examined using kernel densities and summary statistics in Online Appendix II.1.

Because recessions vary in their aggregate severity, we account for these unequal aggregate fluctuations by filtering out aggregate components. Specifically, we regress the logarithm of sectoral outputs on the logarithm of U.S. GDP, take the residuals, compute their first differences, and then calculate the pairwise correlations.⁵ Panel (b) of Figure 1 illustrates the evolution of median conditional correlations. Before 2008, the average conditional correlation decreased to 0.07 from the unconditional average of 0.31. During the Great Recession, the correlation significantly increased, with approximately 35% of the unconditional rise in comovement unexplained by aggregate components. Conversely, the conditional average correlation during the COVID-19 recession is statistically similar to the pre-recession level.⁶ These results suggest that inter-sector linkages played an important role in synchronizing sectoral activity during the Great Recession, whereas the comovement observed during the COVID-19 recession is largely accounted for by aggregate fluctuations.

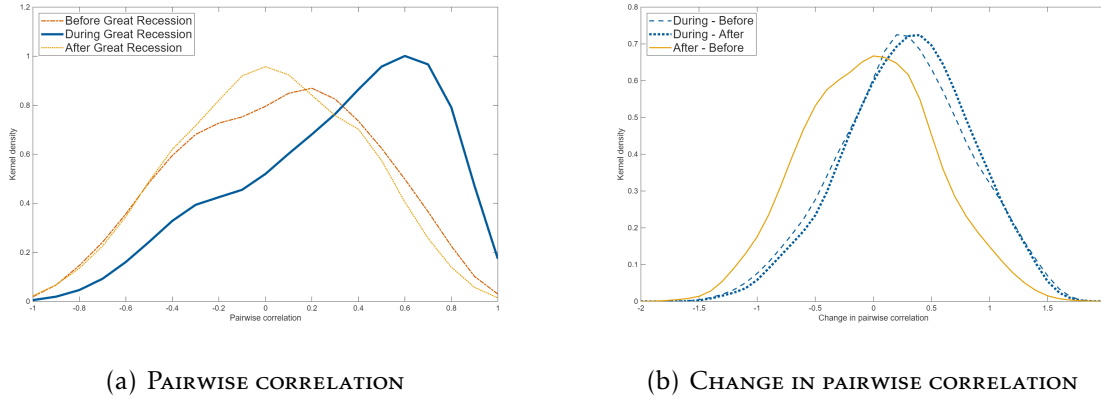
To characterize this synchronization in greater detail, we now turn to quarterly data. Compared with annual observations, quarterly data preserve within-year fluctuations that may otherwise be averaged out, providing a more informative cross-sectional distribution of pairwise correlations during the Great Recession. Panel (a) of Figure 2 displays kernel densities before, during, and after the Great Recession. Following Kahle and Stulz (2013), we define the recession period as 2007Q3-2009Q2, with 2005Q3-2007Q2 and 2009Q3-2011Q2 representing the periods before and after the recession, respectively.⁷ Consistent with annual growth data, we observe a significant rise in sectoral comovement during the Great Recession. The average correlation increases from 0.05 before the recession to 0.31 during the recession, and back to 0.01 afterward.

Panel (b) of Figure 2 further illustrates the heterogeneity in pairwise comovement by plotting the distribution of changes in correlations from before to during the Great Recession. Most sector pairs experienced an increase in comovement, with the distribution centered around 0.31. The correlations of many pairs rose by more than one, indicat-

⁵Our approach implicitly considers the recession duration. However, since the duration of each recession within a fixed time window varies, we adjust the length of our time window to explicitly account for this. Direct comparisons across time windows are not possible due to differing statistical properties, so we standardize each sequence to the pre-Great Recession level. Figure II.1 in Online Appendix II.1 shows that our results remain robust.

⁶Table II.2 in Online Appendix II.1 confirms that the Great Recession is the only recession exhibiting a statistically significant increase in conditional pairwise correlations. While unconditional correlations rise by a similar magnitude during the Great Recession (0.34) and the COVID-19 recession (0.32), the corresponding increases in conditional correlations are 0.12 and 0.01, respectively.

⁷We use a bandwidth of 0.1, as Silverman’s rule of thumb suggests an optimal bandwidth of 0.0956. We also vary the bandwidth values to 0.2 and 0.01, and Figure II.2 in Online Appendix II.1 shows that the kernel density estimates mostly overlap. Additionally, we used different coverage periods and lengths of time windows. All results presented here are robust.



Note: The figure uses BEA real gross output by industry, seasonally adjusted at annual rates, for 57 private non-farm, non-FIRE sectors. Panel (a) plots kernel densities of pairwise sectoral output-growth correlations before, during, and after the Great Recession. Panel (b) plots the distribution of pair-level changes in correlations: during minus before, during minus after, and after minus before. The kernel density is estimated with a bandwidth of 0.1. The periods 2005Q3–2007Q2, 2007Q3–2009Q2, and 2009Q3–2011Q2 represent the windows before, during, and after the Great Recession.

Figure 2
KERNEL DENSITY OF PAIRWISE CORRELATION DURING THE GREAT RECESSION

ing that some sector pairs that previously moved in opposite directions began moving together during the Great Recession.⁸

We conducted the two-sided Kolmogorov-Smirnov (KS) test to investigate whether pairwise correlations in two periods come from the same distribution. A problem with the KS test in our setup is that it assumes that the two samples under comparison are drawn from two independent and identically distributed (iid) distributions. However, a key characteristic of time series data is its persistence and autocorrelation. Indeed, the autocorrelation of the rolling-window pairwise correlation of a given pair fluctuates between 0.5 and 0.9. To correct this problem, we follow [Lanzante \(2021\)](#) and adjust the critical values of the KS test for the presence of autocorrelation. Panel (a) of Table 1 shows that the KS test rejects the null hypotheses at the 5% significance level, even if the underlying autocorrelation is very high. This result indicates that the distribution of pairwise correlations during the Great Recession was statistically different from those before and after the recession.⁹

⁸It is important to note that pairwise output correlations are realized equilibrium moments rather than structural characteristics of sector pairs. As such, they naturally vary over time as underlying shocks and their propagation change. Our objective is therefore not to explain the persistence of individual pairwise correlations, but rather how the cross-sectional distribution of sectoral comovement changes during the Great Recession.

⁹Because each sector appears in multiple pairs, pairwise correlations are not cross-sectionally independent. Consequently, the formal KS significance thresholds serve as a useful heuristic but should be interpreted with some caution.

Table 1
SUMMARY STATISTICS: PAIRWISE CORRELATIONS (QUARTERLY GROWTH)

	Mean	Median	SD	Skewness	KS Stat (CV-KS)
(a) Overall					
Before the Great Recession	0.05	0.07	0.40	-0.11	0.28 (0.09, 0.21)
During the Great Recession	0.31	0.39	0.41	-0.58	
After the Great Recession	0.01	0.01	0.38	-0.07	0.32 (0.09, 0.21)
(b) Two-way trading sectors					
Before the Great Recession	0.10	0.14	0.40	-0.23	0.48 (0.16, 0.39)
During the Great Recession	0.52	0.60	0.31	-1.10	
After the Great Recession	0.02	0.01	0.37	-0.03	0.56 (0.16, 0.39)
(c) One-way trading sectors					
Before the Great Recession	0.02	0.02	0.39	0.00	0.27 (0.13, 0.30)
During the Great Recession	0.25	0.31	0.41	-0.46	
After the Great Recession	0.02	0.03	0.38	-0.12	0.27 (0.13, 0.30)
(d) No trading sectors					
Before the Great Recession	0.05	0.09	0.41	-0.21	0.11 (0.19, 0.46)
During the Great Recession	0.15	0.17	0.41	-0.23	
After the Great Recession	0.00	-0.02	0.38	0.01	0.18 (0.19, 0.46)
(e) Two-way trading and trade credit declined group					
Before the Great Recession	0.05	0.08	0.44	-0.03	0.63 (0.31, 0.74)
During the Great Recession	0.63	0.69	0.24	-1.43	
After the Great Recession	-0.01	-0.03	0.39	0.15	0.72 (0.31, 0.74)
(f) Two-way trading and trade credit unchanged group					
Before the Great Recession	0.10	0.14	0.40	-0.29	0.39 (0.24, 0.58)
During the Great Recession	0.45	0.51	0.33	-0.81	
After the Great Recession	0.02	0.03	0.36	-0.06	0.49 (0.24, 0.58)
(g) One-way trading and trade credit declined group					
Before the Great Recession	0.11	0.17	0.40	-0.26	0.35 (0.35, 0.83)
During the Great Recession	0.39	0.46	0.34	-0.59	
After the Great Recession	-0.02	0.01	0.37	0.08	0.49 (0.35, 0.83)
(h) One-way trading and trade credit unchanged group					
Before the Great Recession	-0.02	-0.04	0.39	0.17	0.33 (0.19, 0.46)
During the Great Recession	0.26	0.35	0.43	-0.53	
After the Great Recession	0.02	0.02	0.38	-0.17	0.29 (0.19, 0.46)

Note: The table reports moments of pairwise sectoral output-growth correlations for the before-, during-, and after-Great-Recession windows, which are represented by the periods 2005Q3–2007Q2, 2007Q3–2009Q2, and 2009Q3–2011Q2, respectively. The KS statistic compares the distribution in the corresponding before or after row with the distribution during the Great Recession; the during row is left blank because it is the reference distribution. Critical values of the KS statistic (CV-KS) are reported in parentheses and adjust for autocorrelation following [Lanzante \(2021\)](#). The first value is the 5% critical value under mild autocorrelation (AR(1) parameter of 0.6), and the second value is the critical value under high autocorrelation (AR(1) parameter of 0.9). Two sectors are classified in the two-way trading group if they are both input supplier and customer to each other, in the one-way trading group if only one sector supplies inputs to the other, and in the no-trading group if no intermediate input is traded between them. A pair is classified in the trade-credit-declined group if the supplier's AR-to-sales ratio and the customer's AP-to-costs ratio both declined by more than the corresponding medians across public firms, respectively -1.6 and -1.0 percentage points; for two-way trading pairs, the condition can hold in either direction.

Additionally, we compute the kernel densities of pairwise correlations using annual growth rates for all recessions, specifically those in 1960, 1970, 1973, 1990, and 2001, as well as the combined 1980 and 1981-1982 recessions, the Great Recession, and the COVID-19 recession (see Online Appendix II.1 for details). Consistent with the implications of Figure 1, the shift in kernel density is only statistically significant during the Great Recession. For all the other recessions, the shift is either insignificant or disappears after controlling for GDP.

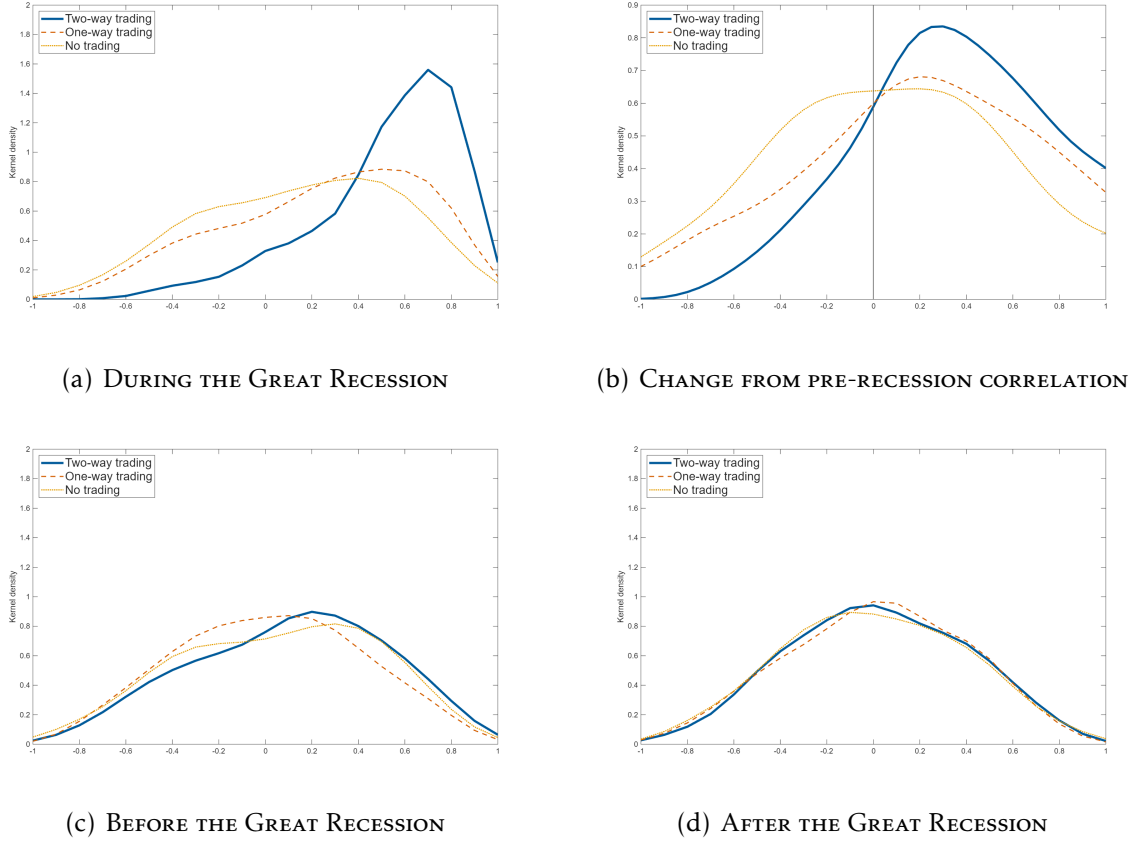
2.2 Observation II: role of intermediate-input linkages

We next examine whether this increase is related to the structure of production linkages across sectors. To identify the intermediate trading relationships between sectors, we adjust the input-output (IO) table of 71 industries from the Bureau of Economic Analysis (BEA) to match the sectors in our sample. We take the average data from 2003 to 2007 to avoid short-term variations and then calculate the IO matrix, where each cell represents the input expenditure share.¹⁰ The pairs are then categorized into three groups based on their level of interconnectedness. Specifically, two sectors are classified into the two-way trading group if they both supply inputs to each other; into the one-way trading group if only one sector supplies inputs to the other; and into the no-trading group if no intermediate inputs are traded between them. The groups contain 464, 796, and 336 pairs, respectively. We maintain this classification throughout the remainder of this section.

The way in which two sectors trade with each other significantly affects sectoral comovement. Panel (a) of Figure 3 displays the kernel densities of the three groups during the Great Recession. The density for two-way trading pairs is shifted furthest to the right. Specifically, the average (median) correlation within the two-way trading group exceeds that of the one-way trading group by 0.19 (0.17) and that of the no-trading group by 0.32 (0.38), as reported in Panels (b)-(d) of Table 1. Panel (b) reinforces this pattern by comparing changes in comovement from before to during the Great Recession. This concentration of increased comovement among trading sectors is difficult to reconcile with a purely aggregate shock, which would predict a broad increase in comovement regardless of sectoral interconnectedness (Online Appendix IV.9). Panels (c) and (d) further show that the three groups exhibit no comparable distributional differences before or after the Great Recession. By contrast, we do not observe similar patterns during other recession episodes (Online Appendix II.2). For example, applying the same classification

¹⁰If the input expenditure share is lower than 0.1%, we set it to 0. We also test different thresholds, such as 0.05% and 0.01%, and find our results robust across these variations.

to the COVID-19 recession yields a much more uniform increase in comovement across all three groups (Figure II.7).



Note: Pairwise correlations are calculated as in Figure 2. Panel (a) compares the distributions during the Great Recession; Panel (b) plots the change in pairwise correlations from the pre-recession window to the Great Recession; Panels (c) and (d) report the corresponding distributions before and after the Great Recession. The kernel density is estimated with a bandwidth of 0.1. Trading groups are defined using the 2003–2007 average input-output table. Two sectors are classified into the two-way trading group if they both supply inputs to each other, into the one-way trading group if only one sector supplies inputs to the other, and into the no-trading group if no intermediate inputs are traded between them. The groups contain 464, 796, and 336 pairs, respectively.

Figure 3
PAIRWISE CORRELATION BY EXTENT OF INTERCONNECTEDNESS

Overall, these findings suggest that the rise in sectoral comovement during the Great Recession was concentrated among sector pairs connected through production networks, yet production linkages alone do not fully account for this pattern. If stronger comovement simply reflected tighter input–output connections, sector pairs with larger input-expenditure shares should exhibit stronger comovement. However, we find little evidence of such a relationship. A linear regression of pairwise correlations on input-expenditure shares and trading-group indicators, reported in Online Appendix II.3, yields a statistically insignificant coefficient on input-expenditure shares. This finding points to

an additional propagation mechanism operating within production networks. We therefore turn to trade credit, which governs the amount of liquidity transferred through supplier–customer payment terms and can either amplify or dampen the transmission of shocks along existing production links, as suggested in [Kiyotaki and Moore \(1997a\)](#).

2.3 Observation III: role of trade credit during the Great Recession

In addition to trading intermediate inputs, firms often allow customers to defer some input payments and receive similar deferrals from their suppliers. Claims against customers are recorded as suppliers’ accounts receivable, while liabilities to their own suppliers are recorded as accounts payable. The Quarterly Financial Report (QFR), a survey of U.S. firms’ financial positions conducted by the U.S. Census Bureau, shows that the average ratios of accounts receivable and accounts payable to total assets for large firms (assets exceeding \$250 million) were 7.9% and 6.3%, respectively, during 2012–2019, while for small firms, these ratios were 23.3% and 12.9%. Among short-term financing sources, large firms have significantly more accounts payable, at least nine times more than short-term bank loans and commercial paper, whereas small firms hold twice as much accounts payable as short-term bank loans.¹¹

Since bilateral trade credit data are unavailable, we construct sector-level measures of changes in trade credit during the Great Recession. Using Compustat, we calculate the ratio of accounts receivable to average sales between the current and previous quarters (the AR-to-sales ratio) to measure the intensity of trade credit provision, and the ratio of accounts payable to average operating cost (the AP-to-costs ratio) to measure the intensity of trade credit reception.¹² For each firm, we take the average over two periods, 2005Q3–2007Q2 and 2008Q3–2009Q1, and then calculate their first difference. Finally, we use the median value across firms within each sector to represent the sectoral change in trade credit usage. Our sample is reduced to 44 sectors, retaining only those with more than three firms. Our measure is in line with that of [Kiyotaki and Moore \(1997a\)](#), who demonstrate that the supplier’s ability to provide liquidity to its shocked customers, rather than the extent to which customers rely on the supplier’s trade credit, is critical for the transmission of liquidity shocks.¹³

We then categorize pairs in the two-way and one-way trading groups into two sub-

¹¹Globally, firms in the Worldscope database typically finance about 20% of their working capital with trade credit, and in 60% of countries, firms use trade credit more than bank credit for short-term financing.

¹²We select the sample following [Kahle and Stulz \(2013\)](#). Please refer to Online Appendix I.3 for details.

¹³We use pre-recession trade credit usage to classify our pairs into trade credit-dependent and independent subgroups. However, we do not find that the trade credit-dependent subgroup significantly comoved more during the Great Recession.

groups based on whether they experienced a decline in trade credit during the Great Recession. Specifically, a pair is assigned to the trade-credit-declined subgroup if the supplier sector's AR-to-sales ratio and the customer sector's AP-to-costs ratio both declined by more than the corresponding medians across all public firms, which are -1.6 and -1.0 percentage points, respectively.¹⁴ Otherwise, the pair is assigned to the trade-credit-unchanged subgroup.¹⁵ In total, 189 pairs in the two-way trading group experienced a trade credit decline and 128 did not, whereas the corresponding numbers for one-way trading pairs are 133 and 326.

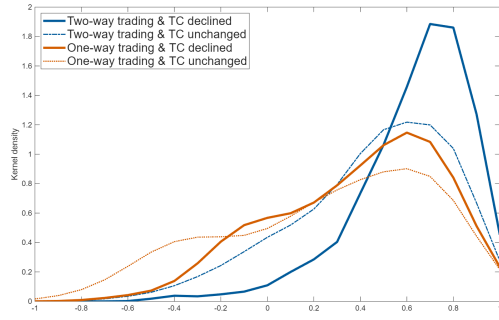
Panel (a) of Figure 4 compares the kernel densities of pairwise correlations during the Great Recession for the four subgroups. Within the two-way trading group, sector pairs that experienced a decline in trade credit exhibit substantially stronger comovement than those whose trade credit remained unchanged. As reported in Panels (e) and (f) of Table 1, the average correlation during the recession is substantially higher for two-way trading pairs with trade credit declines (0.63) compared to those whose trade credit remained unchanged (0.45). The 0.18 difference is equivalent to approximately 43% of the overall increase in pairwise correlations among two-way trading pairs.

Panel (b) reinforces this pattern by comparing changes in pairwise correlations from before to during the Great Recession. The distribution for two-way trading pairs with declining trade credit is shifted furthest to the right, indicating the largest increase in comovement. The contrast is less pronounced for one-way trading pairs. Moreover, Panels (c) and (d) of Figure 4 show that the four subgroups exhibit no meaningful distributional differences either before or after the Great Recession. As a complementary placebo test, we divide the no-trading group into trade-credit-declined and trade-credit-unchanged subgroups. Figure II.8 in Online Appendix II.4 shows no meaningful differences between the two subgroups before, during, or after the Great Recession.

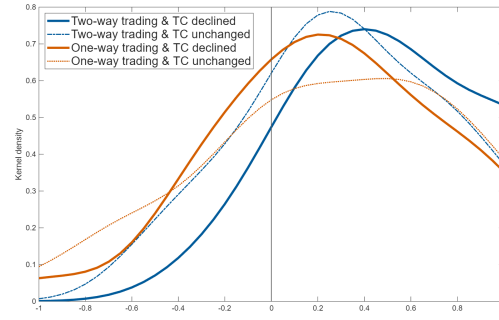
These sector-level facts should be interpreted as suggestive evidence on the relevance of the trade-credit channel rather than as causal evidence. To provide additional evidence on the underlying mechanism, we exploit cross-sectional variation across U.S. public firms during the Great Recession. Specifically, we use the collapse of Lehman Brothers (LB) as a quasi-natural experiment, combining pre-crisis firm-to-firm supply-chain networks with firm-to-bank lending relationships. Restricting firm pairs to suppliers that

¹⁴Note that both ratios should be interpreted as realized trade-credit provision or receipts rather than direct observations of bilateral payment terms. Passive repayment delays would tend to raise accounts receivable and accounts payable, whereas our trade-credit-decline classification requires both the supplier-side AR-to-sales ratio and the customer-side AP-to-costs ratio to fall.

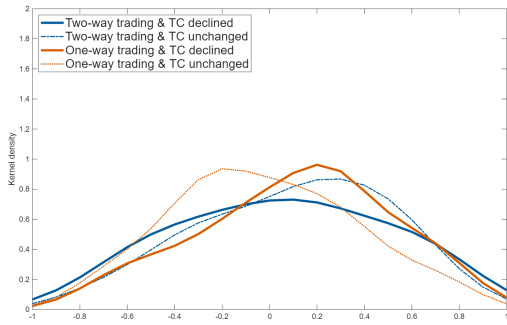
¹⁵Since each sector can be both supplier and customer in the two-way trading group, we classify the pair in the trade-credit-declined subgroup if the trade-credit-decline condition is satisfied in either direction.



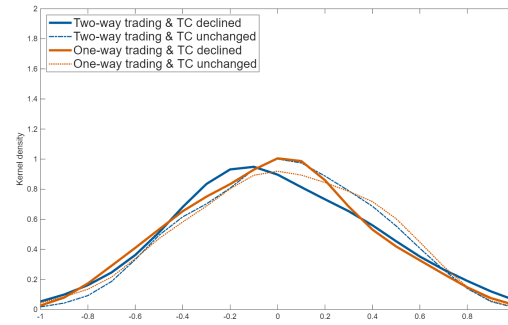
(a) DURING THE GREAT RECESSION



(b) CHANGE FROM PRE-RECESSION CORRELATION



(c) BEFORE THE GREAT RECESSION



(d) AFTER THE GREAT RECESSION

Note: Pairwise correlations are calculated as in Figure 2. The figure restricts the sample to two-way and one-way trading pairs with available sector-level trade-credit measures. Panel (a) compares the distributions during the Great Recession; Panel (b) plots the change in pairwise correlations from the pre-recession window to the Great Recession; Panels (c) and (d) report the corresponding distributions before and after the Great Recession. A pair is classified in the trade-credit-declined subgroup if the supplier sector's AR-to-sales ratio and the customer sector's AP-to-costs ratio both declined by more than the corresponding medians across public firms, respectively -1.6 and -1.0 percentage points. For two-way trading pairs, the condition can hold in either supplier-customer direction. The four groups, two-way declined, two-way unchanged, one-way declined, and one-way unchanged, contain 189, 128, 133, and 326 pairs, respectively. The equal-weight kernel density is estimated with a bandwidth of 0.1.

Figure 4
PAIRWISE CORRELATION BY WHETHER TRADE CREDIT DECLINES

did not borrow from LB before 2008 but had both LB-exposed and unexposed customers, we compare how these common suppliers comoved with the two groups of customers following the collapse. As shown in Online Appendix III, suppliers exhibit stronger comovement with LB-exposed customers, particularly when those customers experienced larger declines in trade credit received after the LB collapse.

2.4 Discussion

Due to delivery delays, adjustment costs, search frictions, and other factors, two sectors may not comove contemporaneously, with one possibly leading the other. Thus, the rise in sectoral comovement during the Great Recession may simply result from synchronization in timing. To address this concern, we follow the method used in [Christiano and Fitzgerald \(1998\)](#), [Hornstein \(2000\)](#), and [Kim and Kim \(2006\)](#), calculating one-period lagged and lead correlations and then taking the maximum value among the three. As displayed in Figure II.9 in Online Appendix II.5, we find that, during the Great Recession, the kernel density of the maximal correlations still shifts significantly to the right. This result suggests that the rise in sectoral comovement is not merely a consequence of lead-lag timing across sectors.

Notably, our sectoral evidence points to the role of supply-side propagation in the rise of comovement during the Great Recession. One alternative explanation is that the pattern reflects common movements in final demand. For example, [Mian et al. \(2013\)](#) argue that the collapse of house prices generated a negative wealth effect that reduced consumption, especially in high-mortgage-debt areas. Investment demand may also have moved sharply during the crisis ([Campello et al., 2010](#); [Duchin et al., 2010](#)). To examine these possibilities, we classify sectors according to the average share of output used as intermediate inputs, final consumption, and investment goods.¹⁶ A sector is classified as input-intensive, consumption-intensive, or investment-intensive if the corresponding share is above the 67th percentile of the cross-sectoral distribution.

In Online Appendix II.5, we compare within-group pairwise correlations. Figure II.10 contrasts input-intensive pairs with consumption-intensive pairs, while Figure II.11 contrasts input-intensive pairs with investment-intensive pairs. In both figures, the distribution for input-intensive pairs shifts furthest to the right during the Great Recession, while consumption-intensive and investment-intensive pairs exhibit much smaller increases in comovement during the recession. Moreover, the distributions of all three groups are similar before and after the recession. These patterns suggest that the rise in sectoral comovement is concentrated among sectors whose output is used intensively as intermediate inputs, consistent with the importance of production-network propagation.

¹⁶Please refer to Table IV.13 for the sector-level values.

3 Model

In this section, we develop a multisector production network model with endogenous trade credit, building on [Bigio and La'O \(2020\)](#). Trade credit arises endogenously because it facilitates transactions under input-quality uncertainty ([Smith, 1987](#); [Antras and Foley, 2015](#)) and helps sustain valuable supplier-customer relationships ([Cuñat, 2007](#); [Brugués, 2026](#)). These benefits are balanced against the financing costs of extending trade credit, allowing trade credit to adjust endogenously to firms' financial conditions.

3.1 Firm, trade credit, and trading relationship

Suppose that the economy consists of n sectors, each populated by a continuum of firms indexed by $h \in [0, 1]$. Firms hire labor and purchase intermediate inputs to produce. Each firm purchases inputs from at most one firm in each supplier sector.¹⁷ To simplify notation, we suppress firm indices. The production function of a representative firm in sector i is

$$y_i = z_i l_i^{\alpha_i} m_i^{\xi_i}, \quad (1)$$

where z_i is sectoral TFP, l_i is labor, and m_i is an intermediate-input bundle.¹⁸ The parameters α_i and ξ_i denote the sectoral labor and intermediate-input shares, with $\alpha_i + \xi_i < 1$. The intermediate-input bundle is

$$m_i = \left(\sum_{j=1}^n \lambda_{ji}^{\frac{1}{\sigma}} m_{ji}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (2)$$

where m_{ji} denotes inputs purchased by a customer in sector i from a supplier in sector j , λ_{ji} is the bilateral input weight on sector- j inputs in sector i 's intermediate-input composite, and $\sigma > 0$ is the elasticity of substitution across intermediate inputs. The weights satisfy $\sum_{j=1}^n \lambda_{ji} = 1$. A value of $\lambda_{ji} = 0$ implies that firms in sector i do not purchase inputs from sector j .

Products can be used either for final consumption or as intermediate inputs. Firms in every sector therefore act both as input customers and input suppliers. Following [Smith \(1987\)](#) and [Antras and Foley \(2015\)](#), we assume that delivered inputs are subject

¹⁷This assumption is not essential; it is imposed only to avoid a coordination problem. We also assume that a firm cannot simultaneously be a supplier and a customer of the same firm.

¹⁸Each sector contains a continuum of ex ante identical firms. In equilibrium, all firms within a sector make identical choices. Accordingly, bilateral terms between suppliers in sector i and customers in sector j are represented by sector-pair objects.

to quality uncertainty.¹⁹ With probability η , a delivered input is suitable for production (the high-quality state); with probability $1-\eta$, it is unsuitable (the low-quality state). This creates a role for trade credit because deferred payment allows customers to condition part of repayment on realized input quality.

Let $tc_{ji} \in [0, 1]$ denote the fraction of the payment for inputs from supplier j to customer i that is deferred through trade credit. The customer pays the nondeferred fraction immediately and repays the deferred fraction only if the input turns out to be of high quality. Hence, for a face price p_j , the *expected repayment price* paid by customer i is

$$(1 - tc_{ji})p_j + \eta tc_{ji}p_j = \left[1 - (1 - \eta)tc_{ji}\right]p_j. \quad (3)$$

For a given face price, higher trade credit lowers the customer's expected repayment price by shifting a larger fraction of payment contingent on input quality. This channel is consistent with the view that suppliers can use payment terms, rather than posted prices alone, to accommodate customer input demand. Moreover, because only a fraction η of delivered inputs is expected to be suitable for production, firm i must purchase m_{ji}/η units from supplier j to obtain m_{ji} effective units for production.

From supplier i 's perspective, m_{ij} denotes the input demand generated by customer j 's production decision. Thus, when supplier i chooses the amount of trade credit extended to customer j , tc_{ij} , it internalizes this effect that the expected repayment price directly affects input demand. As we will show later, a higher level of trade credit lowers the expected repayment price faced by customer j , and thus increases its input demand, m_{ij} . This mechanism is consistent with the pricing-power view of trade credit, which emphasizes that suppliers may optimally subsidize customer financing in order to expand sales or transfer surplus without directly reducing posted input prices (see, among others, [Brennan et al., 1988](#); [Mian and Smith, 1992](#); [Giannetti et al., 2021](#); [Ersahin et al., 2024](#)).

Suppliers also value customer relationships. Empirical evidence suggests that trade credit is not merely a financing arrangement but also an instrument that supports ongoing buyer-supplier relationships (see, for example, [McMillan and Woodruff, 1999](#); [Breza and Liberman, 2017](#); [Ersahin et al., 2024](#)).²⁰ Trade credit and relationship-maintenance

¹⁹Empirical evidence supports this product-quality or seller-performance rationale for trade credit. [Ng et al. \(1999\)](#) show that credit terms vary systematically with information frictions related to product quality and buyer creditworthiness. Using transaction-level contract data, [Antras and Foley \(2015\)](#) show that payment terms in international trade balance importer default risk against exporter performance risk, including the possibility that goods are not delivered as specified.

²⁰In particular, [McMillan and Woodruff \(1999\)](#) show that informal credit is more prevalent in longer and better-informed supplier relationships, especially when network enforcement is important. [Breza and Liberman \(2017\)](#) find that restricting the maturity of trade-credit contracts reduces the likelihood of con-

activities, such as customization, communication, monitoring, and support, can strengthen supplier-customer relationships and increase the surplus generated by continued trade. We summarize the value generated by these activities using the following parsimonious reduced-form measure of relationship surplus:

$$S_{ij}(tc_{ij}, e_{ij}) = \phi tc_{ij}^{1-\kappa} e_{ij}^{\kappa} \frac{p_i m_{ij}}{\eta}, \quad (4)$$

where ϕ governs the overall importance of relationship surplus in the bilateral relationship. The variable tc_{ij} captures financial accommodation through trade credit. Greater trade credit facilitates customer financing, reduces the likelihood of relationship disruption following adverse shocks, and thereby helps preserve trading relationships. The variable e_{ij} denotes nonfinancial relationship-maintenance effort. Such efforts strengthen supplier-customer ties, improve coordination between trading partners, and increase the surplus generated by continued trade. $\kappa \in (0, 1)$ governs the relative importance of effort in generating relationship surplus. Thus, the specification implies that relationship surplus is increasing and concave in both trade credit provision and relationship-maintenance effort. Finally, the term $p_i m_{ij}/\eta$ scales relationship surplus with the size of the trading relationship, reflecting the idea that larger trading relationships generate greater future surplus and therefore stronger incentives for relationship preservation. Because the paper focuses on the macroeconomic implications of trade credit, we do not explicitly model the formation and dissolution of trading relationships. Instead, we take the set of supplier-customer relationships as given within each period and focus on the intensive margin of existing relationships: trade-credit provision, relationship-maintenance effort, and input sales.

Moreover, the Cobb-Douglas form is adopted for parsimony and tractability.²¹ The specification is closely related to the customer-capital and labor-search literature. The customer-capital literature emphasizes that firms invest resources in acquiring and maintaining valuable customer relationships, which constitute an important form of intangible capital (see, for example, [Klemperer, 1987, 1995](#); [Ravn et al., 2006](#); [Gourio and Rudanko, 2014](#)). For example, [Gourio and Rudanko \(2014\)](#) employ a Cobb-Douglas

tinued trade between buyers and suppliers and induces firms to substitute toward internal procurement. More recently, [Ersahin et al. \(2024\)](#) show that suppliers exposed to disruptions extend more trade credit to customers and that relationships are more likely to break when constrained suppliers cannot provide additional credit.

²¹It is increasing in both arguments, exhibits diminishing marginal returns, and captures the idea that financial accommodation and relationship-maintenance effort jointly strengthen supplier-customer relationships. The main mechanism does not rely on this particular functional form. A more general increasing and concave relationship-surplus technology would deliver similar qualitative implications, provided that both instruments increase the value of the relationship and are costly to the supplier.

matching technology to determine the formation of new customer relationships. Similarly, the labor-search literature commonly uses Cobb-Douglas functions to map search inputs into productive worker-firm matches (Mortensen and Pissarides, 1994; Moen, 1997). In the same spirit, Equation (4) combines financial accommodation and relationship-maintenance effort through a Cobb-Douglas relationship-production technology.

Also, maintaining such relationship is costly. Denote the cost of supplier i 's effort maintaining relationship with customer j as

$$\psi_j e_{ij} \frac{p_i m_{ij}}{\eta}$$

where ψ_j governs the marginal cost of maintaining the relationship with customer j . This parameter captures customer-level heterogeneity in the cost of customization, monitoring, coordination, communication, and relationship-specific support.

Although the supplier bears the cost of maintaining the relationship, we assume that the resulting relationship surplus is also valued by the customer. Consequently, a customer remains with its incumbent supplier only if the value of continuing the relationship, which includes both the expected payment and the relationship surplus, is at least as large as the value of switching to an alternative supplier. We summarize the outside option by $-\gamma p_i m_{ij}/\eta$, where a higher γ corresponds to a less attractive outside option, or equivalently a larger cost of switching to a new supplier. The relationship-retention constraint therefore requires that customer j weakly prefer continuing the existing relationship to switching to an alternative supplier:

$$-\left(1 - (1 - \eta)tc_{ij}\right) \frac{p_i m_{ij}}{\eta} + \phi tc_{ij}^{1-\kappa} e_{ij}^{\kappa} \frac{p_i m_{ij}}{\eta} \geq -\gamma \frac{p_i m_{ij}}{\eta}. \quad (5)$$

Equivalently, after dividing by $p_i m_{ij}/\eta$ and rearranging,

$$(1 - \eta)tc_{ij} + \phi tc_{ij}^{1-\kappa} e_{ij}^{\kappa} \geq 1 - \gamma.$$

Trade credit lowers the customer's expected repayment and contributes to relationship surplus, while relationship-maintenance effort increases relationship surplus.

3.2 Timing, financing, and firm problem

Each period consists of two stages. In the first stage, sectoral productivity levels are realized. Firms choose labor, intermediate-input purchases, trade credit provision, and

relationship-maintenance effort. Labor must be paid fully upfront.²² By contrast, payments for intermediate inputs can be split between cash before delivery (CBD) and trade credit. In the second stage, input quality is realized, production takes place, and repayment of trade credit is contingent on realized input quality.

Firms cannot retain earnings across periods. Nevertheless, production requires working capital to finance wage payments, upfront cash payments to suppliers, and relationship-maintenance expenditures. These working-capital needs can be financed either through cash received upfront from customers or through bank borrowing. When working-capital requirements exceed CBD payments received from customers, firms borrow from perfectly competitive banks using output as collateral.²³ Let θ_i denote the liquidation value of firm i 's collateral. Borrowing is therefore subject to the following collateral constraint:

$$b_i = \underbrace{wl_i}_{\text{wage}} + \underbrace{\sum_{j=1}^n (1 - tc_{ji}) \frac{p_j m_{ji}}{\eta}}_{\text{paid CBD}} + \underbrace{\sum_{j=1}^n \psi_j e_{ij} \frac{p_i m_{ij}}{\eta}}_{\text{relationship cost}} - \underbrace{\sum_{j=1}^n (1 - tc_{ij}) \frac{p_i m_{ij}}{\eta}}_{\text{received CBD}} \leq \theta_i p_i y_i. \quad (6)$$

where w denotes the wage rate. Following [Kiyotaki and Moore \(1997b\)](#) and [Jermann and Quadrini \(2012\)](#), sectoral variation in θ_i captures differences in financial conditions, while shocks to θ_i represent sector-level financial shocks.

This constraint highlights the two-sided role of trade credit. Receiving more trade credit (a higher tc_{ji}) reduces firm i 's reliance on external finance by lowering the upfront payments required for intermediate inputs, whereas extending more trade credit (a higher tc_{ij}) increases such reliance by delaying cash receipts from customers. Consequently, trade credit reallocates liquidity across firms within the production network, making suppliers' and customers' financing decisions mutually dependent. A supplier's financial capacity determines its ability to extend trade credit and thereby influences the financing needs of its customers. Conversely, a customer's financial condition affects its demand for intermediate inputs and, in turn, the supplier's incentive to provide trade credit. This two-sided dependence creates a financial linkage through which shocks propagate across production networks and affect real economic activity. This mechanism is consistent with the finance view of trade credit (see, for example, [Petersen and Rajan](#),

²²[Miranda-Pinto and Young \(2022\)](#) show that, in input-output models with working-capital constraints, whether labor is paid upfront has little quantitative impact. Instead, the financing of intermediate-input purchases is the primary channel through which financial frictions propagate and amplify through production networks.

²³For simplicity, we abstract away from a formal bank problem. We assume that intermediaries are perfectly competitive and have 'deep pockets,' providing purely intra-period working capital to firms at a normalized interest rate of zero.

1997; Garcia-Appendini and Montoriol-Garriga, 2013; Jacobson and von Schedvin, 2015; Costello, 2020).²⁴

Including relationship-maintenance effort in the financial constraint is also important because it ties the cost of preserving customer relationships to the supplier's financial condition. When choosing relationship effort, suppliers balance the relationship surplus against both the direct cost of maintaining relationships and the shadow cost of working capital. As a result, trade credit and relationship-maintenance effort depend not only on the relationship surplus but also on the supplier's financial capacity. As shown below, this interaction allows suppliers with different financial conditions to respond asymmetrically to customer financial distress when choosing trade credit.

We now state the representative firm first-stage problem. Before input quality is realized, firms within the same sector are *ex ante* identical and therefore choose the same labor, input purchases, trade credit, and relationship effort. As an input customer, the firm in sector i chooses labor l_i and intermediate-input purchases $\{m_{ji}\}_{j=1}^n$, taking as given the wage w , goods prices $\{p_j\}_{j=1}^n$, and trade-credit terms $\{tc_{ji}\}_{j=1}^n$ offered by suppliers. Expected profits are given by

$$\pi_i = p_i z_i l_i^{\alpha_i} \left(\sum_{j=1}^n \lambda_{ji}^{\frac{1}{\sigma}} m_{ji}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma \xi_i}{\sigma-1}} - \sum_{j=1}^n (1-\eta) tc_{ij} \frac{p_i m_{ij}}{\eta} - w l_i - \sum_{j=1}^n \left(1 - (1-\eta) tc_{ji} \right) \frac{p_j m_{ji}}{\eta}. \quad (7)$$

The first two terms represent expected revenue: production revenue net of expected trade-credit losses arising when customers do not repay for low-quality delivered inputs. The last two terms represent expected production costs, consisting of wage payments and expected payments for intermediate inputs.

In addition, as an input supplier, firm i chooses the trade-credit terms and relationship-maintenance effort offered to each of its customers, $\{tc_{ij}, e_{ij}\}_{j=1}^n$, to maximize its value:

$$v_i = \max_{l_i, m_{ji}, tc_{ij}, e_{ij}} \pi_i - \sum_{j=1}^n \psi_j e_{ij} \frac{p_i m_{ij}}{\eta} + \sum_{j=1}^n \phi tc_{ij}^{1-\kappa} e_{ij}^{\kappa} \frac{p_i m_{ij}}{\eta}, \quad (8)$$

subject to the relationship-retention constraint (5) and the collateral constraint (6). The objective consists of expected operating profits, net of relationship-maintenance costs,

²⁴In particular, Petersen and Rajan (1997) and Garcia-Appendini and Montoriol-Garriga (2013) emphasize that firms with greater access to liquidity provide financing to constrained trading partners. Jacobson and von Schedvin (2015) show that customer defaults on trade-credit obligations impose losses on suppliers and increase suppliers' own default risk, providing direct evidence that trade-credit exposures can transmit distress across firms. Using transaction-level data, Costello (2020) show that disruptions to suppliers' access to credit reduce trade-credit provision and transmit liquidity shocks through supply chains.

plus the relationship surplus generated by preserving customer relationships.

3.3 Households and market clearing

A representative household supplies labor, and consumes final goods. The household chooses consumption and labor supply to maximize expected lifetime utility subject to a sequence of budget constraints:

$$\max_{\{c_t, l_t\}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \left(\log c_t - \chi \frac{l_t^{1+\rho}}{1+\rho} \right) \right] \quad \text{s.t.} \quad p_t c_t \leq w_t l_t + \pi_t, \quad (9)$$

where c_t denotes the household's consumption bundle, l_t denotes labor supply, χ governs the disutility from labor, and ρ is the inverse Frisch elasticity of labor supply. The household earns labor income $w_t l_t$ and obtains aggregate firm profits π_t . Moreover, the consumption bundle is a CES aggregate of goods produced across sectors:

$$c_t = \left(\sum_{i=1}^n \zeta_i^{\frac{1}{\epsilon}} c_{it}^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}}, \quad (10)$$

where ϵ denotes the elasticity of substitution across sectoral goods, and ζ_i represents the preference weight on goods produced in sector i , satisfying $\sum_{i=1}^n \zeta_i = 1$. The corresponding aggregate price index is given by

$$p_t = \left(\sum_{i=1}^n \zeta_i p_{it}^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}}. \quad (11)$$

To close the model, labor supply must equal aggregate labor demand across all firms:

$$l = \sum_{i=1}^n l_i. \quad (12)$$

The market-clearing condition for goods produced in sector i is given by, for $\forall i$,

$$y_i = \sum_{j=1}^n m_{ij}/\eta + c_i, \quad (13)$$

where y_i is defined in Equation (1). The term m_{ij}/η denotes the total amount of sector- i goods ordered by firms in sector j , taking into account that only a fraction η of delivered intermediate inputs is expected to be of high quality. The remaining demand comes from final consumption c_i .

4 Equilibrium Analysis

In this section, we characterize the stationary Nash equilibrium and its properties. A stationary Nash equilibrium is a collection of goods prices $\{p_i\}_{i=1}^n$, wage w , resource allocations $\{y_i, l_i, m_i, \{m_{ji}\}_{j=1}^n\}_{i=1}^n$, final consumption $\{c_i\}_{i=1}^n$, trade-credit arrangements $\{tc_{ij}\}$, and relationship-maintenance efforts $\{e_{ij}\}$ such that firms maximize (8), households maximize (9), and labor and goods markets clear according to (12) and (13). We normalize the aggregate price index to one.

4.1 Production decisions

We first characterize the firm's problem as a producer and, therefore, as a customer of intermediate inputs. Taking as given the wage, the prices, and the trade-credit terms offered by suppliers, firm i chooses labor and intermediate inputs. The following lemma summarizes the optimal production plan.

Lemma 1 (Production Plan) *Define the effective input price and the corresponding input-bundle price index as*

$$\tilde{p}_{ji} = (1 - (1 - \eta)tc_{ji} + \mu_i(1 - tc_{ji}))p_j, \quad \tilde{p}_i = \left(\sum_{j=1}^n \lambda_{ji} \tilde{p}_{ji}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}, \quad (14)$$

where μ_i is the multiplier on the collateral constraint. Given the goods prices $\{p_i\}_{i=1}^n$, wage w , and incoming trade-credit terms $\{tc_{ji}\}_{i,j=1}^n$, the optimal production plan for firm i satisfies

$$(1 + \theta_i \mu_i) \alpha_i p_i y_i = (1 + \mu_i) w l_i, \quad (15)$$

and, for each input from supplier j ,

$$(1 + \theta_i \mu_i) \xi_i \lambda_{ji} \left(\frac{\tilde{p}_{ji}}{\tilde{p}_i} \right)^{1-\sigma} p_i y_i = \frac{\tilde{p}_{ji} m_{ji}}{\eta}. \quad (16)$$

Equilibrium output of firm i is

$$y_i = \left[z_i p_i^{\alpha_i + \xi_i} \left(\frac{\alpha_i (1 + \theta_i \mu_i)}{(1 + \mu_i) w} \right)^{\alpha_i} \left(\frac{\eta \xi_i (1 + \theta_i \mu_i)}{\tilde{p}_i} \right)^{\xi_i} \right]^{\frac{1}{1 - \alpha_i - \xi_i}}. \quad (17)$$

Proof: see Appendix A.2.

Lemma 1 shows how trade credit enters production decisions through effective input prices. Trade credit lowers the expected repayment price of inputs and reduces the

amount of cash that must be paid upfront. Therefore, higher incoming trade credit reduces $\tilde{p}_{ji}$, stimulating demand for intermediate inputs and increasing output.

Financial conditions operate in the opposite direction. A lower θ_i tightens the collateral constraint and raises the multiplier μ_i . This increases the effective cost of inputs that must be financed upfront and depresses both labor and intermediate-input demand and reducing output.

4.2 Optimal trade-credit provision

We now examine the supplier's problem. As shown in Equation (16), trade credit affects the customer's input demand. Therefore, when supplier i chooses tc_{ij} , it internalizes how trade credit affects customer j 's demand for inputs, while also taking into account relationship value and its own financing constraint. We focus on the case in which collateral constraints bind for all firms.

Proposition 1 (Optimal relationship contract) *Consider the case in which $\{\theta_i\}_{i=1}^n$ are sufficiently small so that collateral constraints bind for all firms, i.e. $\mu_i > 0$ for all i . Given the tightness of collateral constraints $\{\mu_i\}_{i=1}^n$, supplier i 's optimal relationship-maintenance effort toward customer j is*

$$e_{ij}^* = \left(\frac{(1 + v_{ij})\kappa\phi}{(1 + \mu_i)\psi_j} \right)^{\frac{1}{1-\kappa}} tc_{ij}^*, \quad (18)$$

where v_{ij} is the multiplier on the relationship-retention constraint. Moreover, trade credit satisfies

$$\begin{aligned} 0 = & \underbrace{(1 - \kappa)\phi tc_{ij}^{*- \kappa} e_{ij}^{* \kappa} - (1 - \eta) - \mu_i + v_{ij}}_{\text{supplier relationship-financing trade-off}} \underbrace{\left(1 - \eta + (1 - \kappa)\phi tc_{ij}^{*- \kappa} e_{ij}^{* \kappa} \right)}_{\text{relationship-retention benefit}} \\ & + \underbrace{\frac{\sigma(1 - \eta + \mu_j)}{\eta + (1 - \eta + \mu_j)(1 - tc_{ij}^*)}}_{\text{customer input-demand sensitivity}} \underbrace{\left(\phi tc_{ij}^{* 1 - \kappa} e_{ij}^{* \kappa} - (1 + \mu_i)\psi_j e_{ij}^* - (1 - \eta)tc_{ij}^* + \mu_i(1 - tc_{ij}^*) \right)}_{\text{supplier net value per unit of input demand}} \quad (19) \end{aligned}$$

Proof: see Appendix A.3.

Proposition 1 summarizes the supplier's trade-off when choosing the trade credit extended to a customer. The first term captures the direct payoff effect of increasing trade credit. A higher tc_{ij} raises the relationship surplus generated with customer j , but it also exposes supplier i to greater expected nonpayment in the low-quality state and reduces the received cash before delivery. When supplier i is financially constrained, this latter cost is amplified by the shadow value of working capital, μ_i .

The second term captures the relationship-retention margin. By lowering the customer's expected repayment and contributing to relationship surplus, trade credit helps satisfy the customer-retention constraint. When this constraint binds, additional trade credit is valuable because it makes it easier for the supplier to keep the customer.

The third term captures the input-demand margin, which reflects how trade credit affects supplier's value through changes in customer input demand. It consists of two components. The first measures the sensitivity of customers' demand for intermediate inputs to trade-credit provision. By lowering the customer's effective input price, trade credit increases customer j 's demand for supplier i 's input. This sensitivity depends on the customer's financial tightness, μ_j , because financially constrained customers benefit more from a reduction in upfront payment requirements.

The second component, the supplier's net value from serving an additional unit of customer input demand, measures whether it is worthwhile for the supplier to accommodate the resulting increase in input demand. This net value compares the benefits from relationship surplus, $\phi tc_{ij}^{1-\kappa} e_{ij}^\kappa$, and upfront cash receipts, $\mu_i(1 - tc_{ij})$, with the costs from expected nonpayment, $(1 - \eta)tc_{ij}$, and relationship-maintenance effort, $(1 + \mu_i)\psi_j e_{ij}$. Moreover, supplier financial tightness, μ_i , affects this trade-off in two opposing ways: a higher μ_i increases the value of receiving cash upfront while also raising the financing cost of relationship maintenance. As a result, the supplier's net value from serving additional customer demand declines with μ_i and can eventually become negative.

4.3 Financial tightness and asymmetric response of trade credit

Next, we discuss the existence and uniqueness of equilibrium and explore how trade credit responds to suppliers' and customers' financial conditions, which constitutes the model's core mechanism. We focus on equilibria in which the relationship-retention constraint (5) binds.²⁵ We begin by providing sufficient conditions for the existence and uniqueness of the equilibrium trade-credit intensity.

Assumption 1 (Sufficient condition for interior trade-credit solution) *The model param-*

²⁵We have also considered the case in which the relationship-retention constraint is slack. In that case, trade credit and relationship-maintenance effort can be reduced without affecting the customer's willingness to continue the relationship. Equilibrium trade credit is then determined by balancing the benefits of stronger customer relationships and higher input demand against the costs of repayment losses and the shadow cost of working capital. It is straightforward to show that trade credit responds monotonically to financial conditions: it increases as the supplier's financial condition improves and as the customer's financial condition deteriorates.

eters satisfy the sufficient regularity conditions

$$\left(\eta \left(\frac{1-\gamma}{(\eta-\gamma)\kappa} - 1 \right) - \sigma(1-\eta) \right) \psi_j \left(\frac{\eta-\gamma}{\phi} \right)^{\frac{1}{\kappa}} < 2\eta(1-\eta) + \sigma(1-\eta)(1-2\eta+\gamma), \quad (20)$$

and

$$\eta > (1-\eta)(2(1-\sigma)\eta + \sigma(1-\gamma)). \quad (21)$$

Condition (20) ensures that even a financially unconstrained supplier does not choose full trade credit, while condition (21) extends this requirement to financially constrained suppliers. Proposition 2 establishes the existence and uniqueness of the equilibrium and characterizes how trade credit responds to the financial conditions of suppliers and customers. This result is central to the paper because it identifies the conditions under which trade credit acts as a cushion or an amplifier for customer financial distress.

Proposition 2 (Response of trade credit to supplier and customer financial tightness)

Suppose that the relationship-retention constraint binds and that Conditions (20) and (21) hold. Then there exists a unique interior level of trade credit, $tc_{ij}^* \in (0, 1)$. Moreover, supplier financial tightness unambiguously reduces trade credit:

$$\frac{\partial tc_{ij}^*}{\partial \mu_i} < 0.$$

The effect of customer financial tightness depends on the supplier's financial tightness:

$$\frac{\partial tc_{ij}^*}{\partial \mu_j} \begin{cases} \geq 0, & \text{if } \mu_i \leq g(tc_{ij}^*, e_{ij}^*), \\ < 0, & \text{if } \mu_i > g(tc_{ij}^*, e_{ij}^*), \end{cases}$$

where the threshold $g(tc_{ij}^*, e_{ij}^*)$ is characterized in Appendix A.4.

Proof: see Appendix A.4.

For a given equilibrium trade credit tc_{ij}^* and relationship-maintenance effort e_{ij}^* , the threshold $g(tc_{ij}^*, e_{ij}^*)$ represents the maximum level of supplier financial tightness at which serving additional customer demand remains valuable to the supplier. We refer to the region $\mu_i \leq g(tc_{ij}^*, e_{ij}^*)$ as the *cushion region*. In this region, serving additional customer demand remains valuable to the supplier, so the supplier weakly expands trade credit when the customer becomes more financially constrained. We refer to the region $\mu_i > g(tc_{ij}^*, e_{ij}^*)$ as the *amplification region*. In this region, the supplier's own financing constraint is sufficiently tight and serving additional customer demand becomes costly to the supplier, so that the supplier reduces trade credit in response to customer financial distress.

The first result in Proposition 2 is that supplier financial tightness unambiguously reduces trade credit. A higher supplier multiplier μ_i raises the shadow value of working capital. Since trade credit delays upfront cash received from customers, a financially constrained supplier requires more upfront payment and reduces the trade credit extended to its customers.

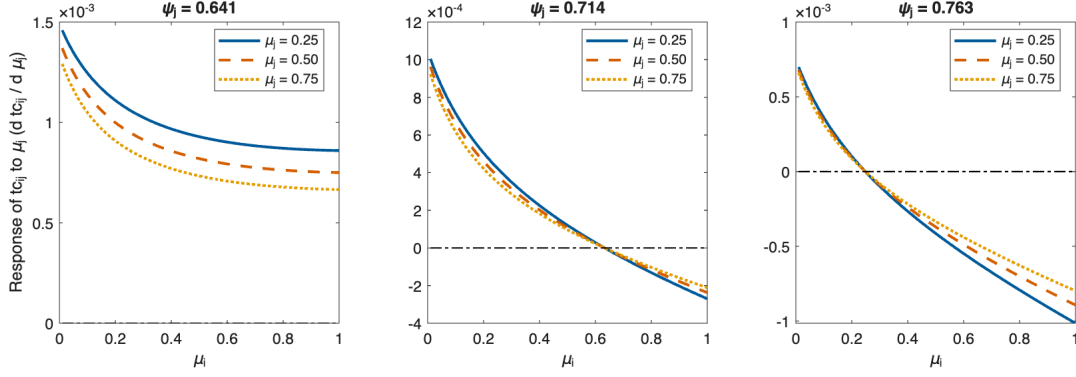
The second result is that customer financial tightness (μ_j) has an asymmetric effect on trade-credit provision, depending on the supplier's financial tightness (μ_i). This asymmetry arises through the input-demand margin in Equation (19). A deterioration in customer financial conditions (a higher μ_j) increases the sensitivity of customer input demand to trade-credit provision, making trade credit more valuable to the customer.

Moreover, how the supplier responds to this stronger demand depends on the supplier's net value from serving the resulting increase in customer demand. Because a higher μ_i raises both the value of receiving cash upfront and the financing cost of relationship maintenance, this net value declines with supplier financial tightness. In the *cushion region*, where $\mu_i \leq g(tc_{ij}^*, e_{ij}^*)$, serving additional customer demand remains valuable to the supplier. Consequently, a deterioration in customer financial conditions induces the supplier to extend trade credit, allowing trade credit to cushion customer financial distress.

In contrast, in the *amplification region*, where $\mu_i > g(tc_{ij}^*, e_{ij}^*)$, serving additional customer demand becomes sufficiently costly that the supplier contracts trade credit when customer financial conditions deteriorate. This increases the customer's upfront payment burden and amplifies customer financial distress.

Figure 5 illustrates the asymmetric response of trade credit to customer financial tightness. For each combination of supplier financial tightness, μ_i , and customer financial tightness, μ_j , we solve for the equilibrium levels of trade credit and relationship-maintenance effort and compute the response of equilibrium trade credit to an increase in the customers' Lagrange multiplier μ_j . Each panel corresponds to a different value of the relationship-maintenance cost parameter, ψ_j , calibrated to the 25th, 50th, and 75th percentiles of its empirical distribution across sectors, as estimated in Section 5.2. The horizontal zero line separates the cushioning and amplification regimes implied by Proposition 2. For each response curve, the intersection with the zero line identifies the threshold function $g(tc_{ij}^*, e_{ij}^*)$, at which the response changes sign.

This pattern illustrates the paper's central mechanism. Customer financial distress increases the demand-side value of trade credit, but whether suppliers accommodate that demand depends on their own financial condition. Relatively unconstrained suppliers find it worthwhile to support distressed customers by extending additional trade credit,



Note: The figure plots $\partial tc_{ij}^* / \partial \mu_j$ against supplier financial tightness μ_i . Each line corresponds to a different value of customer financial tightness μ_j , and each panel corresponds to a different value of the relationship-maintenance cost parameter ψ_j , evaluated at the 25th, 50th, and 75th percentiles of its empirical distribution across sectors, as estimated in Section 5.2. The horizontal dashed line marks zero; values above zero correspond to the cushioning regime, while values below zero correspond to the amplification regime.

Figure 5

RESPONSE OF TRADE CREDIT PROVISION TO CUSTOMER'S FINANCIAL TIGHTNESS

whereas sufficiently financially constrained suppliers do not. The figure also shows that this response depends on relationship-maintenance costs. When ψ_j is low, panel a) of Figure 5, preserving supplier-customer relationships is relatively inexpensive, so the supplier's net value from serving additional customer demand remains positive over a wider range of supplier financial conditions. Suppliers therefore expand trade credit in response to customer distress. As ψ_j increases, panels b) and c) of Figure 5, relationship maintenance becomes more costly, reducing the supplier's net value and shifting the threshold $\mathbf{g}(tc_{ij}^*, e_{ij}^*)$ downward. Consequently, trade credit turns from a cushion into an amplification mechanism at lower levels of supplier financial tightness.

5 Model quantification

We calibrate the model and recover the time paths of exogenous variables in three steps. First, we assign common parameters based on estimates from the existing literature and then directly map a subset of model moments to their empirical counterparts. Second, we estimate the remaining sector-specific parameters by matching model-implied moments to sectoral consumption expenditure shares, input-output linkages, and trade-credit moments under the stationary equilibrium conditions. Finally, using the calibrated model together with observed sectoral output and bond spreads, we recover the time paths of sectoral productivity and financial conditions. Note that sectoral output and sectoral bond spreads in the pre-crisis are used to back out the values of sectoral productivity and sectoral Lagrange multiplier in the stationary equilibrium.

5.1 Calibration

We normalize the labor-disutility parameter, χ , and the scale parameter governing relationship surplus, ϕ , to one. The elasticity of substitution across final consumption goods is set to $\epsilon = 1.5$, while the inverse Frisch elasticity is set to $\rho = 0.55$. Because the model is calibrated at the quarterly frequency, we set the elasticity of substitution across intermediate inputs to $\sigma = 0.01$, an intermediate value between the monthly estimate of [Boehm et al. \(2019\)](#) ($\sigma \approx 0$) and the annual estimate of [Atalay \(2017\)](#) ($\sigma \approx 0.1$).

We calibrate κ using the empirical evidence in [Brugués \(2026\)](#), which links trade-credit usage to discounted prices through the duration of supplier–customer relationships. To map these estimates into our framework, we interpret $\frac{1}{\phi t c_{ij}^{1-\kappa} e_{ij}^{\kappa}}$ as a relationship-specific price discount. Under this interpretation, the elasticity of the discount rate with respect to trade credit is equal to $-(1 - \kappa)$. Using the corresponding estimate reported by [Brugués \(2026\)](#), we obtain $1 - \kappa = 0.21$, implying $\kappa = 0.79$.

Direct evidence on trade-credit delinquency rates in the United States is limited. Using transaction-level payment data, [Boissay and Gropp \(2013\)](#) report that 18.5% of French firms experience at least one trade-credit default per quarter on average, while [Jacobson and von Schedvin \(2015\)](#) estimate a trade-credit failure rate of approximately 10% in Sweden between 1992 and 2005. We therefore take the average of these estimates and set the probability that delivered inputs are suitable for production to $\eta = 0.86$.

The parameter γ governs the cost of the customer’s outside option. A higher value of γ increases the cost of switching to an alternative supplier, making customer-supplier relationships more persistent. We calibrate γ by matching the relationship-retention probability. Specifically, we target a quarterly relationship-retention probability of 0.898, which corresponds to an annual retention probability of approximately 0.65 in [Brugués \(2026\)](#). This calibration yields $\gamma = 0.14$.²⁶

We calibrate the model using 12 sectors corresponding to the two-digit private non-farm, non-FIRE NAICS industries, which together represent the major productive sectors of the U.S. economy.²⁷ Several sector-specific parameters are directly calibrated from the 2005 input–output table (IOT) constructed by the BEA, including sectoral labor shares $\{\alpha_i\}$ and intermediate-input shares $\{\xi_i\}$. Table 2 reports the resulting parameter values.

²⁶Online Appendix IV.1 provides a microfoundation linking γ to the relationship-retention probability. In the appendix, the customer’s switching decision depends on both a deterministic switching cost, governed by γ , and an idiosyncratic stochastic component. The resulting relationship-retention probability is determined jointly by γ and the equilibrium contract. Calibrating the average quarterly retention probability to 0.898 implies $\gamma = 0.14$.

²⁷Online Appendix IV.2 shows that the stylized facts documented in Section 2 remain robust when the analysis is restricted to the 12-sector sample.

5.2 Moment-matched estimation

Next, we estimate the remaining sectoral parameters by matching model-implied moments to their empirical counterparts, subject to the stationary-equilibrium conditions. Under the stationary equilibrium, observed pre-crisis sectoral output and bond spreads uniquely pin down sectoral productivity (z_i) and the sectoral Lagrange multiplier (μ_i). The targeted moments are sectoral consumption expenditure shares, bilateral input expenditure shares, and sectoral accounts-payable-to-costs ratios. The first two sets of moments are constructed from the 2005 IOT, while the accounts-payable-to-costs ratios are computed using the average over 2005Q3–2007Q2, as described in Section 2.3. Although each group of moments is most informative about a particular set of parameters, all moments are jointly determined by equilibrium prices, quantities, and trade credit. Consequently, the remaining sectoral parameters are identified jointly through the model's equilibrium rather than through a one-to-one mapping.

The preference weights $\{\zeta_i\}$ in Equation (10) determine the relative importance of each sector in the household consumption aggregator. Holding relative prices fixed, a larger ζ_i increases the expenditure share of sector i 's good. The corresponding model-implied consumption expenditure share is

$$\text{consumption expenditure share}_i = \frac{p_i c_i}{\sum_{k=1}^n p_k c_k} = \zeta_i p_i^{1-\epsilon}.$$

Similarly, the bilateral input weights $\{\lambda_{ji}\}$ in Equation (2) determine the composition of sector i 's intermediate-input bundle across supplier sectors. Conditional on equilibrium prices, trade credit arrangements, and financing tightness, a larger λ_{ji} raises the share of intermediate-input expenditure, by customer sector i , allocated to supplier sector j . The corresponding model-implied bilateral input expenditure share is

$$\text{input expenditure shares}_{ji} = \frac{p_j m_{ji}/\eta}{\sum_{k=1}^n p_k m_{ki}/\eta} = \frac{\lambda_{ji} \left(1 - (1-\eta)tc_{ji}^* + \mu_i(1 - tc_{ji}^*)\right)^{-\sigma} p_j^{1-\sigma}}{\sum_{k=1}^n \lambda_{ki} \left(1 - (1-\eta)tc_{ki}^* + \mu_i(1 - tc_{ki}^*)\right)^{-\sigma} p_k^{1-\sigma}}.$$

Finally, the relationship-maintenance cost parameters $\{\psi_i\}$ capture heterogeneity in the cost of maintaining customer relationships across sectors. A higher ψ_i implies that customization, monitoring, coordination, communication, and other relationship-specific services are more costly, reducing suppliers' incentives to extend trade credit and maintain customer relationships in sector i . We discipline these parameters using sectoral

accounts-payable-to-costs ratios, whose model counterpart is

$$\text{accounts-payable-to-costs ratio}_i = \frac{\sum_{j=1}^n t c_{ji}^* p_j m_{ji} / \eta}{w l_i + \sum_{j=1}^n p_j m_{ji} / \eta}.$$

Therefore, we choose $\{\zeta_i\}$, $\{\lambda_{ji}\}$, and $\{\psi_i\}$ jointly by minimizing the distance between these model-implied moments and their empirical counterparts:

$$\min_{\{\zeta_i, \lambda_{ji}, \psi_i\}} \sum_k (\text{model}(k) - \text{data}(k))^2, \quad (22)$$

subject to the stationary-equilibrium conditions of the model, $\sum_i \zeta_i = 1$, and $\sum_j \lambda_{ji} = 1$ for every sector i . Table 2 reports the estimated values of $\{\zeta_i\}$ and $\{\psi_i\}$.²⁸

Table 2
CALIBRATION FOR SECTORAL PARAMETERS

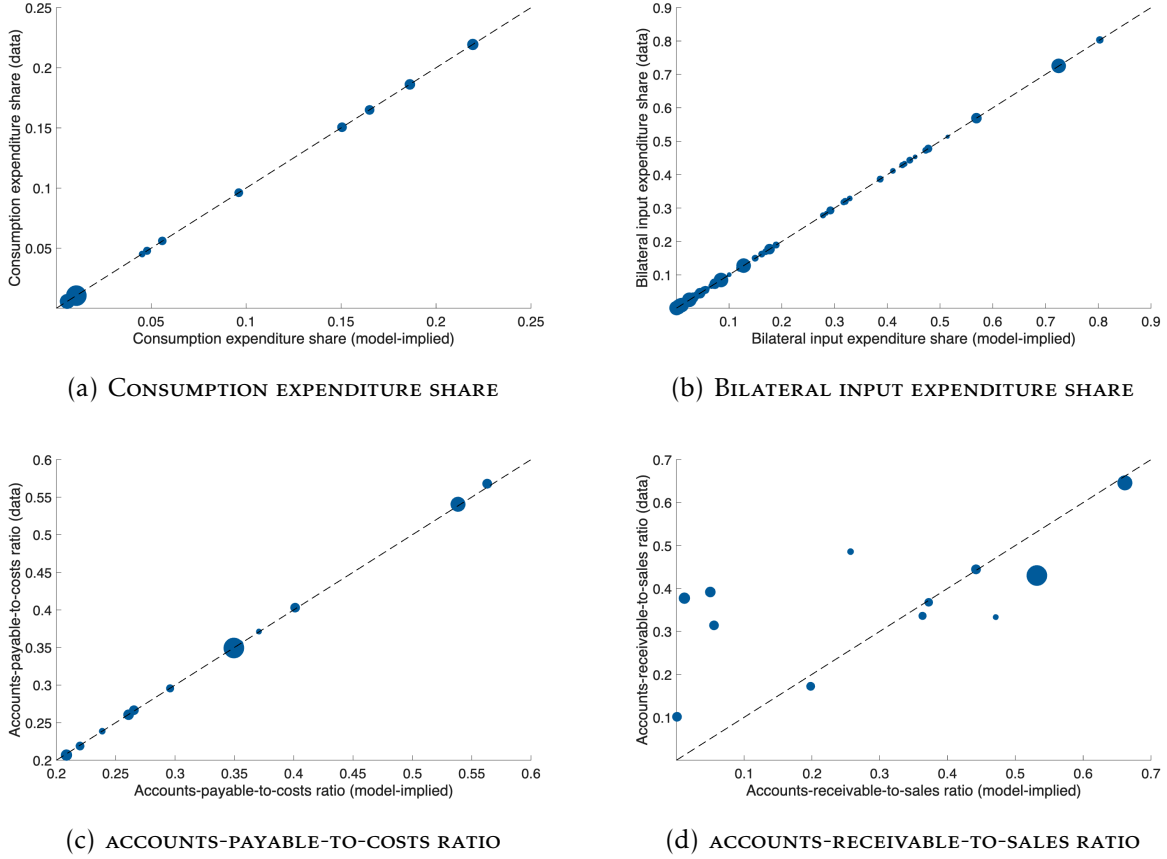
Sectors	ξ	α	ζ	ψ
Mining	0.440	0.537	0.002	1.423
Utilities	0.557	0.414	0.005	1.029
Construction	0.494	0.316	0.131	0.458
Manufacturing	0.650	0.197	0.004	0.641
Wholesale trade	0.370	0.418	0.119	1.057
Retail trade	0.367	0.461	0.227	0.763
Transportation and warehousing	0.509	0.330	0.020	0.637
Information	0.447	0.232	0.023	0.714
Professional and business services	0.372	0.478	0.002	1.396
Educational services and health care	0.387	0.538	0.408	0.458
Arts and recreation services	0.465	0.381	0.042	0.458
Other services	0.382	0.452	0.018	0.667

Note: The table reports sectoral parameters from the calibrated stationary equilibrium. ξ_i is the intermediate-input share, α_i is the labor share, ζ_i is the CES final-demand weight, and ψ_i is the relationship-maintenance cost parameter.

Figure 6 evaluates the fit of the calibrated model. Panels (a)–(c) compare the targeted moments in the model and the data: sectoral consumption expenditure shares, bilateral input expenditure shares, and sectoral accounts-payable-to-costs ratios. In each case, the observations lie close to the 45-degree line, indicating that the calibration closely reproduces the final-demand, input-output, and trade-credit moments used in the estimation. Panel (d) provides an external validation by comparing model-implied accounts-

²⁸Although Assumption 1 establishes sufficient conditions to guarantee a unique interior solution for the analytical characterization in Proposition 2, these conditions are not necessary. Here, some sectoral parameter values exceed these conservative analytical bounds. Our numerical simulation solves the exact nonlinear problem, naturally accommodating interior solutions outside these bounds, as well as instances where trade credit reaches corner solutions or the relationship-retention constraint does not bind.

receivable-to-sales ratios, $\sum_{j=1}^n tc_{ij}^* p_i m_{ij} / (\eta p_i y_i)$, with their empirical counterparts. Although this moment is not targeted in estimation, the model captures a substantial share of the cross-sectoral variation, with a model-data correlation of approximately 0.63.



Note: The figure compares model-implied moments (horizontal axis) with their empirical counterparts (vertical axis). Panel (a) plots consumption expenditure shares, Panel (b) plots bilateral input expenditure shares, Panel (c) plots accounts-payable-to-costs ratios, and Panel (d) plots accounts-receivable-to-sales ratios. Consumption expenditure shares, bilateral input expenditure shares, and accounts-payable-to-costs ratios are targeted in the estimation, while accounts-receivable-to-sales ratios are not targeted. Marker size is proportional to sector size. The dashed line is the 45-degree line.

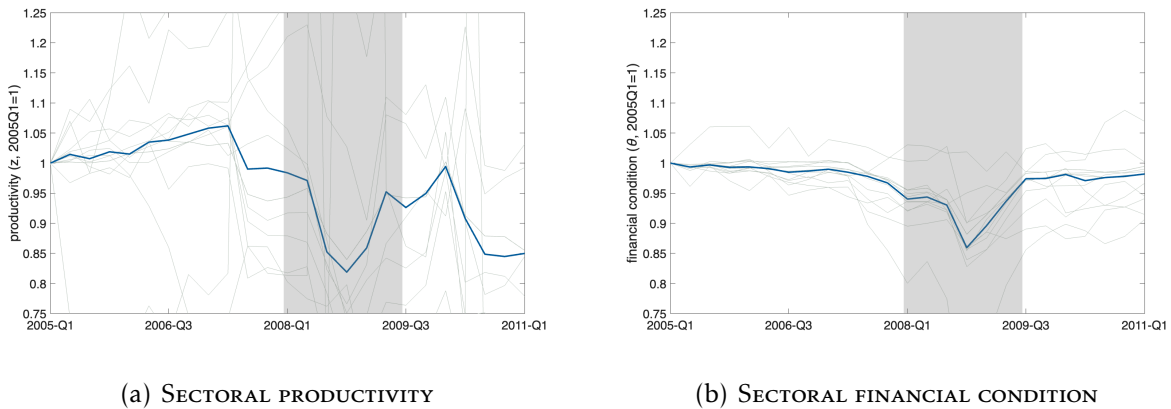
Figure 6
FIT OF ESTIMATED MODEL

5.3 Recover sectoral productivity and financial conditions

We recover two sector-specific exogenous variables in each period: productivity, $\{z_{it}\}$, and financial conditions, $\{\theta_{it}\}$. Their recovery combines two sectoral observables—real gross output and bond spreads—with the equilibrium conditions of the model. Specifically, in each period, we jointly solve the model equilibrium while requiring the model to match observed sectoral real gross output and sectoral bond spreads.

For productivity, we use the chain-type quantity indexes for gross output by industry from the BEA. We assume that the economy starts from the calibrated stationary equilibrium in 2005Q1. The stationary equilibrium determines the initial level of sectoral output, while subsequent sectoral output levels are constructed by applying the observed sector-specific output growth rates from the BEA to these initial values. We then recover the productivity vector $\{z_{it}\}$ by requiring the model-generated sectoral outputs in Equation (17) to match their empirical counterparts in each period. The resulting productivity series rationalizes the observed evolution of sectoral output allocations in equilibrium.

Panel (a) of Figure 7 plots the recovered sectoral productivity levels, z_{it} , normalized to one in 2005Q1. The gray lines show sector-specific paths, while the blue line reports the cross-sector median. The median rises modestly before the Great Recession and then declines sharply during the crisis period. However, the sectoral paths remain highly dispersed throughout the sample, indicating substantial cross-sector heterogeneity. Although the median productivity level recovers after reaching its trough, productivity does not exhibit a smooth or synchronized evolution across sectors.



Note: The figure plots the recovered sectoral productivity z_{it} and financial condition θ_{it} , each normalized to one in 2005Q1. The gray lines show sector-specific paths, and the blue line reports the cross-sector median. The shaded area denotes the Great Recession period. Sectoral productivity and financial conditions are recovered from the model by jointly matching observed sectoral gross output and sectoral bond spreads.

Figure 7
EVOLUTION OF SECTORAL PRODUCTIVITY AND FINANCIAL CONDITIONS

To recover financial conditions, we follow [Bigio and La'O \(2020\)](#) and [Miranda-Pinto and Young \(2022\)](#) by using the sectoral bond spreads constructed by [Gilchrist and Zakrajšek \(2012\)](#).²⁹ In the model, financial frictions create a wedge between the marginal product of internal funds and the cost of external finance. We therefore require that the

²⁹We thank Gilchrist and Zakrajšek for kindly sharing their data.

model-implied borrowing spread equals the observed sectoral bond spread:

$$1 + r_{it} = \frac{1 + \mu_{it}}{1 + \theta_{it}\mu_{it}}.$$

The recovered financial-condition vector $\{\theta_{it}\}$ is the one that reconciles the model-implied borrowing wedge with the observed spreads. A key distinction for our quantitative exercise is that, in the model with endogenous trade credit, the Lagrange multiplier is jointly determined by financial conditions, θ_i , and equilibrium trade credit, tc_{ji} . In contrast, in the fixed-trade-credit model, trade credit does not adjust, so the Lagrange multiplier responds only to changes in financial conditions.

Panel (b) of Figure 7 reports the recovered sectoral financial condition, θ_{it} , also normalized to one in 2005Q1. Financial conditions deteriorate markedly during the Great Recession. The cross-sector median declines throughout 2008, reaches its trough in 2008Q4, and then gradually recovers. Meanwhile, the sector-specific paths remain dispersed, indicating that the deterioration was broad-based but heterogeneous across sectors.

Panel (a) of Table 3 summarizes the pairwise correlations of the recovered productivity and financial shocks before and during the Great Recession in the benchmark economy with endogenous trade credit, where productivity (financial) shocks are measured as the first difference of log productivity (financial conditions).³⁰ The average pairwise correlation of productivity shocks increases from -0.06 before the crisis to 0.15 during the recession. Thus, productivity shocks become somewhat more synchronized during the crisis, but the distribution remains highly dispersed, with an interquartile range of -0.38 to 0.71 .

Financial shocks exhibit a more pronounced increase in comovement. The average pairwise correlation of $\Delta \log \theta_{it}$ rises from 0.25 before the Great Recession to 0.53 during the crisis. However, the lower quartile remains close to zero, indicating substantial heterogeneity across sector pairs. Although financial conditions became more synchronized during the recession, the tightening remained highly uneven across sectors, suggesting that the crisis reflected widespread financial distress rather than a single common financial shock.

This cross-sectional heterogeneity raises an important question: does the observed pattern of sectoral comovement simply reflect more highly correlated underlying shocks? To examine this possibility, Figure 8 plots, for each sector pair, the correlation of output growth against the corresponding correlations of recovered productivity and financial

³⁰Online Appendix IV.3 plots the kernel densities of pairwise correlations of the recovered productivity and financial shocks for both the benchmark economy and the fixed-trade-credit counterfactual, separately for the normal period and the Great Recession.

Table 3
PAIRWISE CORRELATIONS OF RECOVERED SHOCKS

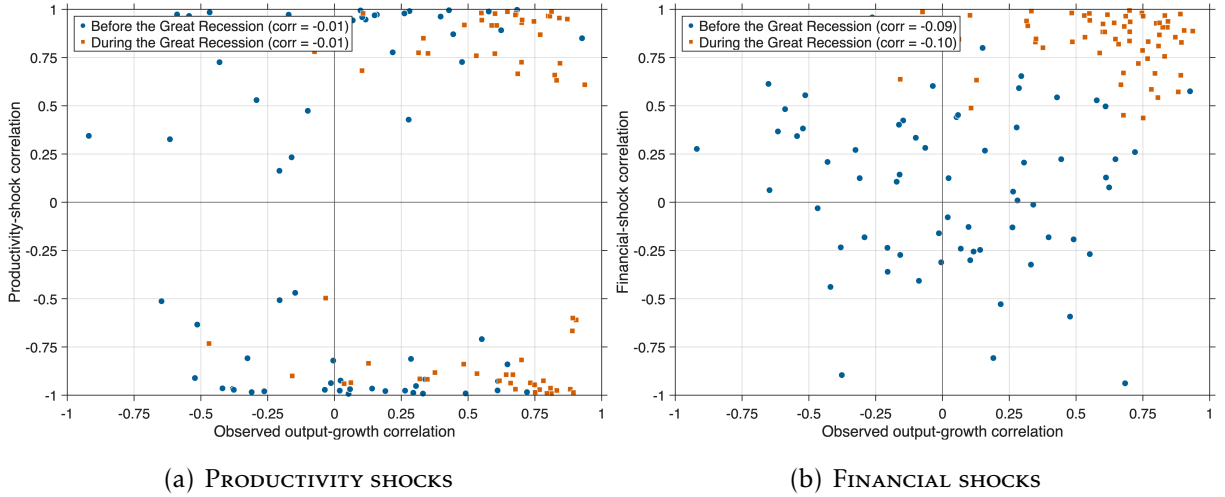
		Mean	SD	Skewness	P25	Median	P75
Panel (a): Endogenous trade credit							
Before the Great Recession	corr ($\Delta \log z_{it}, \Delta \log z_{jt}$)	-0.06	0.46	0.11	-0.41	-0.11	0.26
	corr ($\Delta \log \theta_{it}, \Delta \log \theta_{jt}$)	0.25	0.33	-0.73	0.09	0.27	0.52
During the Great Recession	corr ($\Delta \log z_{it}, \Delta \log z_{jt}$)	0.15	0.57	-0.13	-0.38	0.18	0.71
	corr ($\Delta \log \theta_{it}, \Delta \log \theta_{jt}$)	0.53	0.50	-0.81	0.02	0.80	0.91
Panel (b): Fixed trade credit							
Before the Great Recession	corr ($\Delta \log z_{it}, \Delta \log z_{jt}$)	-0.02	0.47	-0.02	-0.39	-0.02	0.39
	corr ($\Delta \log \theta_{it}, \Delta \log \theta_{jt}$)	0.16	0.41	-0.38	-0.15	0.19	0.49
During the Great Recession	corr ($\Delta \log z_{it}, \Delta \log z_{jt}$)	0.25	0.55	-0.16	-0.33	0.41	0.78
	corr ($\Delta \log \theta_{it}, \Delta \log \theta_{jt}$)	0.53	0.45	-0.90	0.07	0.71	0.87

Note: The table reports moments of the distribution of pairwise correlations of recovered productivity shocks and financial shocks. Sectoral bond spreads and real sectoral outputs are used to recover shocks from the model under endogenous and fixed trade-credit settings. Panel (a) reports the recovered shocks from the benchmark model with endogenous trade credit, while Panel (b) reports the shocks recovered under the fixed trade-credit counterfactual. The windows 2005Q3–2007Q2 and 2007Q3–2009Q2 correspond to the periods before and during the Great Recession.

shocks. Panel (a) shows little systematic relationship between output comovement and productivity-shock comovement. The simple correlation is approximately -0.01 both before and during the Great Recession. Panel (b) presents a similar pattern for financial shocks. Despite the increase in financial-shock correlation during the crisis, sector pairs with stronger output comovement are not systematically those with more highly correlated financial shocks. To quantify these relationships, Online Appendix IV.4 regresses pairwise output-growth correlations on the corresponding correlations of productivity and financial shocks, both in levels and in changes, with and without sector fixed effects for the two sectors in each pair. Across all specifications, the coefficients are small and statistically insignificant. This does not imply that shock comovement is irrelevant; rather, it shows that the cross-sectional ranking of output comovement is not order-preserving in shock correlations.

6 Quantitative exercise

Having quantified the model, we now use it to examine how shocks to recovered sectoral productivity and financial conditions propagate through production linkages and endogenous trade-credit adjustment. We conduct four exercises. First, we use the model-implied bilateral trade credit to examine how trade-credit dynamics are associated with



Note: Each point represents a sector pair. The horizontal axis reports the pairwise correlation of sectoral output growth. The vertical axis reports the corresponding pairwise correlation of recovered productivity shocks in Panel (a) and recovered financial shocks in Panel (b). Blue circles use the pre-Great-Recession window, 2005Q3–2007Q2, and orange squares use the Great Recession window, 2007Q3–2009Q2. The numbers in parentheses in the legend report the simple cross-pair correlation between output-growth comovement and the corresponding recovered-shock correlation.

Figure 8
COMPARISON: CORRELATIONS OF OUTPUT GROWTH VERSUS SHOCKS

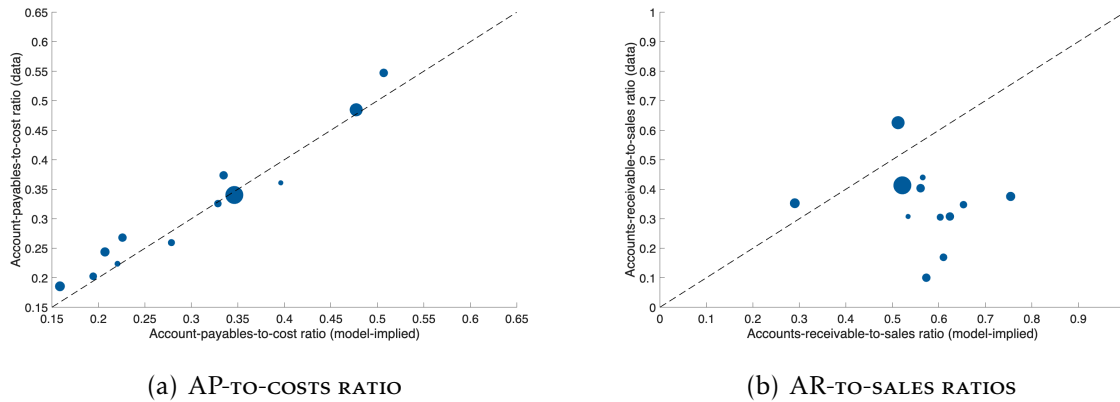
sectoral output comovement. Second, we compare the benchmark economy with a fixed-trade-credit counterfactual to quantify the contribution of endogenous trade credit to sectoral comovement and aggregate consumption dynamics. Third, we recover productivity and financial conditions under a fixed-trade-credit economy to ask what shock structure is needed to match the same sectoral data when trade credit cannot adjust. Finally, we repeat the analysis for the early-1980s and COVID-19 recessions using episode-specific recalibrations, and show how the role of trade credit depends on the nature of the underlying shocks.

6.1 Role of trade credit during the Great Recession

Because bilateral trade-credit provisions are not observed in the data, we first assess the model's ability to reproduce untargeted sector-level trade-credit moments. Figure 9 compares model-implied and empirical trade-credit ratios averaged over 2008Q3–2009Q1.³¹ Panel (a) reports accounts-payable-to-costs ratios, while Panel (b) reports accounts-receivable-to-sales ratios. The model reproduces the cross-sectional distribution of accounts payable

³¹ As an additional out-of-sample validation, Online Appendix IV.5 compares the model-implied bilateral input-expenditure shares during the Great Recession with the corresponding shares from the 2008 BEA input-output table. Despite not targeting these moments in the calibration, the model closely matches the observed bilateral input-use pattern, with a cross-sectional correlation of 0.94.

remarkably well: most sectors lie close to the 45-degree line, indicating that it captures the allocation of trade credit received across sectors during the crisis. The fit for accounts receivable is somewhat less tight, but the model nevertheless generates substantial cross-sectional dispersion and broadly matches the empirical ranking of sectors.



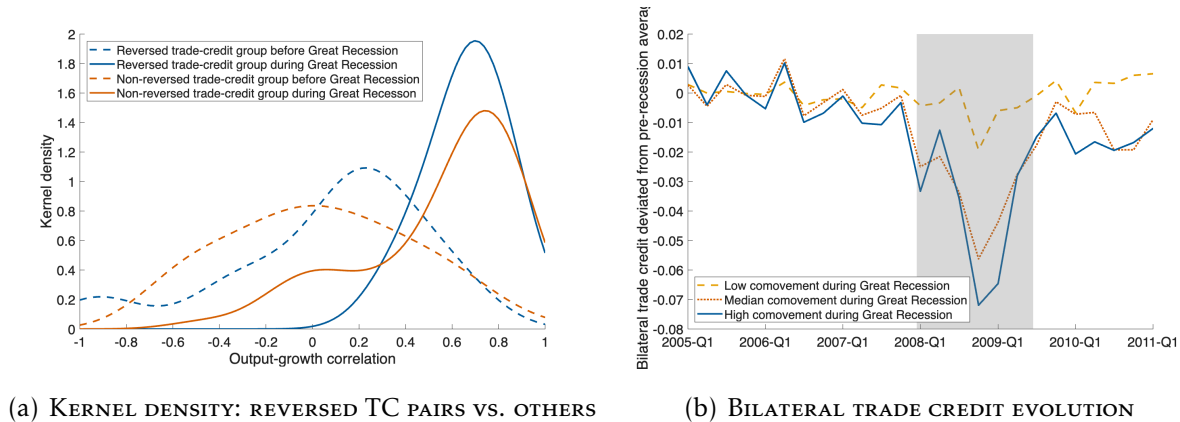
Note: The figure compares model-implied (horizontal axis) and empirical trade-credit ratios (vertical axis) averaged over 2008Q3–2009Q1. Panel (a) plots accounts-payable-to-costs ratios, and Panel (b) plots accounts-receivable-to-sales ratios. The dashed line is the 45-degree line.

Figure 9
FIT OF SECTORAL TRADE CREDIT DURING THE GREAT RECESSION

The model's ability to match untargeted sector-level trade-credit moments provides support for using its implied bilateral trade-credit allocations to study the underlying mechanism. According to Proposition 2, the response of trade credit to customer financial tightness depends on the supplier's financial condition: relatively unconstrained suppliers expand trade credit to support financially distressed customers, whereas sufficiently constrained suppliers contract trade credit, thereby amplifying the transmission of financial shocks through production linkages. To evaluate this mechanism quantitatively, we classify sector pairs according to their model-implied trade-credit responses during the Great Recession. A pair is defined as a *reversed trade-credit pair* if, in either supplier-customer direction, the response of trade credit to customer financial tightness switches from positive before the crisis to negative in any quarter between 2008Q3 and 2009Q1. By this criterion, the model identifies 20 reversed trade-credit pairs.

Panel (a) of Figure 10 compares the kernel densities of pairwise output-growth correlations for reversed and non-reversed trade-credit pairs before and during the Great Recession. Output comovement increases for both groups during the crisis, but the increase is larger for reversed trade-credit pairs, whose density becomes tightly concentrated around high positive correlations. Panels (b) and (c) of Table 4 quantify this pat-

tern. The average output-growth correlation for reversed pairs rises from 0.06 before the recession to 0.66 during the crisis, compared with an increase from 0.03 to 0.52 for non-reversed pairs.³² These results are consistent with Proposition 2: when supplier financial constraints become sufficiently tight, trade-credit provision can reverse from a cushioning mechanism into an amplification channel.



Note: Panel (a) plots kernel densities of pairwise sectoral output-growth correlations before and during the Great Recession, separately for reversed and non-reversed trade-credit pairs. A sector pair is classified as reversed if, in either direction of the bilateral supplier-customer relationship, the model-implied response of trade credit to customer financial tightness switches from positive before the crisis to negative in any quarter between 2008Q3 and 2009Q1. Panel (b) sorts sector pairs into terciles based on their output-growth comovement during the Great Recession and plots the average model-implied bilateral trade credit for each group, expressed as deviations from its pre-recession average over 2005Q3–2007Q2. The shaded area denotes the Great Recession.

Figure 10
ROLE OF TRADE CREDIT DURING THE GREAT RECESSION

Having shown that reversed trade-credit pairs exhibit stronger output comovement, we next examine the mechanism from the opposite perspective: do the sector pairs that comove most strongly during the Great Recession also experience larger contractions in model-implied bilateral trade credit? To answer this question, we sort all sector pairs into terciles based on their output-growth correlations during the Great Recession and trace the evolution of average bilateral trade credit within each group. For each tercile, trade credit is expressed as a deviation from its pre-recession average over 2005Q3–2007Q2.

Panel (b) of Figure 10 shows that all three groups fluctuate around their pre-recession averages before the crisis but diverge during the Great Recession. For each tercile, the plotted average pools all directed trade-credit links (both tc_{ij} and tc_{ji}) corresponding to the sector pairs in that group. The high-comovement group (blue solid line) experiences the largest contraction in bilateral trade credit, concentrated between 2008Q3 and

³²The recovered financial-shock correlations are similar across the two groups during the Great Recession, and are slightly lower for reversed trade-credit pairs. This suggests that the stronger output comovement of reversed pairs is not driven mechanically by more synchronized financial shocks.

Table 4
MODEL-IMPLIED PAIRWISE CORRELATIONS OF SECTORAL OUTPUT GROWTH RATES

	Mean	SD	Skewness	25th	Median	75th	KS stat. (CV-KS)
(a) Data/both shocks with endogenous trade credit							
Before the Great Recession	0.04	0.40	-0.08	-0.26	0.06	0.31	0.59 (0.43, 1.04)
During the Great Recession	0.56	0.32	-1.23	0.38	0.68	0.80	
(b) Reversed trade credit group							
Before the Great Recession	0.06	0.45	-0.84	-0.17	0.17	0.31	0.80 (0.78, 1.89)
During the Great Recession	0.66	0.16	-0.56	0.59	0.69	0.77	
(c) Non-reversed trade credit group							
Before the Great Recession	0.03	0.40	0.11	-0.28	0.02	0.31	0.55 (0.52, 1.24)
During the Great Recession	0.52	0.34	-1.09	0.34	0.67	0.81	
(d) Both shocks with fixed trade credit							
Before the Great Recession	0.12	0.37	-0.13	-0.15	0.08	0.41	0.21 (0.43, 1.04)
During the Great Recession	0.26	0.40	-0.27	0.02	0.25	0.61	
(e) Financial shocks with endogenous trade credit							
Before the Great Recession	0.23	0.59	-0.40	-0.29	0.47	0.72	0.44 (0.43, 1.04)
During the Great Recession	0.55	0.64	-1.47	0.48	0.92	0.97	
(f) Financial shocks with fixed trade credit							
Before the Great Recession	0.31	0.65	-0.61	-0.43	0.53	0.96	0.32 (0.43, 1.04)
During the Great Recession	0.41	0.76	-0.88	-0.45	0.89	0.97	
(g) Productivity shocks with endogenous trade credit							
Before the Great Recession	0.05	0.40	-0.17	-0.29	0.11	0.34	0.29 (0.43, 1.04)
During the Great Recession	0.18	0.53	-0.24	-0.29	0.22	0.68	
(h) Productivity shocks with fixed trade credit							
Before the Great Recession	0.03	0.48	0.08	-0.37	-0.00	0.46	0.18 (0.43, 1.04)
During the Great Recession	0.16	0.53	-0.18	-0.23	0.15	0.65	

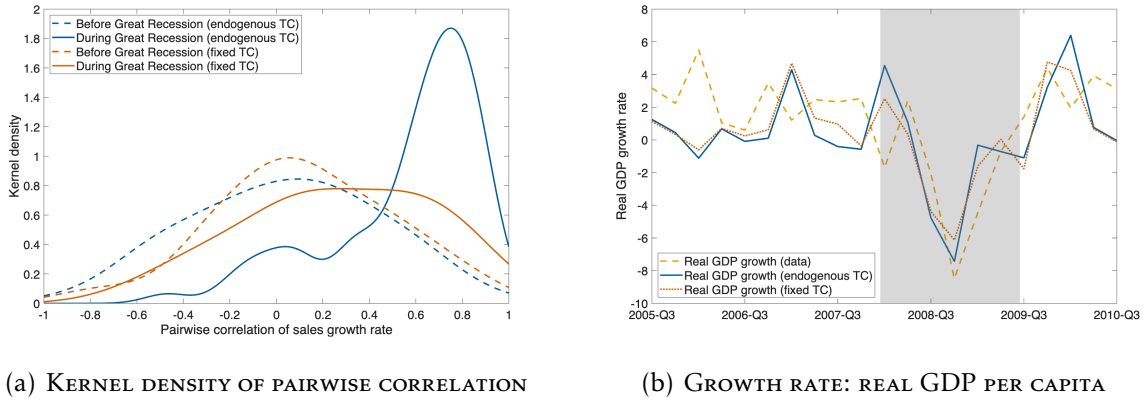
Note: The table reports distributional moments of pairwise sectoral output-growth correlations across the 66 sector pairs. The “Before the Great Recession” window is 2005Q3–2007Q2, and the “During the Great Recession” window is 2007Q3–2009Q2. Panel (a) reports the baseline economy with endogenous trade credit and both productivity and financial conditions recovered in Section 5.3. Panels (b) and (c) split sector pairs according to whether at least one direction of the bilateral supplier-customer relationship enters the reversal region of Proposition 2, where trade credit responds negatively to customer financial tightness, during the crisis period. Panel (d) repeats the baseline exercise while holding bilateral trade credit fixed at its pre-recession average. Panels (e)–(h) report one-shock counterfactuals: in the financial-condition-only exercises, productivity is fixed at its pre-recession sectoral average; in the productivity-only exercises, financial conditions are fixed at their pre-recession sectoral averages. The last column reports the two-sample Kolmogorov–Smirnov statistic comparing the before- and during-Great-Recession distributions within each panel. Critical values in parentheses adjust for autocorrelation following [Lanzante \(2021\)](#); the first value corresponds to mild autocorrelation, $AR(1)=0.6$, and the second to high autocorrelation, $AR(1)=0.9$.

2009Q1. The middle-comovement group (red dotted line) also exhibits a substantial decline, though considerably smaller, whereas the low-comovement group (yellow dashed line) shows only a mild and short-lived reduction. Together with the comparison between reversed and non-reversed trade-credit pairs, these findings provide evidence consistent with the mechanism that contractions in trade credit amplify the propagation of financial shocks through production linkages, thereby strengthening sectoral comovement.

6.2 Counterfactual exercise with fixed trade credit

The analysis in Section 6.1 provides qualitative support for the model's propagation mechanism. We now quantify the importance of this channel. To do so, we compare the benchmark economy with a counterfactual economy in which bilateral trade credit is fixed at its pre-recession average, and then examine how financial shocks, productivity shocks, and both shocks jointly propagate with and without endogenous trade-credit adjustment.

The benchmark economy, blue line in Panel (a) of Figure 11, which incorporates both the recovered shocks and endogenous trade-credit adjustment, reproduces the observed sectoral output dynamics during the Great Recession. Consistent with the empirical evidence in Section 2.1, the distribution of pairwise output-growth correlations shifts sharply to the right during the crisis. As reported in Panel (a) of Table 4, the average pairwise correlation increases from 0.04 before the crisis to 0.56 during the Great Recession.



Note: Panel (a) plots kernel densities of pairwise sectoral output-growth correlations before and during the Great Recession under endogenous and fixed trade credit. In the fixed-trade-credit counterfactual, bilateral trade credit is held at its pre-recession average over 2005Q3–2007Q2. Panel (b) reports real GDP growth in the data, the benchmark model with endogenous trade credit, and the fixed-trade-credit counterfactual. The shaded area denotes the Great Recession.

Figure 11
COMPARISON: ENDOGENOUS VS FIXED TRADE CREDIT WITH BOTH SHOCKS

We then fix bilateral trade credit for all sector pairs at its average pre-recession level (2005Q3–2007Q2) while imposing the same productivity and financial shocks recovered in Section 5.3.³³ This counterfactual preserves the production network, financial constraints, and recovered sectoral shocks, but eliminates the response of trade credit to financial shocks.

³³We impose this restriction economy-wide because trade credit is jointly determined in equilibrium. Fixing trade credit for only a subset of sector pairs would induce endogenous adjustments in the remaining trade-credit relationships through changes in production, prices, and firms' financial conditions.

In the fixed-trade-credit counterfactual, shown by the red line in Panel (a) of Figure 11, the distribution of pairwise output-growth correlations also shifts to the right, indicating that the recovered productivity and financial shocks, together with the production network, generate some increase in sectoral comovement. Thus, the underlying shocks themselves account for an important part of the observed rise in comovement. However, the shift is considerably smaller than in the benchmark economy. As reported in Panel (d) of Table 4, the average pairwise correlation increases from 0.12 before the Great Recession to 0.26 during the crisis, compared with an increase from 0.04 to 0.56 in the benchmark economy.

These results indicate that the recovered productivity and financial shocks account for part of the rise in sectoral comovement, but endogenous trade-credit adjustment substantially amplifies their propagation. Under the fixed-trade-credit counterfactual, suppliers cannot adjust payment terms in response to changes in customers' financial conditions, so shocks are transmitted only through production linkages and firms' borrowing constraints. In the benchmark economy, by contrast, deteriorating financial conditions push some supplier–customer pairs from the positive- to the negative-response region characterized in Proposition 2. Once in this region, suppliers contract rather than expand trade credit as their customers become more financially constrained. The resulting reduction in trade credit raises customers' upfront financing needs, causing them to cut intermediate-input demand more sharply. This decline in demand feeds back to suppliers' sales and production through production linkages, making the output of the two sectors respond more similarly to the same underlying shocks. Endogenous trade-credit adjustment therefore amplifies the transmission of the recovered shocks, leading to substantially stronger sectoral comovement during the Great Recession.

We examine the implications of endogenous trade-credit adjustment for the growth rate of real GDP per capita, which corresponds to aggregate consumption growth in the model. Panel (b) of Figure 11 compares the benchmark economy (blue solid line) with the counterfactual economy in which trade credit is fixed at its pre-recession average (red dotted line), together with the data (yellow dashed line). The benchmark economy closely reproduces the sharp decline in real GDP per capita observed during the Great Recession, particularly between 2008Q3 and 2009Q1. The counterfactual economy, however, experiences a noticeably milder contraction. At the trough of the recession (2008Q4), aggregate consumption growth declines by 7.4% in the benchmark economy, compared with 6.1% in the counterfactual economy—a 21% smaller decline. The intuition is straightforward. When trade credit is held fixed, suppliers can no longer endogenously tighten payment terms in response to deteriorating financial conditions. Consequently, customers' financ-

ing constraints are not further tightened by reductions in trade credit provision, mitigating production distortions and dampening the decline in aggregate output relative to an economy with endogenous trade credit adjustment.

We next isolate the roles of financial and productivity shocks by introducing one set of shocks at a time, while holding the other at its pre-recession average (2005Q3–2007Q2). Financial shocks generate a large increase in sectoral comovement. With endogenous trade credit, the average pairwise output-growth correlation rises from 0.23 before the Great Recession to 0.55 during the crisis. The upper part of the distribution becomes especially concentrated at high positive correlations, with the median rising from 0.47 to 0.92. When trade credit is fixed at its pre-recession level, financial shocks still generate some comovement, but the increase is smaller: the average correlation rises from 0.31 to 0.41. These results indicate that while financial shocks alone generate some increase in sectoral comovement, allowing trade credit to adjust endogenously markedly amplifies their propagation through production networks.

Productivity shocks also raise sectoral comovement, but the effect is more moderate. With endogenous trade credit, the average pairwise correlation increases from 0.05 before the Great Recession to 0.18 during the crisis. With fixed trade credit, the corresponding increase is from 0.03 to 0.16. The similarity between the endogenous- and fixed-trade-credit cases indicates that productivity shocks affect sectoral comovement mainly through the production network itself, rather than through endogenous adjustments in trade credit. Overall, the one-shock exercises show that both productivity and financial shocks matter, but the trade-credit mechanism is especially important for amplifying financial shocks.

6.3 Recalibrating sectoral shocks in the fixed trade credit model

In our benchmark calibration, productivity and financial conditions are recovered from the model’s equilibrium conditions. Consequently, their estimated correlation structure depends on the propagation mechanisms embedded in the model. To isolate the contribution of endogenous trade-credit adjustment as a propagation mechanism, we re-solve the model under a fixed-trade-credit regime, holding bilateral trade credit constant at its pre-recession average in the benchmark economy, and recover the productivity and financial conditions required to match the same sectoral output and bond-spread data. This fixed-trade-credit economy closely resembles the environment in [Bigio and La’O \(2020\)](#): production linkages and working-capital constraints remain operative, but trade credit no longer adjusts endogenously to suppliers’ and customers’ financial conditions.

Panel (b) of Table 3 reports the pairwise correlations of recovered productivity and financial shocks in the fixed-trade-credit economy. The comparison shows that the fixed-trade-credit model requires a larger increase in recovered shock comovement from the pre-crisis period to the Great Recession. For financial shocks, the average pairwise correlation rises from 0.16 before the crisis to 0.53 during the crisis, an increase of 0.37. In the endogenous-trade-credit benchmark, the corresponding increase is from 0.25 to 0.53, or 0.28. Thus, the fixed-trade-credit model requires about a 32% larger increase in financial-shock comovement. A similar pattern holds for productivity changes: the average correlation increases by 0.27 in the fixed-trade-credit model, compared with 0.21 in the benchmark, about 23% larger.

This comparison highlights how endogenous trade credit changes the attribution of sectoral comovement. When trade credit is fixed and the model is required to match the observed sectoral output and bond-spread data, the increase in sectoral comovement is rationalized by a larger increase in the correlation of the recovered productivity and financial shocks. In the benchmark economy, by contrast, part of the increase in comovement is generated endogenously through trade-credit adjustment. As financially constrained suppliers reduce trade credit to distressed customers, payment terms tighten along supplier-customer relationships, amplifying the transmission of financial distress through the production network and increasing sectoral comovement.

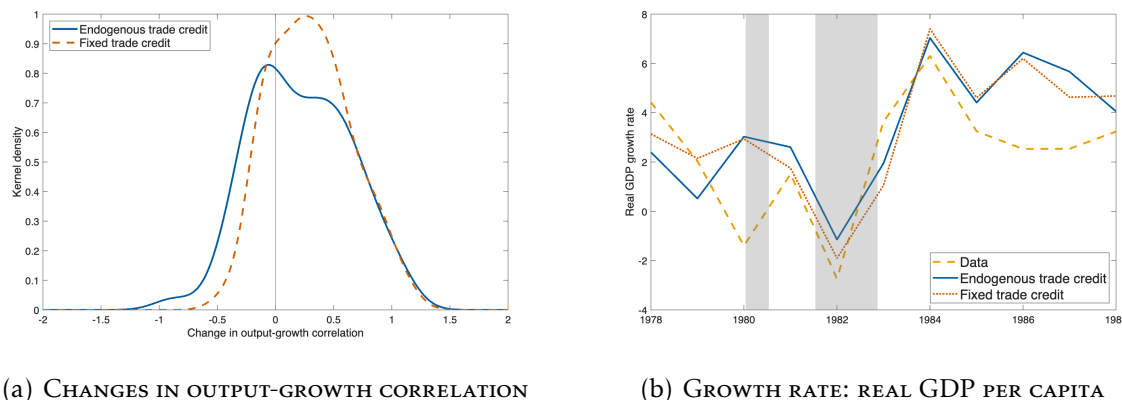
6.4 The early-1980s recessions

We next examine the early-1980s recessions as a contrast to the Great Recession. Following Chari et al. (2007), this episode provides a useful benchmark because, despite a severe aggregate downturn, sectoral comovement did not increase to the same extent as during the Great Recession. To ensure that the comparison reflects the economic environment prevailing at the time, we recalibrate the model's stationary equilibrium using 1978 data. Based on this recalibrated economy, we recover annual sectoral productivity and financial conditions by requiring the model to match observed sectoral output and bond spreads. We then compare the benchmark economy with a counterfactual in which bilateral trade-credit allocations are fixed at their 1978 levels.³⁴

Panel (a) of Figure 12 plots the distribution of changes in pairwise output-growth correlations during the early-1980s recessions relative to the post-recession comparison period. The distribution under fixed trade credit is shifted to the right relative to that under endogenous trade credit, indicating a broader increase in sectoral comovement.

³⁴Online Appendix IV.7 provides details of the recalibration, the fit of the recalibrated stationary equilibrium, and the recovered paths of productivity and financial conditions.

The average increase is 0.30 when trade credit is fixed, compared with 0.22 when trade credit adjusts endogenously. Thus, allowing trade credit to adjust endogenously leaves a larger fraction of sector pairs with little or even no increase in comovement, consistent with endogenous trade credit cushioning shock transmission during the early-1980s recessions.



Note: The model is recalibrated using 1978 input-output and trade-credit moments. Sectoral productivity and financial conditions are recovered annually to match sectoral output and bond spreads. Panel (a) plots kernel densities of changes in pairwise sectoral output-growth correlations under endogenous and fixed trade credit, where the change is defined as the correlation during the early-1980s recession window, 1978–1985, minus the correlation in the post-recession comparison window, 1983–1989. Panel (b) reports real GDP per capita growth in the data, the benchmark model with endogenous trade credit, and the fixed-trade-credit counterfactual. The shaded area denotes the early-1980s recessions.

Figure 12
THE EARLY-1980s RECESSIONS: ENDOGENOUS VS FIXED TRADE CREDIT

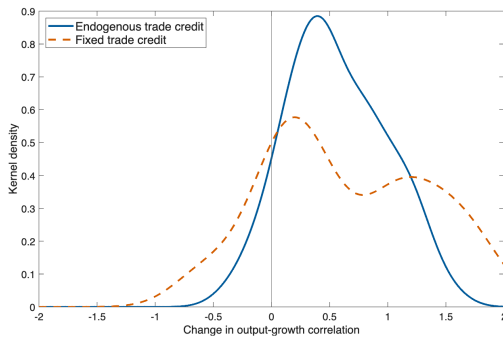
Panel (b) compares the growth rate of real GDP per capita in the data with the model-implied paths. Although the benchmark model understates the magnitude of the observed contraction, it captures the broad dynamics of the early-1980s recessions. In the data, real GDP per capita growth reaches a trough of approximately -2.7% in 1982. The benchmark economy produces a milder trough of -1.1%, whereas the fixed-trade-credit counterfactual generates a substantially deeper contraction of -1.9%. Taken together with the evidence on sectoral comovement in Panel (a), these findings suggest that, in contrast to its amplifying role during the Great Recession, endogenous trade credit may have acted as a cushion during the early-1980s recessions, dampening both the rise in sectoral comovement and the aggregate contraction.

6.5 The COVID-19 recession

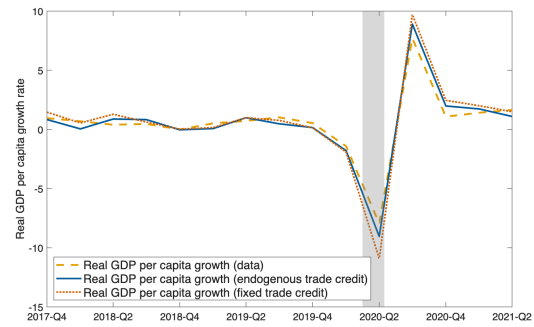
We next apply the same framework to the COVID-19 recession. This episode provides a useful contrast to the Great Recession because it generated a sharp aggregate contrac-

tion and a large increase in unconditional sectoral comovement, but the underlying disruption was not primarily a financial crisis. To make the comparison consistent, we re-estimate the stationary economy using the 2017 input-output table and pre-COVID trade-credit moments, while keeping the common structural parameters fixed. We then recover quarterly sectoral productivity and financial conditions so that the model matches sectoral output and bond spreads. The pre-COVID window is 2017Q2–2019Q1, and the COVID window is 2019Q2–2021Q1. In the fixed-trade-credit counterfactual, bilateral trade credit is fixed at its pre-COVID averages over 2017Q2–2019Q1.³⁵

Panel (a) of Figure 13 plots the kernel density of changes in pairwise output-growth correlations during the COVID-19 recession relative to the pre-recession period. Sectoral comovement rises substantially under both cases. The average increase is 0.56 with endogenous trade credit and 0.64 when trade credit is fixed. The more pronounced difference, however, lies in the upper tail of the distribution. Under fixed trade credit, the 75th percentile reaches 1.21, compared with 0.85 under endogenous trade credit. The thicker upper tail indicates that a larger fraction of sector pairs experiences exceptionally large increases in comovement. These results suggest that, although the pandemic induces a broad rise in sectoral comovement regardless of the trade-credit regime, endogenous trade credit helps attenuate the most extreme increases.



(a) CHANGE IN OUTPUT-GROWTH CORRELATIONS



(b) REAL GDP PER CAPITA GROWTH

Note: The model is recalibrated using 2017 input-output and sectoral trade-credit moments over 2017Q2–2019Q1. Sectoral productivity and financial conditions are recovered quarterly to match sectoral output and bond spreads. Panel (a) plots kernel densities of changes in pairwise sectoral output-growth correlations under endogenous and fixed trade credit, where the change is defined as the correlation during the COVID-19 recession window, 2019Q2–2021Q1, minus the correlation in the pre-COVID window, 2017Q2–2019Q1. Panel (b) reports real GDP per capita growth in the data, the benchmark model with endogenous trade credit, and the fixed-trade-credit counterfactual. The shaded area denotes the COVID-19 recession.

Figure 13
COVID-19 RECESSION: SECTORAL COMOVEMENT AND AGGREGATE GROWTH

³⁵Online Appendix IV.8 provides details of the recalibration, the fit of the recalibrated stationary equilibrium, and the recovered paths of productivity and financial conditions.

Panel (b) compares the growth rate of real GDP per capita in the data with the model-implied paths. The benchmark model captures the sharp contraction in 2020Q2, when real GDP per capita growth falls by 7.9% in the data and by 9% in the model. The fixed-trade-credit counterfactual generates a deeper contraction of 10.9% in the same quarter. Thus, during the COVID-19 recession, endogenous trade-credit adjustment appears to cushion the aggregate decline rather than amplify it.

7 Conclusion

We document a distinctive feature of sectoral comovement over the U.S. business cycle. The distribution of sectoral output-growth correlations, conditional on aggregate GDP, is relatively stable across most recessions but shifts sharply to the right during the Great Recession. Sectoral evidence shows that this increase is concentrated among production-linked sectors and is stronger for pairs experiencing trade-credit contractions, suggesting that trade credit is a relevant financial linkage within production networks.

We then develop a multisector model with endogenous trade-credit adjustment to study this mechanism. The model emphasizes the asymmetric role of trade credit. When suppliers retain sufficient financial capacity, they can extend trade credit to customers facing tighter financial conditions, cushioning shock transmission. When suppliers are themselves sufficiently financially constrained, however, trade-credit provision contracts and can amplify distress along supplier-customer relationships. The quantitative exercises suggest that this mechanism contributes meaningfully to the rise in sectoral comovement and to the aggregate downturn during the Great Recession. Applying the same framework to the early-1980s and COVID-19 recessions, the counterfactuals suggest that trade credit instead acts as a cushion, dampening sectoral spillovers and aggregate contractions.

More generally, our results suggest that endogenous trade-credit adjustment is an important state-dependent financial linkage in production networks. This does not imply that trade credit is the only driver of sectoral comovement during financial crises. Rather, the empirical evidence and the quantitative model point to a mechanism through which supplier liquidity and trade credit can shape the propagation of financial shocks across sectors. This perspective suggests that policies affecting supplier liquidity may also influence the transmission of shocks through production networks. A full evaluation of such policies, however, would require a broader welfare analysis and lies beyond the scope of this paper.

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Appendix

A Model details: proof of propositions and lemmas

A.1 Lagrangian

Let μ_i denote the multiplier on the collateral constraint and let v_{ij} denote the relationship-retention multiplier, normalized by the scale of the trading relationship. The Lagrangian terms relevant for firm i 's choices can be written as

$$\begin{aligned} \mathcal{L}_i = & p_i z_i l_i^{\alpha_i} \left(\sum_{j=1}^n \lambda_{ji}^{\frac{1}{\sigma}} m_{ji}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma \xi_i}{\sigma-1}} - \sum_{j=1}^n (1-\eta) t c_{ij} \frac{p_i m_{ij}}{\eta} - w l_i - \sum_{j=1}^n (1 - (1-\eta) t c_{ji}) \frac{p_j m_{ji}}{\eta} \\ & - \sum_j \psi_j e_{ij} \frac{p_i m_{ij}}{\eta} + \sum_j \phi t c_{ij}^{1-\kappa} e_{ij}^{\kappa} \frac{p_i m_{ij}}{\eta} + \sum_j v_{ij} \frac{p_i m_{ij}}{\eta} \left((1-\eta) t c_{ij} + \phi t c_{ij}^{1-\kappa} e_{ij}^{\kappa} - (1-\gamma) \right) \\ & + \mu_i \left(\theta_i p_i z_i l_i^{\alpha_i} \left(\sum_{j=1}^n \lambda_{ji}^{\frac{1}{\sigma}} m_{ji}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma \xi_i}{\sigma-1}} + \sum_j (1 - t c_{ij}) \frac{p_i m_{ij}}{\eta} - w l_i - \sum_j (1 - t c_{ji}) \frac{p_j m_{ji}}{\eta} - \sum_j \psi_j e_{ij} \frac{p_i m_{ij}}{\eta} \right). \end{aligned} \quad (23)$$

A.2 Proof of Lemma 1

Proof. We first differentiate Equation (23) with respect to l_i and m_{ji} . Combining the resulting first-order conditions with the CES cost-minimization condition yields the optimality conditions in Equations (15) and (16). Combining the labor and aggregate-input first-order conditions further implies

$$l_i = \frac{\alpha_i (1 + \theta_i \mu_i) p_i y_i}{(1 + \mu_i) w}, \quad m_i = \frac{\eta \xi_i (1 + \theta_i \mu_i) p_i y_i}{\tilde{p}_i}.$$

Substituting these expressions into the production function, $y_i = z_i l_i^{\alpha_i} m_i^{\xi_i}$, yields Equation (17). ■

A.3 Proof of Proposition 1

Proof. We characterize supplier i 's optimal relationship policy toward customer j . From Lemma 1, customer j 's demand for supplier i 's input depends on trade credit through the customer's effective input price. The elasticity of demand with respect to trade credit is therefore

$$\frac{\partial \log m_{ij}}{\partial t c_{ij}} = \frac{\sigma(1 - \eta + \mu_j)}{\eta + (1 - \eta + \mu_j)(1 - t c_{ij})}.$$

We then differentiate Equation (23) with respect to e_{ij} and $t c_{ij}$. The first-order condition for e_{ij} implies Equation (18), while the first-order condition for $t c_{ij}$ yields the optimal trade-credit condition in Equation (19). ■

A.4 Proof of Proposition 2

Proof.

Under a binding relationship-retention constraint, effort can be written as a function of trade credit:

$$e_{ij} = \mathbf{e}(tc_{ij}) = \left(\frac{1 - \gamma - (1 - \eta)tc_{ij}}{\phi tc_{ij}^{1-\kappa}} \right)^{1/\kappa}.$$

The condition $\eta > \gamma$ implies that $\mathbf{e}(1) > 0$, so full trade credit does not eliminate the need for relationship-maintenance effort.

Substituting $\mathbf{e}(tc_{ij})$ into the supplier's payoff gives a one-dimensional problem in tc_{ij} . Let $\mathbf{H}(tc_{ij}; \mu_i, \mu_j)$ denote the first-order condition for this reduced problem, with $\mathbf{H}(tc_{ij}^*; \mu_i, \mu_j) = 0$. We have

$$\begin{aligned} \mathbf{H}(tc_{ij}; \mu_i, \mu_j) = & \sigma(1 - \eta + \mu_j)(\mu_i + 1 - \gamma) - (2(1 - \eta) + \mu_i)(1 + \mu_j) + (2(1 - \eta) + \mu_i)(1 - \eta + \mu_j)(1 - \sigma)tc_{ij} \\ & + (1 + \mu_i)\psi_j \mathbf{e}(tc_{ij}) \left(\frac{1 + \mu_j - (1 - \eta + \mu_j)tc_{ij}}{\kappa} \left(\frac{1 - \eta}{1 - \gamma - (1 - \eta)tc_{ij}} + \frac{1 - \kappa}{tc_{ij}} \right) - \sigma(1 - \eta + \mu_j) \right). \end{aligned}$$

Condition (20) implies that $\mathbf{H}(1; 0, \mu_j) < 0$. Condition (21) strengthens this requirement by guaranteeing that $\mathbf{H}(1; \mu_i, \mu_j) < 0$ for any $\mu_i > 0$. Moreover, under the maintained regularity conditions, $\mathbf{H}$ is strictly decreasing in tc_{ij} , and $\lim_{tc_{ij} \downarrow 0} \mathbf{H}(tc_{ij}; \mu_i, \mu_j) > 0$. Therefore, there exists a unique interior solution $tc_{ij}^* \in (0, 1)$.

The comparative statics follow from the implicit function theorem:

$$\frac{\partial tc_{ij}^*}{\partial \mu_i} = -\frac{\mathbf{H}_{\mu_i}}{\mathbf{H}_{tc_{ij}}}, \quad \frac{\partial tc_{ij}^*}{\partial \mu_j} = -\frac{\mathbf{H}_{\mu_j}}{\mathbf{H}_{tc_{ij}}}.$$

A higher supplier multiplier μ_i raises the shadow cost of delaying cash receipts and of financing relationship-maintenance effort, which implies $\mathbf{H}_{\mu_i} < 0$. Hence $\frac{\partial tc_{ij}^*}{\partial \mu_i} < 0$.

For the customer-side response, $\mathbf{H}_{\mu_j}$ is affine in the supplier multiplier μ_i . In particular, it can be written as

$$\mathbf{H}_{\mu_j} \propto \mathbf{g}(tc_{ij}^*, e_{ij}^*) - \mu_i,$$

where the threshold is defined as

$$\mathbf{g}(tc_{ij}^*, e_{ij}^*) = \frac{-2(1 - \eta) + \sigma(1 - \gamma) + 2(1 - \sigma)(1 - \eta)tc_{ij}^* + \mathbf{x}(tc_{ij}^*)\psi_j e_{ij}^*}{(1 - \sigma)(1 - tc_{ij}^*) - \mathbf{x}(tc_{ij}^*)\psi_j e_{ij}^*}, \quad (24)$$

and

$$\mathbf{x}(tc_{ij}^*) = \left(\frac{1}{tc_{ij}^*} - 1 \right) \left(\frac{1 - \gamma}{(1 - \gamma - (1 - \eta)tc_{ij}^*)\kappa} - 1 \right) - \sigma. \quad (25)$$

Since $\partial tc_{ij}^* / \partial \mu_j = -\mathbf{H}_{\mu_j} / \mathbf{H}_{tc}$ and $\mathbf{H}_{tc} < 0$, the sign of the trade-credit response is the same as the sign of $\mathbf{H}_{\mu_j}$. Therefore, $\frac{\partial tc_{ij}^*}{\partial \mu_j} \geq 0$ if $\mu_i \leq \mathbf{g}(tc_{ij}^*, e_{ij}^*)$; and $\frac{\partial tc_{ij}^*}{\partial \mu_j} < 0$ if $\mu_i > \mathbf{g}(tc_{ij}^*, e_{ij}^*)$. ■

**Online Appendix for
Trade Credit and Sectoral Comovement during Recessions**

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I Data Description

I.1 Sectors' characteristics

For the annual outputs, we use the nominal gross output by industry, provided by the Bureau of Economic Analysis (BEA), adjusted for sectoral chain-type price indexes. For the quarterly outputs, we use the real gross output by industry (seasonally adjusted at annual rates).

The quarterly series began in 2005Q1, while the annual series began in 1947. After excluding the agriculture, forestry, fishing and hunting (AFFH), finance, insurance and real estate (FIRE), and public sectors, our sample consists of 57 sectors and 1596 pairs.

The list and characteristics of the sectors are shown in Table IV.13. All pairwise correlations of output growth are calculated using Equation (I.1). The periods 2005Q3-2007Q2, 2007Q3-2009Q2, and 2009Q3-2011Q2 represent the times before, during, and after the Great Recession, respectively. $\overline{corr}_{before}$, $\overline{corr}_{in}$, and $\overline{corr}_{after}$ are the average correlations for the corresponding sector with all others.

The two-way or one-way linkages indicate the number of sectors with which the corresponding sector has a trading relationship. Two sectors are classified as part of the two-way trading group if they are both input suppliers and customers to each other. They are classified as part of the one-way trading group if only one sector supplies inputs to the other.

The consumption share for each sector is equal to the share of personal consumption expenditure over the sum of total intermediate input and personal consumption expenditure.

The ratio of accounts receivable to average sales between the current and past quarters (the AR-to-sales ratio) measures the intensity of trade credit provision, while the ratio of accounts payable to average operating cost (the AP-to-costs ratio) measures the intensity of trade credit received from suppliers. We take the mean value of both ratios for each firm from Compustat over 2005Q3–2007Q2 and 2008Q3–2009Q1, calculate the first dif-

ference between these two windows, and use the cross-firm median within each sector as the sector-level measure, restricting attention to sectors with more than three firms. The median declines in the AR-to-sales ratio and the AP-to-costs ratio are -1.6 and -1.0 percentage points, respectively.

I.2 Measurement for sectoral comovements

The correlation of real GDP growth between two countries is commonly used to study business cycle comovement across countries (see, for example, [Frankel and Rose, 1998](#); [Clark and van Wincoop, 2001](#)). In this study, we apply a similar measure, the pairwise correlation of gross output growth between two sectors, to examine intersectoral comovement. We calculate the growth rates of sectoral outputs and then determine the correlation of these growth rates between any pair of sectors over a specified time window, as follows:

$$\text{corr}(\Delta y_i, \Delta y_j) = \frac{\sum_{t \in \mathcal{T}} (\Delta y_{it} - \overline{\Delta y_i})(\Delta y_{jt} - \overline{\Delta y_j})}{(\#\mathcal{T} - 1) \text{std}(\Delta y_i) \text{std}(\Delta y_j)}, \quad (\text{I.1})$$

where i and j with $i \neq j$ represent two different sectors; $\mathcal{T}$ is the time window; Δy_{it} is the output growth rate from the previous period at time t ; and $\overline{\Delta y_i}$ and $\text{std}(\Delta y_i)$ are, respectively, the sample mean and standard deviation over $\mathcal{T}$. For the quarterly analysis around the Great Recession, we use eight consecutive quarters before, during, and after the recession. For the annual recession comparisons, we use an eight-year window unless otherwise specified, and we also report robustness checks using six- and ten-year windows. [Li and Martin \(2019\)](#) use the same measure of sectoral comovement but over a different time window. The main results are robust across these alternative choices.

Previous literature, such as [Christiano and Fitzgerald \(1998\)](#), [Hornstein \(2000\)](#), and [Kalemli-Ozcan et al. \(2013\)](#), introduce another approach to measuring comovements. They regress one sector's employment on another's and use the R^2 value to measure comovement, assessing how much one sector's employment can be explained by the other's. Both approaches capture the strength of bilateral relationships, but the regression R -squared drops the directional sign of the association because it effectively squares the correlation in a bivariate setting. In contrast, our correlation-based approach preserves the sign, which is crucial for distinguishing between synchronous comovement and sectoral divergence.

I.3 Compustat

Following [Kahle and Stulz \(2013\)](#), we use the Compustat Database and create our firm-level sample by filtering out:

- observations with negative total assets (atq), negative sales (saleq), negative cash and marketable securities, or cash and marketable securities greater than total assets;
- firms not incorporated in the U.S.;
- all financial firms (firms with standard industrial classification (SIC) codes between 6000 and 6999);
- firms with a market capitalization less than \$50 million and with a book value of assets less than \$10 million;
- firms with quarterly asset or sales growth greater than 100% at some point during the sample period.

Table [I.1](#) displays the summary statistics of all selected firms.

Table I.1
SUMMARY STATISTICS OF SELECTED COMPUSTAT FIRMS

	Obs	Before		During		Difference	
		Mean	SD	Mean	SD	Mean	SD
<i>AR/Sales</i>	1246	63	80.2	60.8	79.6	-2.3	27.5
<i>AP/Cost</i>	1248	66.9	184.0	59.2	173.9	-7.7	118.6
<i>Investment/TA</i>	1249	1.4	1.5	1.1	1.3	-0.2	0.9
<i>Cash/TA</i>	1249	17.2	18.3	15.9	16.2	-1.3	9.9
<i>Short – term debt/TA</i>	1235	2.3	4.4	2.8	5.1	0.5	4.4
<i>Long – term debt/TA</i>	1243	16.5	17.8	19.2	19.5	2.6	11.8
<i>OIBDP/TA</i>	1241	3.7	2.7	2.9	3.2	-0.7	2.4
<i>Tobin's Q</i>	1249	1.86	0.71	1.44	0.60	-0.43	0.47
<i>Inventory/TA</i>	1241	11.5	12.5	11.4	11.6	-0.1	4.3
<i>g_{sales}</i>	1249	2.9	4.3	-2.9	7.4	-5.8	8.1
<i>g_{assets}</i>	1249	2.1	2.7	-1.1	3.7	-3.2	4.2
<i>log(TA)</i>	1249	7.2	1.6	7.3	1.6	0.1	0.3

I.4 Syndicated loan from Dealscan

Following [Chodorow-Reich \(2014\)](#), we use the Dealscan Database and create our firm-bank connection by filtering out

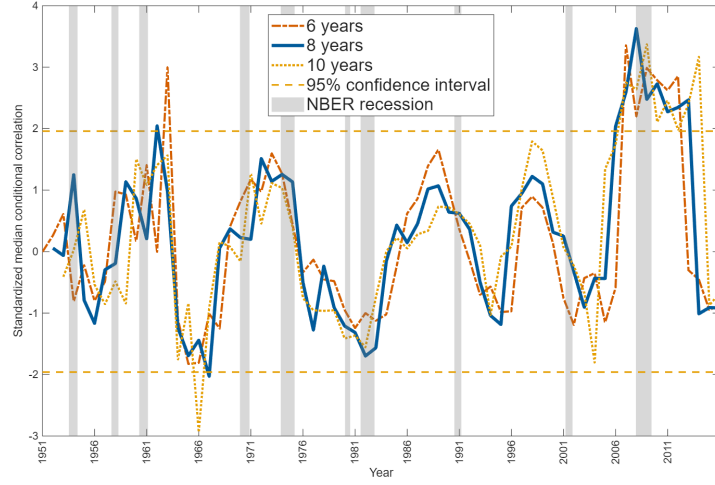
- firms not incorporated in the U.S.;
- all financial firms (firms with standard industrial classification (SIC) codes between 6000 and 6999);
- loans due before October 2008;
- loans whose main purpose is neither working capital nor general corporate purposes.

II Additional Sectoral Evidence

II.1 Pairwise correlations during U.S. postwar recessions

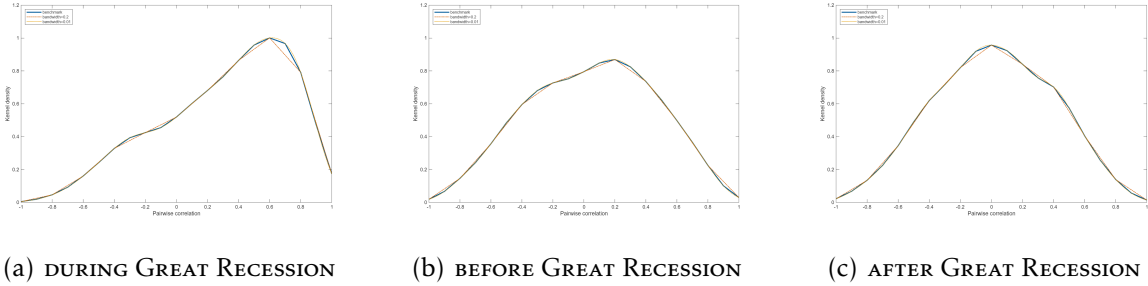
By regressing the logarithm of sectoral output on the logarithm of U.S. GDP, our baseline approach removes the aggregate component of sectoral activity. Because recessions differ in duration, we also vary the length of the rolling window used to compute pairwise correlations. Direct comparisons across windows of different lengths are difficult because the sampling properties of correlations differ with the number of observations. We therefore standardize each sequence relative to its pre-Great-Recession level. Figure [II.1](#) shows that the Great Recession remains the main episode in which conditional sectoral comovement rises sharply, consistent with Panel (b) of Figure [1](#). The increase is visible under six-, eight-, and ten-year rolling windows, indicating that the result is not driven by the particular window length used in the benchmark analysis.

To estimate the kernel density, we use Silverman’s rule of thumb for bandwidth selection, given by $1.06\hat{\sigma}n^{-1/5}$, where $\hat{\sigma} = 0.4$ is the standard deviation of the pairwise correlations before the Great Recession, and $n = 1596$ is the number of pairs in our sample. As a robustness check, we also use a wider bandwidth, equal to the benchmark bandwidth plus 0.1, and a narrow bandwidth of 0.01. Figure [II.2](#) shows that the qualitative pattern is insensitive to these choices. In particular, the recession-period distribution remains shifted toward high positive correlations under all three bandwidths, while the pre- and post-recession distributions are much less concentrated at high correlations. Narrower bandwidths add local variation but do not change the main conclusion.



Note: The figure plots the standardized median of conditional pairwise sectoral output-growth correlations using six-, eight-, and ten-year rolling windows. Conditional correlations are computed after removing the aggregate component by regressing log sectoral output on log U.S. GDP. Each series is standardized relative to its pre-Great-Recession distribution. The horizontal lines mark the 95% confidence interval, and shaded areas denote NBER recessions.

Figure II.1
MEDIAN OF PAIRWISE CORRELATION STANDARDIZED TO PRE-GR

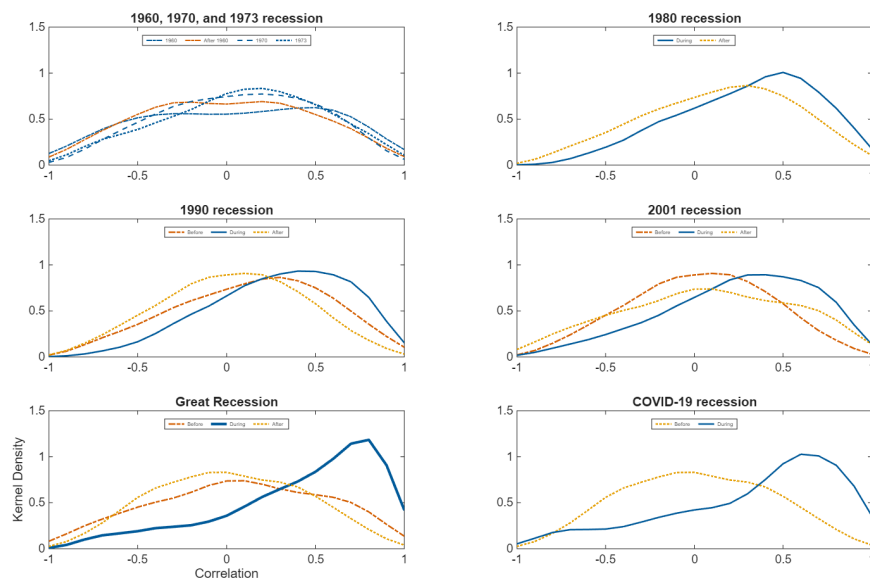


Note: The figure reports kernel densities of pairwise sectoral output-growth correlations around the Great Recession under alternative bandwidth choices. Panel (a) uses the Great Recession window, Panel (b) uses the pre-recession window, and Panel (c) uses the post-recession window. The exercise shows that the rightward shift in the recession-period distribution is not driven by a particular bandwidth choice.

Figure II.2
KERNEL DENSITY DURING GR WITH DIFFERENT BANDWIDTHS

As shown in Figure 2 in Section 2.1, we compare the kernel density of pairwise correlations for nine U.S. recessions after WWII: the 1960, 1970, 1973, 1990, and 2001 recessions, the combined 1980 and 1981-1982 recessions, the Great Recession, and the COVID-19 recession. These correlations are calculated with annual growth over eight years, starting three years before each recession. For comparison, we use the periods 1962-1968, 1983-1989, 1992-1999, 2002-2007, and 2011-2018 to represent the pre-1970, post-1980 (pre-1990), post-1990 (pre-2001), post-2001 (pre-Great Recession), and post-Great Recession (pre-COVID-19) periods, respectively. These time windows are selected

to avoid any overlap with recession years. Unfortunately, we could not find a suitable time window for comparison with the 1973 recession.



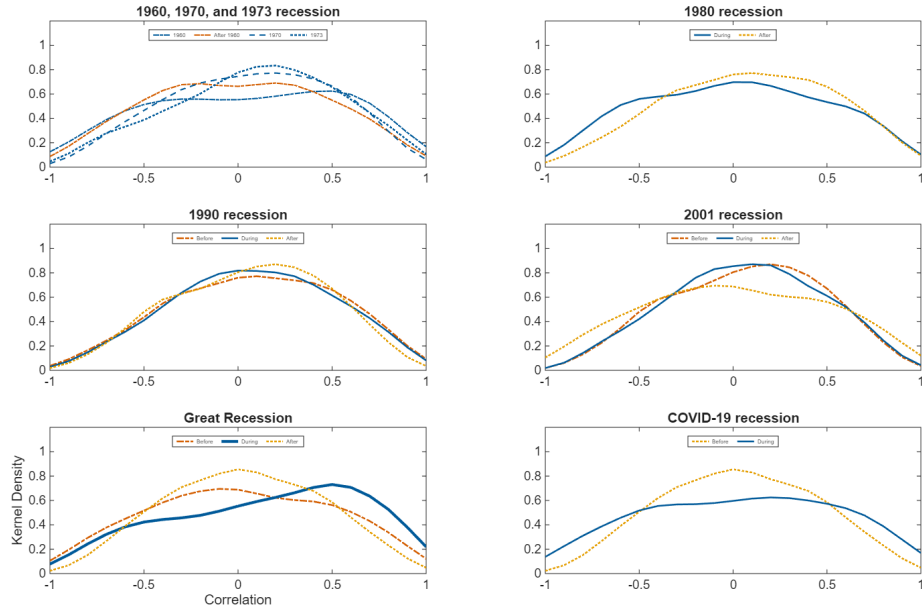
Note: The figure plots kernel densities of unconditional pairwise sectoral output-growth correlations for U.S. recessions. Correlations are computed using annual sectoral growth rates over eight-year windows around each recession. The comparison highlights how much sectoral comovement rises before removing the aggregate component.

Figure II.3

KERNEL DENSITY OF PAIRWISE UNCONDITIONAL CORRELATION IN U.S. RECESSIONS

Figure II.3 shows the kernel density of unconditional pairwise correlations, while Figure II.4 displays the kernel density after removing the aggregate component. Table II.2 presents the corresponding summary statistics and KS statistics. The unconditional correlations rise in several recessions, as aggregate downturns mechanically induce common movements across sectors. The conditional correlations provide a sharper test. Once the aggregate component is removed, most recession episodes show little change in the distribution of pairwise correlations. The Great Recession stands out: the conditional distribution shifts substantially to the right, indicating that the increase in comovement during this episode is not driven only by aggregate fluctuations.

Following Chari et al. (2007), we examine the early-1980s recessions as a useful contrast with the Great Recession because the aggregate decline in GDP was sizable but the financial environment differed. In 1982, real GDP declined by 1.9%, with the steepest quarterly contraction of 6.5% in 1982Q1. By comparison, real GDP declined by 2.7% in 2008, with the largest quarterly contraction of 8.2% in 2008Q4. Despite this comparable aggregate decline, the early-1980s recessions do not generate a similarly pronounced



Note: The figure plots kernel densities of conditional pairwise sectoral output-growth correlations for the same recession windows as in Figure II.3. Conditional correlations are computed after removing the aggregate component from sectoral output. Relative to the unconditional comparison, this figure isolates the residual comovement that is not explained by economy-wide fluctuations.

Figure II.4

KERNEL DENSITY OF PAIRWISE CONDITIONAL CORRELATION IN U.S. RECESSIONS

increase in conditional sectoral comovement. After removing the aggregate component, the density remains much less concentrated in positive correlations, and some sector pairs move in opposite directions. The KS test does not indicate a statistically significant distributional shift. This contrast suggests that the Great Recession involved a stronger intersectoral propagation component than earlier aggregate downturns of comparable size.

During the COVID-19 recession, unconditional sectoral comovement also rises sharply, increasing from 0.01 before the recession to 0.33 during it. However, after controlling for GDP, this increase largely disappears. The COVID-19 episode therefore looks more like a common aggregate shock, whereas the Great Recession displays a residual increase in comovement after the aggregate component is removed.

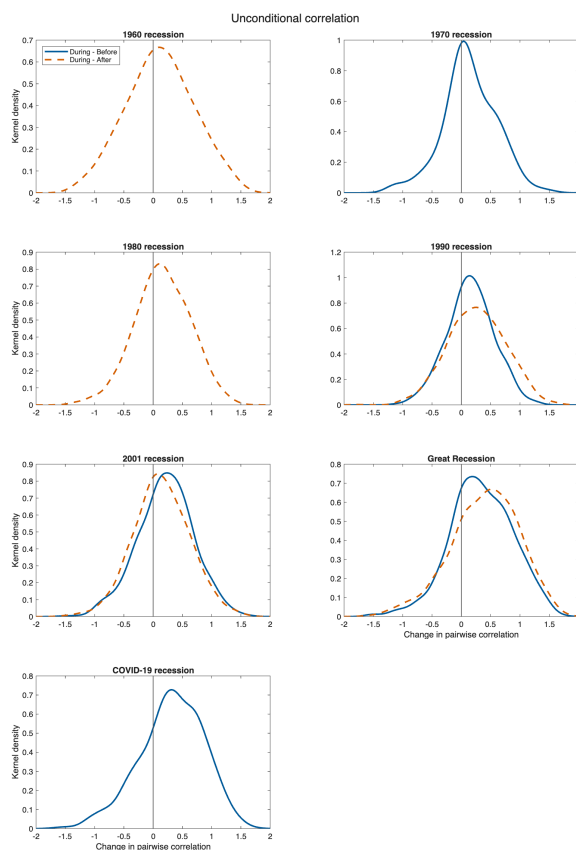
Figure II.5 plots the distribution of changes in unconditional pairwise output-growth correlations around postwar recessions. For each recession, the change is computed as the correlation during the recession minus the correlation in the comparison window before or after the recession, whenever such a window is available. The figure shows that pairwise correlations are not fixed objects: they move across recession windows, and several episodes display changes centered close to zero with substantial dispersion. However, the Great Recession and the COVID-19 recession stand out in the unconditional

Table II.2
PAIRWISE CORRELATIONS AROUND OTHER RECESSIONS

	Unconditional			Conditional		
	Mean	Median	KS Stat (CV)	Mean	Median	KS Stat (CV)
The 1960 recession						
During	0.27	0.32		0.05	0.09	
After	0.16	0.19	0.09(0.09,0.21)	0.00	0.01	0.08 (0.09,0.21)
The 1970 recession						
Before	0.16	0.19	0.08(0.09,0.21)	0.00	0.01	0.08 (0.09,0.21)
During	0.24	0.28		0.07	0.08	
The 1973 recession						
During	0.36	0.39		0.09	0.12	
The early-1980s recessions						
During	0.30	0.35		0.01	0.01	
After	0.14	0.17	0.17(0.09,0.21)	0.09	0.09	0.08 (0.09,0.21)
The 1990 recession						
Before	0.14	0.17	0.15(0.09,0.21)	0.09	0.09	0.04 (0.09,0.21)
During	0.29	0.33		0.08	0.08	
After	0.03	0.05	0.27(0.09,0.21)	0.06	0.09	0.04 (0.09,0.21)
The 2001 recession						
Before	0.03	0.05	0.16(0.09,0.21)	0.06	0.09	0.05 (0.09,0.21)
During	0.24	0.29		0.06	0.07	
After	0.07	0.08	0.11(0.09,0.21)	0.02	0.00	0.07 (0.09,0.21)
The Great Recession						
Before	0.07	0.08	0.33(0.09,0.21)	0.02	0.00	0.14 (0.09,0.21)
During	0.41	0.52		0.14	0.21	
After	0.01	0.00	0.41(0.09,0.21)	0.03	0.03	0.18 (0.09,0.21)
The COVID-19 recession						
Before	0.01	0.00	0.36(0.09,0.21)	0.03	0.03	0.08 (0.09,0.21)
During	0.33	0.46		0.04	0.05	

Note: The table reports means and medians of pairwise sectoral output-growth correlations around each recession. “Unconditional” refers to raw pairwise correlations, while “Conditional” refers to correlations after removing aggregate movements. The KS statistic is the two-sample Kolmogorov–Smirnov distance comparing the distribution in the before or after period with the recession-period distribution for the same episode. Critical values for a 5% significance level are reported in parentheses and adjust for autocorrelation following [Lanzante \(2021\)](#). The first value corresponds to mild autocorrelation, with AR(1) parameter 0.6; the second corresponds to high autocorrelation, with AR(1) parameter 0.9.

correlations, with distributions shifted more clearly to the right. This pattern is consistent with the fact that both episodes involved large aggregate disturbances that raised sectoral comovement broadly.



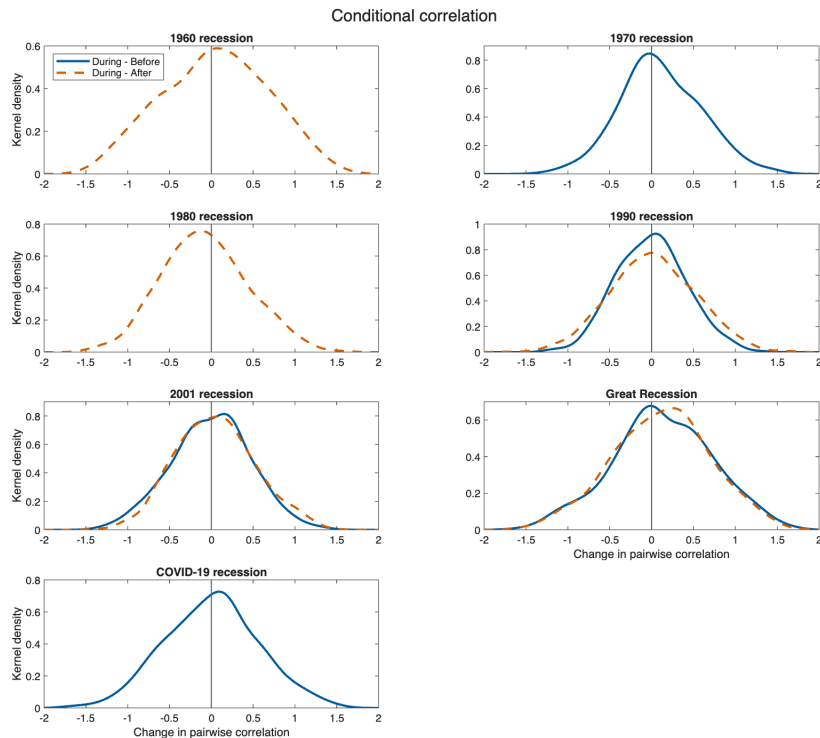
Note: The figure plots kernel densities of changes in unconditional pairwise sectoral output-growth correlations around postwar U.S. recessions. The blue solid line reports the change between the recession window and the pre-recession comparison window, and the orange dashed line reports the change between the recession window and the post-recession comparison window, whenever the corresponding comparison window is available. The vertical line marks zero change.

Figure II.5

CHANGES IN UNCONDITIONAL PAIRWISE CORRELATIONS AROUND POSTWAR RECESSIONS

Figure II.6 repeats the exercise using conditional pairwise correlations, after removing the aggregate component. The contrast is informative. Once aggregate movements are removed, the COVID-19 shift becomes much less pronounced, while the Great Recession still displays a visible rightward shift relative to its comparison windows. Other recessions show weaker and less systematic changes, with distributions mostly centered near zero. Thus, although pairwise correlations vary over time, the Great Recession remains

distinctive because the increase in comovement survives after controlling for aggregate fluctuations. This supports the main-text focus on mechanisms that propagate shocks across sectors, rather than on aggregate shocks alone.



Note: The figure plots kernel densities of changes in conditional pairwise sectoral output-growth correlations around postwar U.S. recessions. Conditional correlations are computed after removing the aggregate component from sectoral output growth. The blue solid line reports the change between the recession window and the pre-recession comparison window, and the orange dashed line reports the change between the recession window and the post-recession comparison window, whenever the corresponding comparison window is available. The vertical line marks zero change.

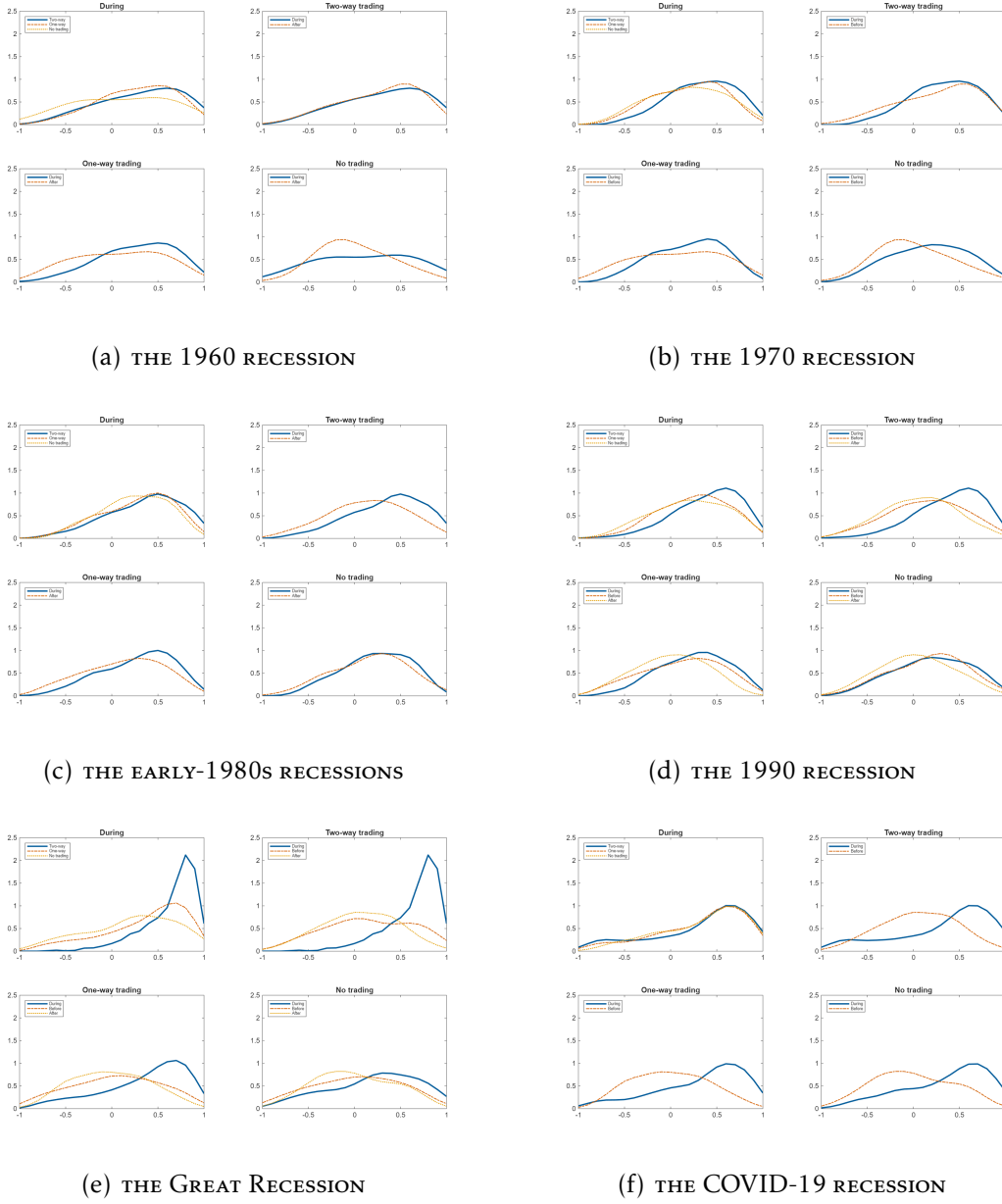
Figure II.6

CHANGES IN CONDITIONAL PAIRWISE CORRELATIONS AROUND POSTWAR RECESSIONS

II.2 Role of IO linkage in the other recessions

We apply the same sector-pair classification as in Section 2.2 to the other recessions. Figure II.7 presents the kernel densities for two-way trading, one-way trading, and no-trading pairs around each recession.

The comparison reveals several patterns. First, the distributions for the three groups change little during the 1960 and 1970 recessions. Second, the two-way and one-way trading groups shift modestly to the right during the early-1980s and 1990 recessions,



Note: The figure applies the input-output classification used in the main text to other recession episodes. Two-way trading pairs are sector pairs that buy from and sell to each other, one-way trading pairs have a supplier-customer relationship in only one direction, and no-trading pairs have no recorded intermediate-input linkage. In each subfigure, the first panel compares the three groups during the recession, while the remaining panels compare the before- and after-recession distributions within each group.

Figure II.7

KERNEL DENSITY OF UNCONDITIONAL CORRELATION BY INTERCONNECTEDNESS

but the shift is much smaller than during the Great Recession. Third, the Great Recession exhibits the same ordering observed in the quarterly data: two-way trading pairs display the strongest comovement, followed by one-way trading pairs, while no-trading pairs comove the least. This ordering is informative because it links the rise in comovement

to the structure of input-output relationships rather than to a uniform increase across all pairs. Fourth, during the COVID-19 recession, all three groups shift to the right by a similar amount. This last pattern is more consistent with a broad aggregate component in the COVID-19 recession, whereas the Great Recession generated a rise in comovement that was more concentrated among production-linked sectors.

II.3 Input-output intensity and sectoral comovement

This appendix examines whether the increase in sectoral comovement during the Great Recession can be explained solely by the intensity of input-output connections. If a purely real production-network mechanism were the main driver, sector pairs with larger bilateral input shares or larger Leontief-inverse exposure should exhibit stronger comovement. We test this implication using directed supplier-customer observations.

Because pairwise output-growth correlations are symmetric while input-output linkages are directed, we include two observations for each unordered sector pair, one for each supplier-customer direction. The dependent variable is the same across the two directions, while the bilateral input share and Leontief-inverse exposure vary by direction. We estimate variants of the following regression:

$$y_{ij} = \alpha + \beta_1 \text{Input Share}_{ij} + \beta_2 \text{Leontief}_{ij} + \beta_3 \mathbf{1}_{ij}^{\text{Two-way}} + \beta_4 \mathbf{1}_{ij}^{\text{One-way}} + \varepsilon_{ij}, \quad (\text{II.2})$$

where y_{ij} is either the pairwise output-growth correlation during the Great Recession or the change in pairwise correlation from the pre-recession period to the Great Recession. Input Share_{ij} is the standardized directed bilateral input-expenditure share from supplier sector i to customer sector j , and Leontief_{ij} is the standardized directed Leontief-inverse exposure. $\mathbf{1}_{ij}^{\text{Two-way}}$ and $\mathbf{1}_{ij}^{\text{One-way}}$ are indicators for two-way trading and one-way trading groups. The omitted category for the trading indicators is sector pairs with no direct input-output linkage. Standard errors are clustered at the unordered sector-pair level.

The results show that the extensive margin of input-output linkages is strongly associated with sectoral comovement, whereas the continuous intensity of these linkages is not. Column (1) of Table II.3 shows that the directed bilateral input share is positively related to output-growth comovement when entered alone, but the explanatory power is very small. Once we control for whether the sector pair is connected through one-way or two-way trade, the coefficient on the input share becomes small and statistically insignificant. The Leontief-inverse exposure is also statistically insignificant throughout.

By contrast, the trading-type indicators are large and precisely estimated. Relative

Table II.3
INPUT-OUTPUT INTENSITY AND SECTORAL COMOVEMENT

	Correlation during the Great Recession			Change in correlation
	(1)	(2)	(3)	(4)
Input Share _{ij}	0.0382*** (0.0111)		-0.0053 (0.0078)	-0.0036 (0.0115)
Leontief _{ij}	-0.0167 (0.0114)		-0.0010 (0.0101)	-0.0045 (0.0136)
$\mathbf{1}_{ij}^{\text{Two-way}}$		0.367*** (0.0265)	0.371*** (0.0269)	0.323*** (0.0380)
$\mathbf{1}_{ij}^{\text{One-way}}$		0.0971*** (0.0267)	0.0991*** (0.0267)	0.136*** (0.0365)
Observations	3,192	3,192	3,192	3,192
Adjusted R^2	0.005	0.117	0.117	0.042

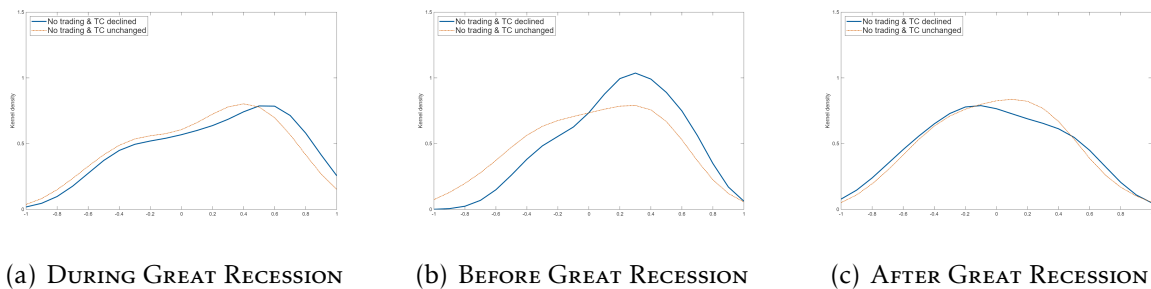
Note: The table reports variants of Equation (II.2) using directed supplier-customer observations. Columns (1)–(3) use the pairwise output-growth correlation during the Great Recession as the dependent variable. Column (4) uses the change in pairwise output-growth correlation from the pre-recession period to the Great Recession. The directed bilateral input share and directed Leontief-inverse exposure are standardized. Two-way and one-way trading indicators are defined using direct input-output linkages, with no direct trading as the omitted group. Standard errors, clustered by unordered sector pair, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

to sector pairs with no direct input-output linkage, two-way trading pairs have output-growth correlations about 0.37 higher during the Great Recession, while one-way trading pairs have correlations about 0.10 higher. The same pattern holds for the change in correlation: two-way trading pairs experience an increase about 0.32 larger than no-trading pairs, and one-way trading pairs experience an increase about 0.14 larger. These findings support the interpretation in the main text: sectoral comovement during the Great Recession is concentrated among sectors connected through production networks, but it is not well explained by measured input-output intensity alone. This helps motivate the paper’s focus on trade credit as a financial linkage embedded in supplier-customer relationships.

II.4 Role of trade credit in the no-trading group

Following the categorization in Section 2.3, a pair is considered to have experienced a trade credit decline during the Great Recession (henceforth, the trade credit-declined group) if both the supplier’s AR-to-sales ratio and the customer’s AP-to-costs ratio declined more than the corresponding median values across all public firms, which are -1.6 and -1.0 percentage points, respectively. Otherwise, the pair is categorized as part of

the unchanged group (henceforth, the trade credit-unchanged group). For pairs in the no-trading group, there is no observed input-output direction. We therefore classify a no-trading pair as trade credit declined if the decline condition is met in either possible direction. Figure II.8 reports the results. Unlike the two-way and one-way trading groups in the main text, the no-trading group does not exhibit a meaningful difference between the trade credit-declined and trade credit-unchanged subgroups before, during, or after the Great Recession. This placebo exercise suggests that the trade credit result is tied to production relationships rather than to sector-level trade credit contractions alone.

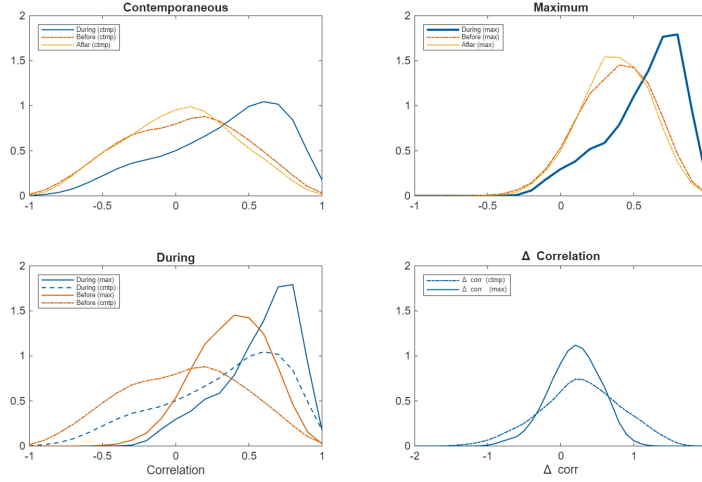


Note: The figure restricts the sample to sector pairs without an observed input-output linkage. A no-trading pair is classified as trade credit declined if the trade-credit-decline condition is satisfied in either possible direction, since there is no observed supplier-customer direction for these pairs. The absence of a clear difference between the trade-credit-declined and unchanged groups serves as a placebo check for the trade-credit mechanism.

Figure II.8
KERNEL DENSITY FOR NO TRADING GROUP BY THE TRADE CREDIT SUBGROUP

II.5 Robustness check

Due to delivery delays, adjustment costs, search frictions, and other factors, two sectors may not comove contemporaneously; instead, one sector may lead the other. The rise in contemporaneous comovement during the Great Recession could therefore reflect changes in timing rather than stronger economic linkages. To address this concern, we calculate contemporaneous, one-period lagged, and one-period lead correlations and take the maximum value among the three. Figure II.9 reports the results before, during, and after the Great Recession. The maximal correlations also rise sharply during the Great Recession. Although maximal correlations are mechanically higher than contemporaneous correlations in all periods, the recession-period distribution still shifts to the right relative to the before- and after-recession distributions. The increase in comovement is therefore not merely a consequence of allowing sectors to lead or lag one another by one quarter.



Note: The figure compares the distribution of contemporaneous pairwise correlations with the distribution obtained after allowing one sector to lead or lag the other by one quarter. For each sector pair, the maximal-correlation measure takes the maximum of the contemporaneous, one-quarter lead, and one-quarter lag correlations. The rightward shift during the Great Recession remains visible under this alternative timing convention.

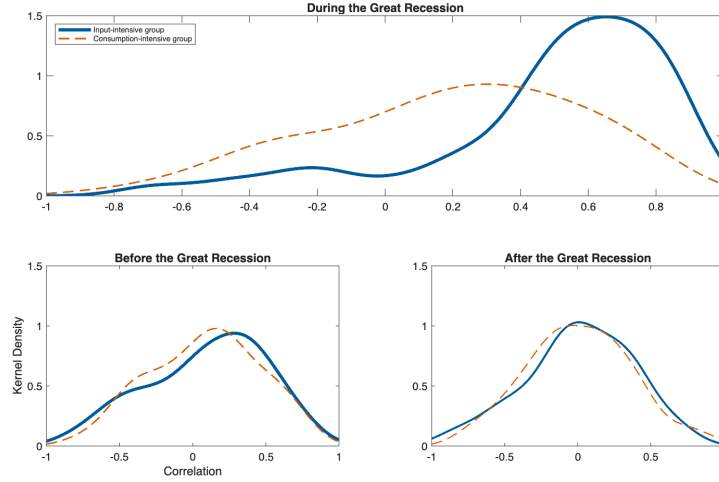
Figure II.9

ROBUSTNESS CHECKS: MAXIMUM AND CONTEMPORANEOUS CORRELATIONS

We further examine whether the rise in sectoral comovement during the Great Recession is driven more by production-network linkages than by common demand for final consumption or investment goods. Using the input-output table, we compute, for each sector, the average shares of output allocated to intermediate inputs, final consumption, and investment goods over 2003–2007. A sector is classified as input-intensive, consumption-intensive, or investment-intensive if its corresponding output-use share exceeds the 67th percentile of the cross-sectoral distribution. We then compare pairwise output correlations within each group, focusing on sector pairs in which both sectors belong to the same intensive category.

Figure II.10 compares input-intensive pairs with consumption-intensive pairs. Before and after the Great Recession, the two distributions are relatively close to each other. During the Great Recession, however, the distribution for input-intensive pairs shifts sharply to the right and becomes concentrated at high positive correlations. The distribution for consumption-intensive pairs also moves to the right, but it is more dispersed and places substantially less mass at very high correlations. This pattern suggests that the rise in comovement is not primarily driven by common movements in final consumption demand. Instead, the increase is especially strong among sectors whose output is used intensively as intermediate inputs, consistent with propagation through production linkages.

Figure II.11 performs the same comparison between input-intensive and investment-



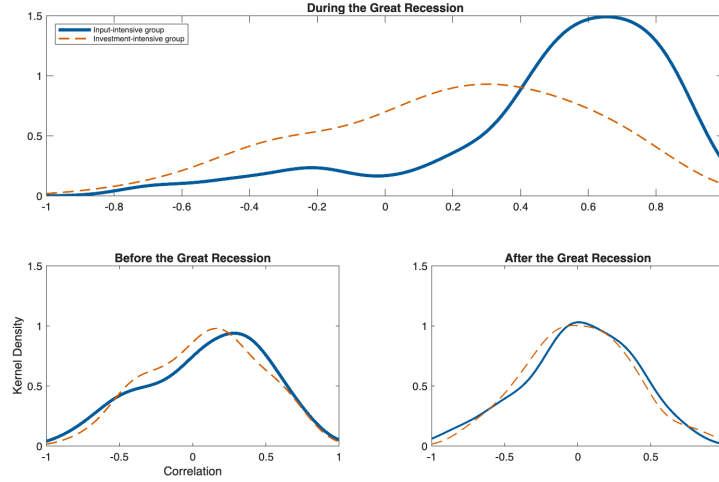
Note: The figure plots kernel densities of pairwise output-growth correlations for input-intensive and consumption-intensive sector pairs before, during, and after the Great Recession. A sector is classified as input-intensive or consumption-intensive if the corresponding output-use share is above the 67th percentile of the cross-sectoral distribution. The densities are computed for within-group pairs.

Figure II.10

ROBUSTNESS CHECKS: INPUT PROVIDERS VERSUS CONSUMPTION PROVIDERS

intensive pairs. This exercise addresses the concern that investment-goods sectors may be especially sensitive during financial crises and may therefore account for the increase in comovement. The results again show that input-intensive pairs exhibit the strongest rightward shift during the Great Recession. Investment-intensive pairs display some increase in comovement, but their distribution is less concentrated at high positive correlations than the distribution for input-intensive pairs. Before and after the Great Recession, the two distributions are much closer. Thus, while investment demand may have contributed to the recessionary downturn, the cross-sectional pattern of comovement is more pronounced among sectors connected through intermediate-input use.

Taken together, these results reinforce the main empirical interpretation. The increase in sectoral comovement during the Great Recession is not concentrated among sectors most exposed to final consumption or investment demand. Rather, it is strongest among sectors whose output is intensively used as intermediate inputs. While this sector-level classification does not isolate direct bilateral trade by itself, it makes it difficult to attribute the pattern solely to common final-demand shocks. By complementing our direct pair-level evidence, this finding further reinforces the view that the Great Recession involved propagation through production linkages, creating the environment in which trade credit can operate as an additional financial transmission margin.



Note: The figure plots kernel densities of pairwise output-growth correlations for input-intensive and investment-intensive sector pairs before, during, and after the Great Recession. A sector is classified as input-intensive or investment-intensive if the corresponding output-use share is above the 67th percentile of the cross-sectoral distribution. The densities are computed for within-group pairs.

Figure II.11

ROBUSTNESS CHECKS: INPUT PROVIDERS VERSUS INVESTMENT-GOOD PROVIDERS

III Firm-level evidence on trade-credit transmission

The sector-level evidence points to a role for trade credit along production linkages. We complement that evidence with firm-level variation around the collapse of Lehman Brothers (LB), one of the central financial disruptions of the Great Recession. This exercise examines whether financially exposed customers experienced changes in trade credit and whether those changes were associated with stronger comovement with their suppliers.

III.1 Data and sample selection

We first construct a firm-to-firm production network using Form 10-K, following the method in [Garcia-Appendini and Montoriol-Garriga \(2013\)](#). We identify 641 supplier-customer pairs, comprising 426 suppliers and 176 customers. To establish relationships between listed firms and LB, we use syndicated loan data from DealScan. Following [Ivashina and Scharfstein \(2010\)](#), we select firms for which LB led or joined a syndicated loan before its collapse. Additionally, in line with the literature on financial networks, such as [Allen and Gale \(2000\)](#), [Elliott et al. \(2014\)](#), and [Acemoglu et al. \(2015\)](#), among others, we identify lenders that share a direct financial network with LB through the syndicated loan market. Thus, a firm is considered indirectly connected to LB if it did

not borrow directly from LB but from lenders connected to LB. Overall, we have 19 out of 426 suppliers directly connected to LB; 237 indirectly connected through their lenders; and 170 without a connection to LB through the syndicated loan market. Among the 176 customers, 40 borrowed directly from LB; 120 were indirectly connected; and 16 had no relationship with LB. It is important to note that we classify these firms based only on information from the syndicated loan market and do not exclude any other potential financial connections they may have had with LB.

We use Compustat to obtain financial variables for listed firms, with summary statistics reported in Table III.4. Pairwise firm-level correlations increase by 0.17 during the Great Recession, consistent with the sector-level evidence. We measure financial variables using firm-level averages over 2005Q3–2007Q2 and 2008Q3–2009Q1 to represent the pre-recession and recession periods, respectively. Before the recession, suppliers in our sample are smaller, extend less trade credit, and hold more cash than the average Compustat firm, while customers are larger, receive less trade credit, and hold less cash. This pattern is consistent with the way supplier-customer relationships are observed in Form 10-K filings: suppliers report major customers, which tend to be relatively large. Consistent with Kahle and Stulz (2013), standard balance-sheet ratios such as investment, cash, and short- and long-term debt do not change significantly for suppliers and customers across the two windows. In contrast, profitability and growth decline substantially during the recession, and market valuation falls as reflected in Tobin's Q.

III.2 Trade credit provision and reception during the Great Recession

We use data from U.S. public firms in Compustat to study the evolution of trade credit, calculating the AR-to-sales and AP-to-costs ratios as defined in Section 2.3. These ratios are then adjusted for seasonality using moving-average methods at the firm level. Figure III.12 shows the evolution of the median values for both ratios from 1980Q3 to 2016Q3. The ratios exhibit modest fluctuations over time, even during the 1990 and 2001 recessions. During the Great Recession, both ratios initially increased and then fell sharply starting in 2008Q3, with the decline especially pronounced for AR-to-sales. This pattern is consistent with a broad tightening of trade-credit terms: suppliers deferred a smaller share of new sales as trade credit, while some existing receivables may also have been written off. This finding is consistent with evidence from Costello (2019) for the U.S. during the Great Recession and Love et al. (2007) for the Mexican crisis in 1994 and the Asian flu in 1997.

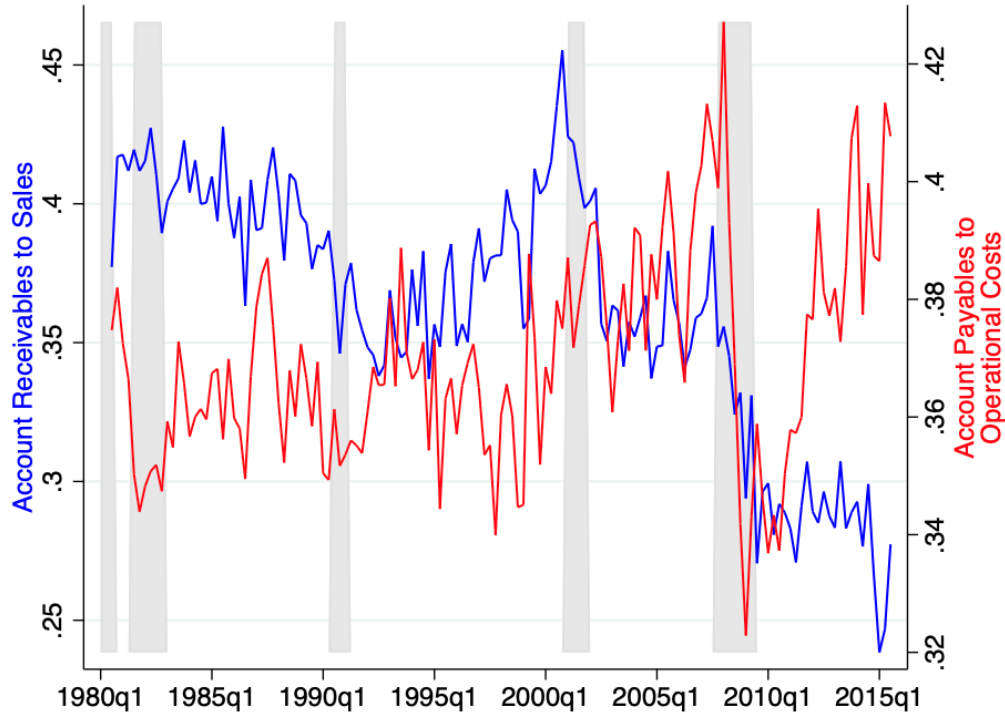
We then test how the customers' financial positions responded to the LB shock. In

Table III.4
SUMMARY STATISTICS: PAIRED SUPPLIER AND CUSTOMER

	Obs	Before		During		Difference	
		Mean	SD	Mean	SD	Mean	t-stats
corr_{ij}	641	.035	.44	.20	.46	0.17***	(7.14)
Suppliers							
<i>AR/Sales</i>	426	59.92	24.78	59.34	24.74	-0.58	(-0.34)
<i>AP/Cost</i>	426	58.16	53.18	54.28	36.11	-3.88	(-1.25)
<i>Investment/TA</i>	426	1.46	1.82	1.33	1.62	-0.12	(-1.04)
<i>Cash/TA</i>	426	20.00	19.78	18.00	18.32	-1.99	(-1.53)
<i>Short – term debt/TA</i>	425	2.90	4.73	3.30	5.94	0.41	(1.10)
<i>Long – term debt/TA</i>	426	16.39	18.81	18.23	20.65	1.85	(1.36)
<i>OIBDP/TA</i>	423	3.33	3.03	2.23	3.86	-1.10***	(-4.61)
<i>Tobin's Q</i>	423	1.88	0.73	1.36	0.57	-0.52***	(-11.48)
<i>Inventory/TA</i>	426	11.98	10.42	12.50	10.61	0.52	(0.73)
<i>g_{sales}</i>	426	2.94	3.45	0.24	3.71	-2.71***	(-11.03)
<i>g_{assets}</i>	426	2.89	3.91	0.18	3.88	-2.71***	(-10.13)
<i>log(TA)</i>	426	6.48	1.69	6.61	1.73	0.13	(1.09)
<i>log(sales)</i>	426	5.07	1.73	5.18	1.73	0.11	(0.92)
CR	452	1.07	0.86				
customers							
<i>AR/Sales</i>	176	51.67	39.94	52.69	42.66	1.02	(-0.23)
<i>AP/Cost</i>	176	62.74	49.32	62.1	47.62	-0.63	(-0.12)
<i>Investment/TA</i>	176	1.55	1.48	1.49	1.53	-0.06	(-0.35)
<i>Cash/TA</i>	176	11.75	13.7	10.28	11.88	-1.47	(-1.07)
<i>Short – term debt/TA</i>	176	3.95	5.91	4.55	8.16	0.6	(-0.79)
<i>Long – term debt/TA</i>	176	20.32	17.43	23.59	18.48	3.28	(-1.71)
<i>OIBDP/TA</i>	176	3.83	2.08	3.31	2.36	-0.52*	(-2.19)
<i>Tobin's Q</i>	175	1.82	0.64	1.46	0.5	-0.36***	(-5.88)
<i>Inventory/TA</i>	176	13.21	12.99	13.23	12.95	0.02	(-0.01)
<i>g_{sales}</i>	176	2.61	2.77	0.48	3.59	-2.12***	(-6.21)
<i>g_{assets}</i>	176	2.69	3.16	0.94	2.84	-1.75***	(-5.48)
<i>log(TA)</i>	176	9.04	1.59	9.2	1.56	0.16	(-0.92)
<i>log(sales)</i>	176	7.72	1.6	7.83	1.58	0.11	(-0.63)
CR	185	0.66	.71				

particular, we regress the change in financial measures on whether the customers were directly or indirectly connected to LB before its collapse, along with other control variables, as

$$\Delta y_j = \alpha_0 + \alpha_1 \mathbf{1}_{j,dir}^{LB} + \alpha_2 \mathbf{1}_{j,indir}^{LB} + \delta' \Delta X_j + \beta y_{j,before} + \epsilon_j, \quad (\text{III.3})$$



Note: The figure plots the median AR-to-sales and AP-to-costs ratios for U.S. public firms in Compustat. The AR-to-sales ratio measures trade credit provision to customers, while the AP-to-costs ratio measures trade credit received from suppliers. Firm-level ratios are adjusted for seasonality using moving averages before the cross-firm median is computed. Shaded areas denote NBER recessions.

Figure III.12
EVOLUTION OF INTENSITIES OF TRADE CREDIT PROVISION AND RECEPTION

where j is an index for a customer; y is the financial variable of interest; Δ stands for the first difference; $\mathbf{1}_{j,dir}^{LB}$ is an indicator variable that takes the value of 1 when the customer is directly connected to LB in the syndicated loan market; $\mathbf{1}_{j,indir}^{LB}$ is an indicator variable that takes the value of 1 when the customer is indirectly connected to LB in the syndicated loan market; X is the control variables listed in Table III.4; and ΔX is the first difference between the in and pre-recession median of X .

Table III.5 presents the point estimates for equation (III.3), with robust standard errors reported in parentheses. Column (1) shows that in the regression of the AP-to-costs ratio, the coefficient for the indicator of a direct connection to LB is negative and statistically significant, indicating a 6.5-percentage-point decrease in the ratio for LB-connected customers. This suggests that, during the recession, LB-connected customers either settled their existing trade credit obligations more quickly or their suppliers deferred a smaller proportion of new sales as trade credit. The coefficient for the indirect connection is negative but not statistically significant. This may be because other lenders connected to LB

absorbed the shocks, preventing systematic transmission to their own borrowers.

Table III.5
REGRESSION RESULTS OF EQUATION (III.3)

	$\Delta \frac{AP_j}{OC_j}$	$\Delta \frac{AR_j}{sales_j}$	$\Delta \frac{debt_j}{TA_j}$	$\Delta \frac{cash_j}{TA_j}$
	(1)	(2)	(3)	(4)
$\mathbf{1}_{j,dir}^{LB}$	-6.52*** (.956)	-.981 (3.15)	3.15 (2.73)	-8.05*** (1.68)
$\mathbf{1}_{j,indir}^{LB}$	-6.55 (4.81)	3.2 (2.93)	3.07 (2.21)	-6.55*** (1.26)
obs	58	58	58	58
adjusted R^2	.477	.352	.321	.416

Notes: * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

In columns (2)-(4), we analyze other short-term financial measures, such as the AR-to-sales ratio, short-term debt, and the cash ratio. The LB shock shows no significant effects on the AR-to-sales and short-term debt ratios. However, while LB-unrelated firms substantially increased their cash holdings during the recession (by roughly 8.7 percentage points, as indicated by the constant term), LB borrowers accumulated significantly less cash. This relative inability to build liquidity buffers highlights the severe financial strain on LB-connected customers, preventing them from engaging in the precautionary cash hoarding typically observed among constrained firms during crises (e.g., [Kahle and Stulz \(2013\)](#)).

III.3 Transmission of the LB Shock

We investigate whether and how the pairwise correlation between two firms is associated with LB exposure and trade-credit contraction. Specifically, we calculate the first difference of pairwise correlations between two periods to eliminate fixed effects associated with suppliers, customers, and pairs. By comparing the change in the correlation of a common supplier with different customers, our approach resembles a difference-in-difference framework, reducing the likelihood that time-varying unobservables, other than the connection to LB, affect the change in comovement over time. Our empirical strategy is specified as follows:

$$\Delta \text{corr}_{ij} = \alpha_0 + \alpha_1 \mathbf{1}_j^{LB} + \alpha_2 \mathbf{1}_j^{LB} \times \Delta \frac{AP_j}{OC_j} + \delta_i + \beta' \Delta X_j + \epsilon_{ij}, \quad (\text{III.4})$$

where i and j are, respectively, indexes for the supplier and customer; Δcorr_{ij} is the change in the pairwise correlation before and during the recession; $\mathbf{1}_j^{LB}$ is the indicator that customer j is either directly or indirectly connected to LB; δ_i denotes supplier fixed effects; and X are characteristics of customer j , including the first differences of the financial variables listed in Table III.4. To highlight the role of trade credit in propagating and amplifying the LB shock, we include the interaction between the LB indicator and the change in the AP-to-costs ratio. Since LB-connected customers experienced a contraction in trade credit received, on average, a negative coefficient on this interaction implies that a larger decline in trade credit is associated with a larger increase in supplier-customer comovement. Supplier fixed effects absorb supplier-level changes in comovement over the recession window.

Table III.6 reports the point estimates. Column (1) presents the benchmark specification in Equation (III.4). During the Great Recession, for customers experiencing no change in their AP-to-costs ratio, the correlation between a supplier and a customer directly connected to LB increased by 0.88 more than for an unconnected customer. This baseline increase is large relative to the unconditional sample average increase of 0.17. Furthermore, the coefficient on the interaction between LB exposure and the change in the AP-to-costs ratio is negative and statistically significant. Because a decline in the AP-to-costs ratio corresponds to a reduction in trade credit received, the negative interaction implies that LB-exposed customers experiencing larger trade credit contractions comoved even more strongly with their suppliers.

We also test whether customers that relied more on trade credit before the Great Recession comoved more during the Great Recession. In doing so, we replace the change in AP-to-costs ratio in Equation (III.4) with the pre-recession ratio AP_j^{pre}/OC_j^{pre} . The point estimates in column (2) show that the level of the pre-recession ratios fails to predict the change in pairwise correlations. Lastly, we perform a ‘placebo test’ by comparing the change in correlations between two regular periods, namely 2003Q3-2005Q2 versus 2005Q3-2007Q2. As shown in column (3), we do not find evidence that the control and the treatment group had pre-existing differences in comovement before the Great Recession.

Table III.6
POINT ESTIMATES FOR EQUATION (III.4)

	Δcorr_{ij}	placebo test	
	(1)	(2)	(3)
$\mathbf{1}_{j,dir}^{LB}$	.875* (.345)	-.172 (1.12)	-.167 (.296)
$\mathbf{1}_{j,indir}^{LB}$	.687 [†] (.36)	-.501 (1.15)	-.0439 (.3)
$\mathbf{1}_{j,dir}^{LB} \times \Delta \frac{AP_j}{OC_j}$	-.393** (.143)		
$\mathbf{1}_{j,indir}^{LB} \times \Delta \frac{AP_j}{OC_j}$	-.392** (.142)		
$\mathbf{1}_{j,dir}^{LB} \times \frac{AP_j^{pre}}{OC_j^{pre}}$		.00736 (.0177)	
$\mathbf{1}_{j,indir}^{LB} \times \frac{AP_j^{pre}}{OC_j^{pre}}$		.00952 (.018)	
obs	148	148	150
adjusted R^2	.125	.0881	.0522

Notes: [†] $p < 0.10$ * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

IV More Results from Quantitative Analysis

IV.1 Microfoundation for the calibration of γ

This appendix provides a microfoundation for the calibration of the outside-option parameter γ . In the model, customer j 's payoff from staying with supplier i is

$$-\left(1 - (1 - \eta)tc_{ij}\right) \frac{p_i m_{ij}}{\eta} + \phi tc_{ij}^{1-\kappa} e_{ij}^{\kappa} \frac{p_i m_{ij}}{\eta}.$$

Suppose that customer j also receives a stochastic outside option. Let ε_{ij} denote the idiosyncratic attractiveness of the alternative supplier, measured in units of the transaction value. The payoff from switching is

$$(-\gamma + \varepsilon_{ij}) \frac{p_i m_{ij}}{\eta},$$

where γ captures the deterministic component of the switching cost. A higher ε_{ij} makes

the outside option more attractive. The customer remains with the incumbent supplier if

$$-\left(1 - (1 - \eta)tc_{ij}\right) \frac{p_i m_{ij}}{\eta} + \phi tc_{ij}^{1-\kappa} e_{ij}^\kappa \frac{p_i m_{ij}}{\eta} \geq (-\gamma + \varepsilon_{ij}) \frac{p_i m_{ij}}{\eta}.$$

After dividing by $p_i m_{ij}/\eta$, this condition becomes

$$\varepsilon_{ij} \leq (1 - \eta)tc_{ij} + \phi tc_{ij}^{1-\kappa} e_{ij}^\kappa - (1 - \gamma).$$

Therefore, if ε_{ij} has distribution function $\mathbf{F}_\varepsilon$, the model-implied retention probability for relationship (i, j) is

$$\mathbf{F}_\varepsilon \left((1 - \eta)tc_{ij} + \phi tc_{ij}^{1-\kappa} e_{ij}^\kappa - (1 - \gamma) \right).$$

We calibrate γ using the average relationship in the stationary equilibrium. For a candidate value of γ , let $tc_{ij}^*(\gamma)$, $e_{ij}^*(\gamma)$, $p_i^*(\gamma)$, and $m_{ij}^*(\gamma)$ denote the corresponding stationary-equilibrium objects. Define the transaction-weighted average incumbent relationship benefit as

$$\bar{\mathcal{B}}(\gamma) = \frac{\sum_{i=1}^n \sum_{j=1}^n \left((1 - \eta)tc_{ij}^*(\gamma) + \phi \left(tc_{ij}^*(\gamma) \right)^{1-\kappa} \left(e_{ij}^*(\gamma) \right)^\kappa \right) \frac{p_i^*(\gamma) m_{ij}^*(\gamma)}{\eta}}{\sum_{i=1}^n \sum_{j=1}^n \frac{p_i^*(\gamma) m_{ij}^*(\gamma)}{\eta}}.$$

This object summarizes the average value of staying with the incumbent supplier, weighted by the size of supplier-customer transactions.

To close the calibration, we assume that the idiosyncratic component of the customer's outside option is mean zero:

$$\varepsilon_{ij} \sim \mathcal{U}[-\bar{\varepsilon}, \bar{\varepsilon}].$$

Under this assumption, $\mathbf{F}_\varepsilon^{-1}(R) = \bar{\varepsilon}(2R - 1)$, where R denotes the quarterly retention probability. We match the model-implied retention probability to its empirical counterpart:

$$R = \mathbf{F}_\varepsilon \left(\bar{\mathcal{B}}(\gamma) - (1 - \gamma) \right),$$

where $\bar{\mathcal{B}}(\gamma)$ is the equilibrium average relationship benefit. Equivalently, γ is backed out from the fixed-point equation

$$\gamma = 1 - \bar{\mathcal{B}}(\gamma) + \bar{\varepsilon}(2R - 1).$$

Following [Brugués \(2026\)](#), we set the quarterly retention probability of supplier-customer relationships to $R = 0.898$, which corresponds to an annual retention probability of about 0.65. Given $\bar{\varepsilon} = 0.1$, we solve the fixed-point equation using the equilibrium average relationship benefit and obtain the calibrated value $\gamma = 0.14$.

IV.2 Stylized facts at the 12-sector aggregation

The empirical analysis in Section 2 is conducted at a more disaggregated sectoral level. Because the quantitative model is calibrated to 12 sectors, we also replicate the main stylized facts at the same level of aggregation used in the model. This exercise is not intended to replace the more disaggregated empirical evidence; rather, it verifies that the cross-sectional patterns motivating the model remain visible after aggregating the economy to the sectors used in the quantitative analysis.

We first classify sector pairs according to model-implied input-output linkages. A directed input-output link is active if the average input expenditure share over 2005Q3–2007Q2 is at least 0.5 percent. Sector pairs are then classified as two-way trading pairs, one-way trading pairs, or no-trading pairs. Panel (a)–(c) of Table IV.7 report summary statistics for pairwise output-growth correlations before, during, and after the Great Recession. During the Great Recession, connected sector pairs comove substantially more than no-trading pairs. The average correlation is 0.66 for two-way trading pairs and 0.54 for one-way trading pairs, compared with 0.19 for no-trading pairs. The increase from the pre-recession period is also concentrated among connected pairs: the average correlation rises by 0.69 for two-way trading pairs and 0.47 for one-way trading pairs, while it is essentially unchanged for no-trading pairs. Thus, even at the 12-sector aggregation, the rise in sectoral comovement is concentrated among sectors connected through input-output linkages.

We then examine whether trade-credit adjustment is associated with stronger comovement among trading pairs. Since bilateral trade credit is directed, we compute the change in model-implied bilateral trade credit for each active supplier-customer link from the pre-recession average (2005Q3–2007Q2) to the average in the Great Recession over 2008Q3–2009Q1. A directed link is classified as experiencing a trade-credit decline if this change is below the median change among all active directed links. For two-way trading pairs, a pair is assigned to the trade-credit-decline group if either direction satisfies this condition; for one-way trading pairs, the active direction is used.

Panel (d)–(g) of Table IV.7 report the results. Among two-way trading pairs, those experiencing a decline in model-implied trade credit exhibit stronger comovement during the Great Recession. Their average output-growth correlation is 0.70, compared with 0.56 for two-way trading pairs without a trade-credit decline. The increase from the pre-recession period is also larger, 0.79 versus 0.44. The distinction is more muted among one-way trading pairs, where the corresponding Great Recession correlations are 0.56 and 0.52. Overall, the 12-sector evidence supports the same qualitative message as the disaggregated empirical analysis: the rise in comovement is concentrated among

Table IV.7
PAIRWISE OUTPUT-GROWTH CORRELATIONS OF 12-SECTOR SAMPLE

	Obs	Mean	SD	Skewness	P25	Median	P75
Panel (a): Two-way trading group							
Before the Great Recession	29	-0.03	0.43	0.06	-0.30	-0.10	0.33
During the Great Recession	29	0.66	0.22	-2.03	0.61	0.70	0.80
After the Great Recession	29	0.04	0.37	0.68	-0.26	-0.07	0.24
Panel (b): One-way trading group							
Before the Great Recession	31	0.08	0.38	-0.07	-0.16	0.10	0.28
During the Great Recession	31	0.54	0.32	-0.79	0.32	0.65	0.80
After the Great Recession	31	0.07	0.40	-0.01	-0.25	0.09	0.35
Panel (c): No trading group							
Before the Great Recession	6	0.18	0.38	-0.30	0.02	0.15	0.58
During the Great Recession	6	0.19	0.47	-0.01	-0.16	0.20	0.49
After the Great Recession	6	0.24	0.41	-0.46	-0.18	0.36	0.54
Panel (d): Two-way trading with declined TC							
Before the Great Recession	21	-0.09	0.42	0.00	-0.35	-0.15	0.31
During the Great Recession	21	0.70	0.12	-0.84	0.64	0.70	0.80
After the Great Recession	21	0.05	0.37	0.67	-0.26	0.03	0.24
Panel (e): Two-way trading without declined TC							
Before the Great Recession	8	0.12	0.45	0.15	-0.24	0.09	0.51
During the Great Recession	8	0.56	0.38	-0.87	0.29	0.68	0.85
After the Great Recession	8	0.02	0.40	0.71	-0.25	-0.09	0.28
Panel (f): One-way trading with declined TC							
Before the Great Recession	16	0.10	0.39	0.12	-0.16	0.11	0.30
During the Great Recession	16	0.56	0.32	-0.59	0.32	0.67	0.83
After the Great Recession	16	0.11	0.43	-0.34	-0.17	0.21	0.41
Panel (g): One-way trading without declined TC							
Before the Great Recession	15	0.05	0.38	-0.32	-0.26	0.06	0.27
During the Great Recession	15	0.52	0.33	-0.99	0.37	0.59	0.77
After the Great Recession	15	0.04	0.38	0.40	-0.26	-0.13	0.34

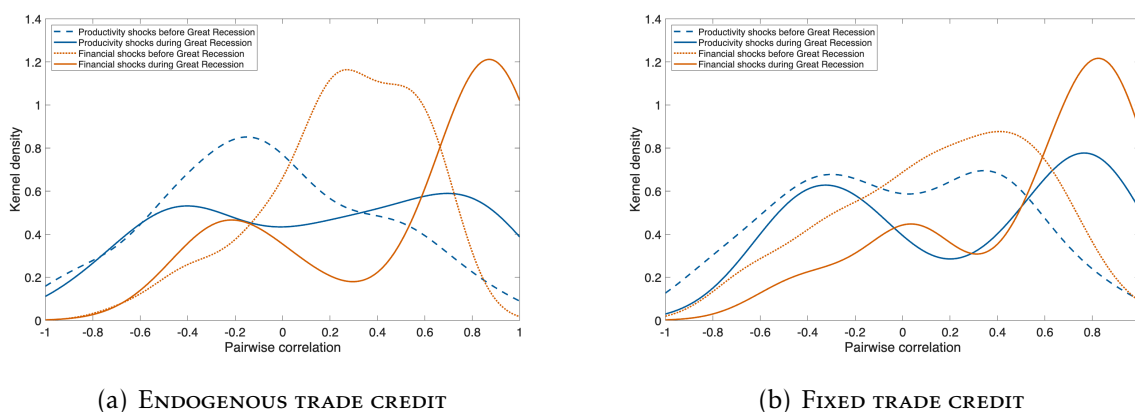
Note: The table reports moments of the distribution of pairwise output-growth correlations at the 12-sector aggregation. Sector pairs are classified using model-implied input-output linkages. A directed link is active if the average input expenditure share over 2005Q3–2007Q2 is at least 0.5 percent. Two-way trading pairs have active links in both directions, one-way trading pairs have an active link in only one direction, and no-trading pairs have no active link in either direction. Among trading pairs, a directed supplier-customer link is classified as experiencing a trade-credit decline if the change in its model-implied bilateral trade credit from 2005Q3–2007Q2 to 2008Q3–2009Q1 is below the median change among active directed links. For two-way trading pairs, the pair is classified as trade-credit declined if either direction satisfies this condition; for one-way trading pairs, the active direction is used. “Before,” “during,” and “after” refer to 2005Q3–2007Q2, 2007Q3–2009Q2, and 2009Q3–2011Q2, respectively.

production-linked sectors, and within these links, reductions in trade credit are associated with stronger comovement, especially for mutually connected sector pairs.

IV.3 Pairwise correlations of recovered productivity and financial shocks

Figure IV.13 plots the kernel densities of pairwise correlations of the recovered productivity and financial shocks. Pairwise correlations are computed for every sector pair separately over the normal period (2005Q3–2007Q2) and the Great Recession (2007Q3–2009Q2).

Panel (a) reports the recovered shocks from the benchmark model with endogenous trade credit, while Panel (b) reports the recovered shocks from the counterfactual model in which bilateral trade credit is fixed at its pre-recession average. In each panel, the blue lines correspond to productivity shocks and the orange lines correspond to financial shocks. Dashed lines denote the normal period, and solid lines denote the Great Recession.



Note: The figure plots kernel densities of pairwise correlations of the recovered productivity and financial shocks. Correlations are computed for all sector pairs separately over the normal period (2005Q3–2007Q2) and the Great Recession (2007Q3–2009Q2). Panel (a) uses the benchmark model with endogenous trade credit. Panel (b) uses the counterfactual model in which bilateral trade credit is fixed at its pre-recession average.

Figure IV.13

KERNEL DENSITIES: CORRELATION OF PRODUCTIVITY AND FINANCIAL SHOCKS

IV.4 Sectoral comovement and correlations of recovered shocks

This appendix examines whether sector pairs with stronger output comovement are simply those with more highly correlated recovered productivity or financial shocks. Using the recovered productivity $\{z_{it}\}$ and financial conditions $\{\theta_{it}\}$ from Section 5.3, we compute pairwise correlations of sectoral output growth, productivity shocks, and financial

shocks over both the pre-recession period and the Great Recession. We first estimate the following regression in levels:

$$\rho_{ij,\tau}^y = a_\tau + \beta_\tau^z \rho_{ij,\tau}^z + \beta_\tau^\theta \rho_{ij,\tau}^\theta + \delta_i + \delta_j + u_{ij,\tau}, \quad (\text{IV.5})$$

where $\rho_{ij,\tau}^x = \text{corr}_\tau(\Delta \log x_{it}, \Delta \log x_{jt})$, for $x \in \{y, z, \theta\}$, and τ denotes either the pre-recession period (2005Q3–2007Q2) or the Great Recession (2007Q3–2009Q2). The terms δ_i and δ_j denote sector fixed effects for the two sectors in each pair. We then estimate the analogous specification in first differences:

$$\Delta \rho_{ij}^y = a + \beta^z \Delta \rho_{ij}^z + \beta^\theta \Delta \rho_{ij}^\theta + \delta_i + \delta_j + u_{ij}, \quad (\text{IV.6})$$

where $\Delta \rho_{ij}^x \equiv \rho_{ij,\text{GR}}^x - \rho_{ij,\text{Pre-GR}}^x$, for $x \in \{y, z, \theta\}$.

Table IV.8 shows little evidence of an order-preserving relationship between output comovement and the correlation of recovered shocks. Without sector fixed effects, the coefficients are small and the R^2 values are close to zero in both levels and changes. Adding sector fixed effects increases the explanatory power, especially during the Great Recession, but the coefficients on productivity and financial-shock correlations remain small and still not statistically significant. Thus, sector pairs with more highly correlated recovered productivity shocks or financial shocks are not necessarily the pairs with higher output-growth comovement. This supports the interpretation that the rise in sectoral comovement reflects propagation through production and trade-credit linkages, rather than a simple reordering of sector pairs by the correlation of recovered fundamentals.

IV.5 Input-expenditure shares during the Great Recession

This appendix provides an additional validation of the production-network structure used in the quantitative analysis. In the benchmark calibration, the CES input-bundle weights $\{\lambda_{ji}\}$ are estimated using pre-Great-Recession input-output moments. Under the CES specification, however, bilateral input-expenditure shares are not fixed by these weights alone. They also depend on equilibrium input prices, trade-credit terms, and sectoral financial conditions. Therefore, the model generates time-varying bilateral input-expenditure shares even though the underlying technological input weights are held fixed.

To assess whether the model generates a reasonable crisis-period input-expenditure structure, Figure IV.14 compares model-implied bilateral input-expenditure shares during the Great Recession with the corresponding shares constructed from the 2008 BEA input-output table. The observations are concentrated around the 45-degree line, es-

Table IV.8
OUTPUT COMOVEMENT AND CORRELATIONS OF RECOVERED SHOCKS

	Before the Great Recession		During the Great Recession		Change	
	(1)	(2)	(3)	(4)	(5)	(6)
Productivity shocks	-0.007 (0.058)	-0.007 (0.061)	0.007 (0.047)	-0.047 (0.026)	0.087 (0.286)	-0.128 (0.235)
Financial shocks	-0.091 (0.125)	-0.087 (0.130)	-0.235 (0.296)	-0.218 (0.284)	-0.046 (0.167)	-0.069 (0.156)
Sector fixed effects	No	Yes	No	Yes	No	Yes
Observations	66	66	66	66	66	66
R^2	0.009	0.276	0.010	0.793	0.003	0.505

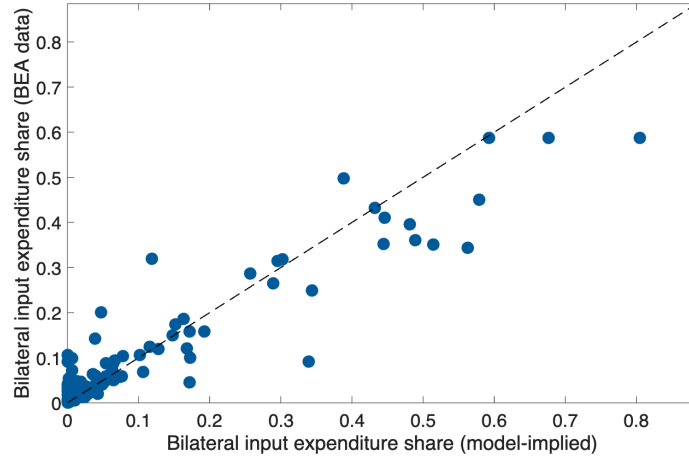
Note: The table reports regressions of pairwise output-growth correlations on pairwise correlations of recovered productivity growth and changes in recovered financial conditions. The first two columns use the pre-Great-Recession window, the next two columns use the Great-Recession window, and the last two columns use the change from the pre-recession period to the Great Recession. Sector fixed effects are sectoral fixed effects for the two sectors in each unordered pair. Standard errors are reported in parentheses.

pecially for smaller and medium-sized bilateral input links. The correlation between the model-implied and BEA bilateral input-expenditure shares is 0.94. This high correlation is reassuring because the 2008 input-output table is not directly targeted in the benchmark calibration. The comparison therefore shows that the model captures the broad cross-sectional pattern of bilateral input use during the crisis while preserving the distinction between pre-crisis technological input weights and endogenous crisis-period adjustments.

This exercise helps address the concern that the results are mechanically driven by imposing a fixed pre-crisis input-output structure. The calibrated CES input weights discipline the technological component of bilateral input demand, while equilibrium prices, financial conditions, and trade-credit terms generate time variation in expenditure shares. The comparison with the 2008 input-output table suggests that this structure provides a reasonable approximation to the crisis-period production network.

IV.6 Correlated versus uncorrelated shocks

Figure 8 shows that sector pairs with stronger output comovement during the Great Recession are not simply those with more highly correlated recovered productivity or financial conditions. A remaining concern, however, is that the underlying shock-generating process itself may have changed during the Great Recession. To examine this possibility, we conduct a simulation exercise that separates the role of the crisis-period shock



Note: The figure compares model-implied bilateral input-expenditure shares during the Great Recession with bilateral input-expenditure shares constructed from the 2008 BEA input-output table. The dashed line is the 45-degree line.

Figure IV.14

BILATERAL INPUT-EXPENDITURE SHARES DURING THE GREAT RECESSION

environment from that of cross-sector shock correlations.

First, for $\mathbf{x} \in \{z, \theta\}$, we estimate a diagonal VAR process for recovered productivity and financial conditions from Section 5.3:

$$\log \mathbf{x}_t = \mathbf{D}_\rho^x \log \mathbf{x}_{t-1} + \boldsymbol{\epsilon}_t^x + \mathbf{1}_{\text{GR},t} \boldsymbol{\nu}_t^x, \quad (\text{IV.7})$$

where $\mathbf{x}_t = [x_{1t}, \dots, x_{nt}]^T$ denotes the vector of recovered productivity, $\{z_{it}\}$, or financial conditions, $\{\theta_{it}\}$. The matrix $\mathbf{D}_\rho^x$ is diagonal, with sector-specific persistence parameters on the diagonal.³⁶ The shock $\boldsymbol{\epsilon}_t^x$ captures normal-period residual variation, while $\boldsymbol{\nu}_t^x$ captures the additional Great-Recession-period component as in Li and Martin (2019). We use the residuals from this specification to estimate covariance matrices for the normal period and the Great Recession.

The estimated covariance matrices indicate that the Great Recession shock environment differs from the normal period in both volatility and cross-sector dependence. The diagonal elements are larger during the Great Recession, implying that sector-specific productivity and financial shocks became more volatile. The off-diagonal elements also point to stronger cross-sector correlation. For productivity shocks, the average off-diagonal correlation rises from -0.06 in the normal period to 0.15 during the Great Recession. The

³⁶The model can be extended to allow for cross-sector autoregressive terms, which would generate correlated sectoral productivity or financial conditions through lagged cross-sector dependence. We use a diagonal specification to keep the comparison between correlated and uncorrelated shocks transparent.

increase is more pronounced for financial shocks: the average off-diagonal correlation rises from 0.25 to 0.53. Thus, the Great Recession features not only larger sectoral disturbances, but also more correlated underlying innovations, especially on the financial side.

Starting from the stationary equilibrium, we simulate productivity and financial conditions over an eight-quarter window and solve the model to compute pairwise correlations of sectoral output growth. In the benchmark simulation, shocks are drawn using the full estimated covariance matrices. In the counterfactual simulation, we preserve the same sector-specific shock variances but set all off-diagonal covariance terms to zero, thereby eliminating cross-sector shock correlation while holding constant the volatility of shocks within each sector. Within each period, we use the same underlying random draws for the full-covariance and diagonal-covariance simulations, so the comparison isolates the role of cross-sector covariance. The simulation is repeated 1,000 times.

Panel (a) of Table IV.9 reports the simulated distribution of pairwise output-growth correlations. Under the full covariance matrices, the average correlation rises sharply, from approximately zero in the normal-period simulation to 0.48 under the Great-Recession shock environment. This confirms that the covariance structure of the recovered productivity and financial innovations can generate a substantial increase in sectoral comovement.

The counterfactual simulations in Panel (b) of Table IV.9 show, however, that cross-sector shock covariance is not necessary for the model to generate strong output comovement. When the off-diagonal covariance terms are set to zero, the average pairwise correlation still increases substantially, from near zero in the normal-period simulation to 0.43 under the Great-Recession shock environment. This result indicates that even when underlying shocks are independent across sectors, our model can still transform heterogeneous sectoral shocks into synchronized movements in output.

IV.7 Recalibration for the early-1980s recessions

This appendix describes the recalibration of the model for the quantitative analysis of the early-1980s recessions. We follow the same procedure as in Section 5. First, common structural parameters are held fixed at their benchmark values. Sectoral intermediate-input shares, $\{\xi_i\}$, are recalibrated using the 1978 input-output table published by the BEA. Because comparable sectoral labor-compensation data are unavailable in the 1978 input-output accounts, we retain the benchmark labor shares, $\{\alpha_i\}$. The remaining sector-specific parameters are then re-estimated using 1978 moments together with the stationary-

Table IV.9
PAIRWISE CORRELATIONS OF SECTORAL OUTPUT GROWTH RATES

	Mean	SD	Skewness	25th	Median	75th	KS stat. (CV-KS)
(a) Simulation with correlated shocks							
Before the Great Recession	-0.00	0.03	0.77	-0.02	-0.00	0.02	
During the Great Recession	0.48	0.19	-0.15	0.35	0.49	0.62	
(b) Simulation with uncorrelated shocks							
Before the Great Recession	0.00	0.11	0.09	-0.08	0.01	0.08	
During the Great Recession	0.43	0.12	-0.26	0.35	0.43	0.51	
(c) Early-1980s: data/both shocks with endogenous trade credit							
During the early 1980s recessions	0.35	0.37	-0.35	0.11	0.32	0.62	0.27 (0.47, 1.03)
After the early 1980s recessions	0.13	0.38	-0.13	-0.09	0.12	0.43	
(d) Early-1980s: both shocks with fixed trade credit							
During the early 1980s recession	0.52	0.39	-0.55	0.15	0.70	0.86	0.29 (0.47, 1.03)
After the early 1980s recession	0.34	0.41	-0.54	0.06	0.43	0.69	
(e) COVID-19: data/both shocks with endogenous trade credit							
Before the COVID-19 recession	0.23	0.36	-0.38	-0.05	0.24	0.49	0.79 (0.47, 1.03)
During the COVID-19 recession	0.83	0.14	-0.88	0.74	0.87	0.94	
(f) COVID-19: both shocks with fixed trade credit							
Before the COVID-19 recession	0.45	0.53	-0.96	0.15	0.58	0.95	0.59 (0.47, 1.03)
During the COVID-19 recession	0.87	0.10	-0.81	0.81	0.91	0.95	

Note: The table reports distributional moments of pairwise sectoral output-growth correlations. Panels (a) and (b) report simulation results. In these panels, each observation is the average pairwise output-growth correlation from one valid simulated economy; moments are computed across valid simulations. The correlated-shock simulation uses the full estimated cross-sector covariance matrices of productivity and financial-condition innovations, while the uncorrelated-shock simulation preserves sector-specific variances but sets all off-diagonal covariance terms to zero. Panels (c)–(f) report moments across the 66 sector pairs in the early-1980s and COVID-19 counterfactual exercises. For the early-1980s recessions, 1978–1985 is the recession window and 1983–1989 is the post-recession comparison window. For the COVID-19 exercise, 2017Q2–2019Q1 is the pre-COVID window and 2019Q2–2021Q1 is the COVID window. In the endogenous-trade-credit cases, recovered productivity and financial conditions match sectoral output growth by construction. In the fixed-trade-credit cases, bilateral trade credit is fixed at its pre-recession averages. The last column reports the two-sample Kolmogorov–Smirnov statistic comparing the two windows within each panel when applicable. Critical values in parentheses adjust for autocorrelation following [Lanzante \(2021\)](#); the first value corresponds to mild autocorrelation, $AR(1)=0.6$, and the second to high autocorrelation, $AR(1)=0.9$.

equilibrium conditions. Specifically, bilateral input-expenditure shares and final-consumption expenditure shares are constructed from the 1978 input–output table. The sectoral accounts-payable-to-costs ratios are computed by first averaging firm-level observations within 1978 and then taking the median across firms within each sector. These empirical moments are jointly used to estimate the remaining sectoral parameters, including the final-demand weights $\{\zeta_i\}$, bilateral input weights $\{\lambda_{ji}\}$, and relationship-maintenance cost parameters $\{\psi_i\}$. As in the benchmark calibration, the parameter estimates are obtained by matching the stationary equilibrium to the corresponding empirical moments. Table IV.10 reports the resulting parameter values of the recalibrated stationary economy.

Figure IV.15 compares the model-implied and empirical moments. The re-estimated

Table IV.10
RE-ESTIMATION FOR THE EARLY-1980s RECESSIONS

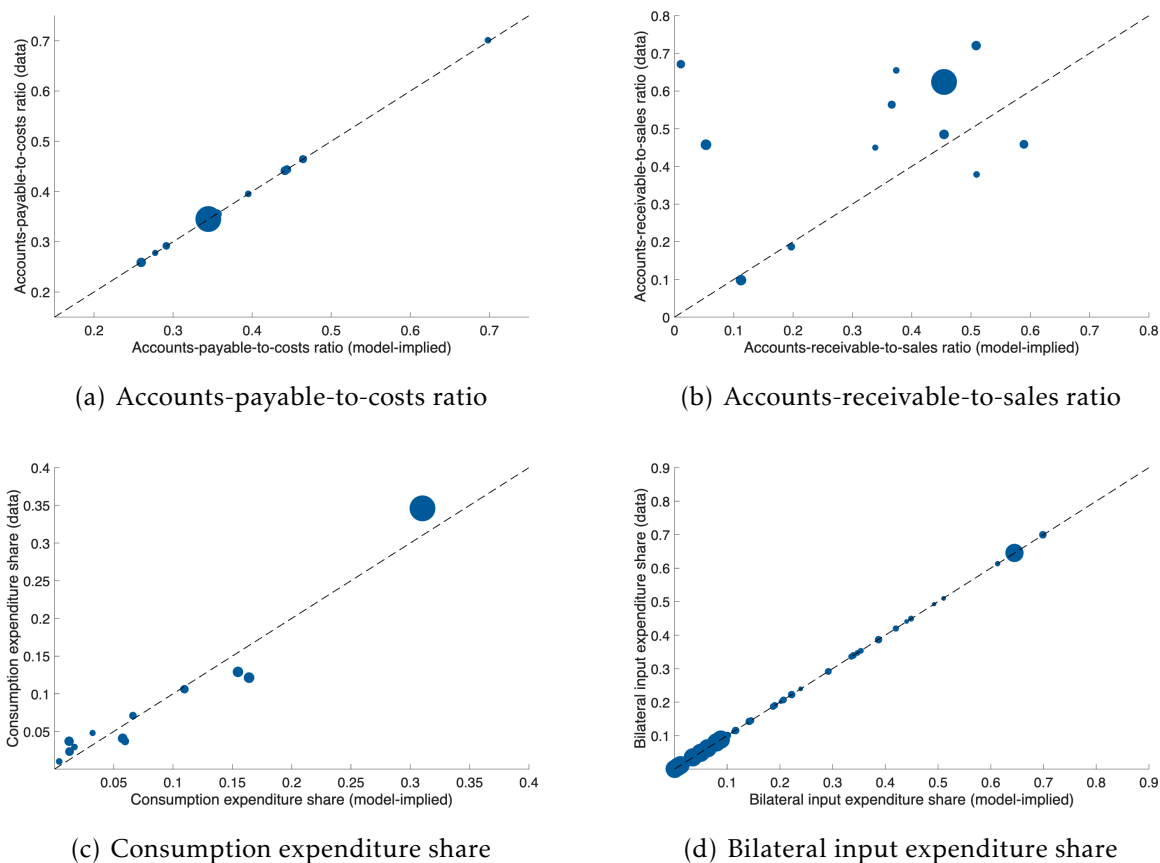
Sectors	ξ	α	ζ	ψ
Mining	0.381	0.537	0.002	1.635
Utilities	0.443	0.414	0.010	1.305
Construction	0.534	0.316	0.300	0.410
Manufacturing	0.578	0.197	0.170	0.410
Wholesale trade	0.214	0.418	0.036	1.301
Retail trade	0.218	0.461	0.190	1.339
Transportation and warehousing	0.408	0.330	0.007	1.077
Information	0.295	0.232	0.040	1.119
Professional and business services	0.260	0.478	0.007	0.946
Educational services and health care	0.281	0.538	0.173	1.498
Arts and recreation services	0.402	0.381	0.046	0.410
Other services	0.291	0.452	0.019	0.943

Note: The table reports the sectoral parameters used in the early-1980s recessions. The intermediate-input shares ξ_i are constructed from the 1978 input-output table, while labor shares α_i are kept at their benchmark values. The final-demand weights ζ_i , and relationship-maintenance cost parameters ψ_i are estimated jointly by matching the stationary equilibrium to the targeted 1978 moments.

stationary economy closely matches the targeted accounts-payable-to-costs ratios and bilateral input-expenditure shares. The consumption expenditure shares are also closely aligned with the data. The accounts-receivable-to-sales ratios are not targeted and therefore provide a more demanding external check; the fit is weaker for this moment, but the model still generates meaningful cross-sectoral dispersion.

We then use the recalibrated economy to recover sectoral productivity and financial conditions over the early-1980s recessions. Figure IV.16 plots the recovered sectoral paths, normalized to one in 1977. Productivity does not exhibit a broad-based collapse during the early-1980s recessions; the cross-sector average rises gradually, although individual sectors remain heterogeneous. Financial conditions also do not display the sharp deterioration observed during the Great Recession. The average financial condition moves only mildly during the recession window and remains relatively stable compared with the Great Recession.

Figure IV.17 reports the kernel densities of pairwise correlations of recovered productivity and financial shocks. Productivity shocks are weakly correlated: the average pairwise correlation is -0.05 during the early-1980s recessions and 0.03 after the recession. Financial shocks are positively correlated in both periods, with average correlations of 0.61 and 0.75 , respectively. Despite these positively correlated recovered financial-condition changes, the main-text counterfactual shows that endogenous trade-credit ad-

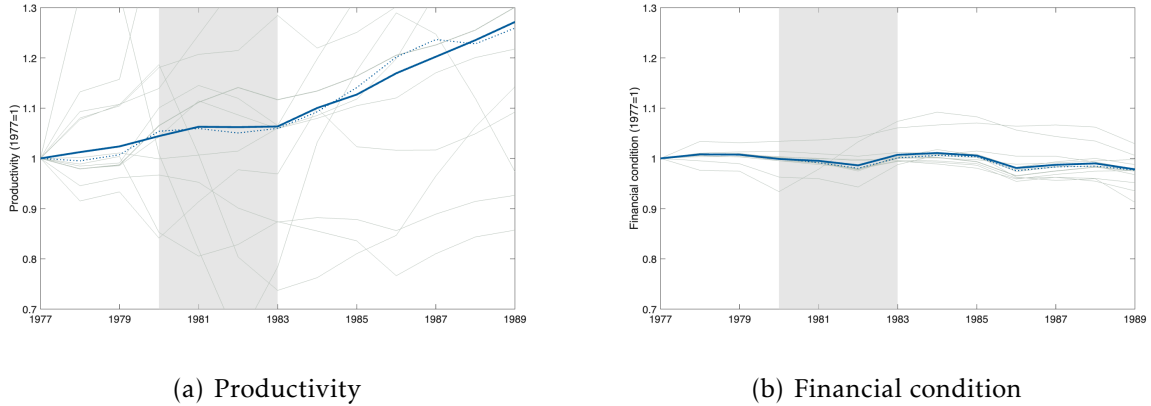


Note: The figure compares model-implied moments with their empirical counterparts in the 1978 re-estimation. The dashed line is the 45-degree line. Accounts-payable-to-costs ratios, consumption expenditure shares, and bilateral input expenditure shares are targeted. Accounts-receivable-to-sales ratios are not targeted.

Figure IV.15
MODEL FIT FOR THE 1978 RE-ESTIMATION

justment dampens, rather than amplifies, sectoral comovement in this episode.

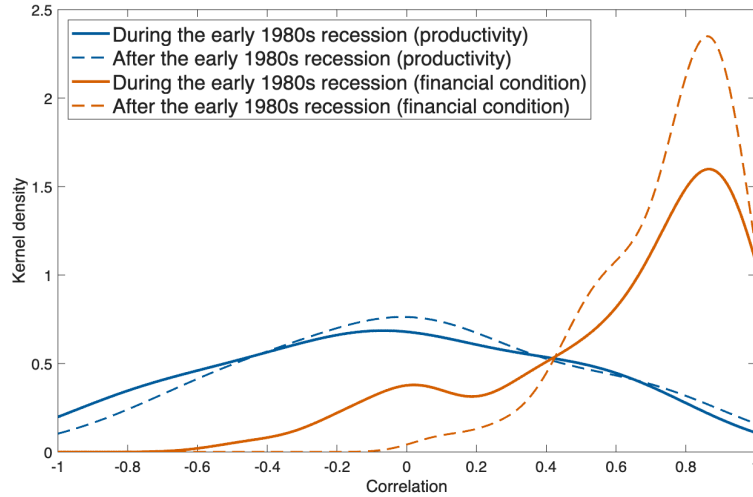
As shown in Panel (a)-(b) of Table IV.11, for the early-1980s recessions, sectoral comovement rises relative to the post-recession comparison window under both regimes, but the increase is larger when trade credit is fixed. With endogenous trade credit, the average pairwise correlation is 0.35 during the recession and 0.13 afterward. With fixed trade credit, the corresponding means are 0.54 and 0.24. The median comparison gives the same message: 0.32 versus 0.12 under endogenous trade credit, compared with 0.60 versus 0.28 under fixed trade credit. Thus, endogenous trade credit does not amplify the early-1980s episode; instead, relative to the fixed-trade-credit economy, it dampens the rise in sectoral comovement.



Note: The figure plots recovered sectoral productivity and financial conditions for the early-1980s recessions, normalized to one in 1977. Thin gray lines denote sector-specific paths. The solid blue line denotes the cross-sector average, and the dotted blue line denotes the median. The shaded area marks the early-1980s recession window.

Figure IV.16

RECOVERED PRODUCTIVITY AND FINANCIAL CONDITIONS: THE EARLY-1980s RECESSIONS



Note: The figure plots kernel densities of pairwise correlations of recovered productivity and financial-condition changes. The early-1980s recession window is 1978–1985, and the post-recession comparison window is 1983–1989.

Figure IV.17

PAIRWISE CORRELATIONS OF RECOVERED SHOCKS: THE EARLY-1980s RECESSIONS

IV.8 Recalibration for the COVID-19 recession

This appendix reports the re-estimation used for the COVID-19 recession exercise. We follow the same procedure as in Section 5, while using data appropriate for the COVID period. The common structural parameters are kept at their benchmark values. We con-

Table IV.11
PAIRWISE OUTPUT-GROWTH CORRELATIONS IN OTHER RECESSIONS

	Mean	SD	Skewness	P25	Median	P75
Panel (a): Early-1980s recession, endogenous trade credit						
During the early-1980s recessions	0.35	0.37	-0.35	0.11	0.32	0.62
After the early-1980s recessions	0.13	0.38	-0.13	-0.09	0.12	0.43
Panel (b): Early-1980s recession, fixed trade credit						
During the early-1980s recessions	0.54	0.25	-0.62	0.35	0.60	0.74
After the early-1980s recessions	0.24	0.44	-0.40	-0.01	0.28	0.59
Panel (c): COVID-19 recession, endogenous trade credit						
Before the COVID-19 recession	0.28	0.37	-0.41	-0.05	0.34	0.59
During the COVID-19 recession	0.84	0.14	-1.36	0.74	0.89	0.95
Panel (d): COVID-19 recession, fixed trade credit						
Before the COVID-19 recession	0.20	0.64	-0.34	-0.35	0.45	0.78
During the COVID-19 recession	0.83	0.27	-1.72	0.88	0.96	0.98

Note: The table reports moments of the distribution of pairwise sectoral output-growth correlations. For the early-1980s recessions, the recession window is 1978–1985 and the post-recession comparison window is 1983–1989. For the COVID-19 recession, the pre-recession window is 2017Q2–2019Q1 and the recession window is 2019Q2–2021Q1. Each row uses the 66 unordered sector pairs among the 12 sectors.

struct sectoral input shares $\{\xi_i\}$, labor shares $\{\alpha_i\}$, bilateral input-expenditure shares, and consumption expenditure shares from the 2017 input-output table. For trade credit, we target sectoral accounts-payable-to-costs ratios computed from quarterly firm-level data over 2017Q2–2019Q1. Within each sector, we use the median pre-COVID accounts-payable-to-costs ratio. We then estimate the remaining sectoral parameters, including final-demand weights $\{\zeta_i\}$, bilateral input weights $\{\lambda_{ji}\}$, and relationship-maintenance cost parameters $\{\psi_i\}$, by matching the stationary equilibrium to the pre-COVID moments.

Table IV.12 reports the sectoral parameters used in the COVID-19 recalibration. We construct the input shares $\{\xi_i\}$ and labor shares $\{\alpha_i\}$ from the 2017 input-output table, and estimate the final-demand weights $\{\zeta_i\}$ and relationship-maintenance cost parameters $\{\psi_i\}$ by matching the stationary equilibrium to pre-COVID moments. The table shows that the re-estimation preserves substantial cross-sector heterogeneity in production structure and final demand. Manufacturing has the largest intermediate-input share, while the education, health care, and social-assistance receives the largest final-demand weight. The estimated relationship-maintenance cost parameter is broadly similar across sectors, except for retail trade, where the higher value indicates a larger cost of maintain-

ing customer relationships in the recalibrated stationary equilibrium.

Table IV.12
SECTORAL PARAMETERS FOR THE COVID-19 RECALIBRATION

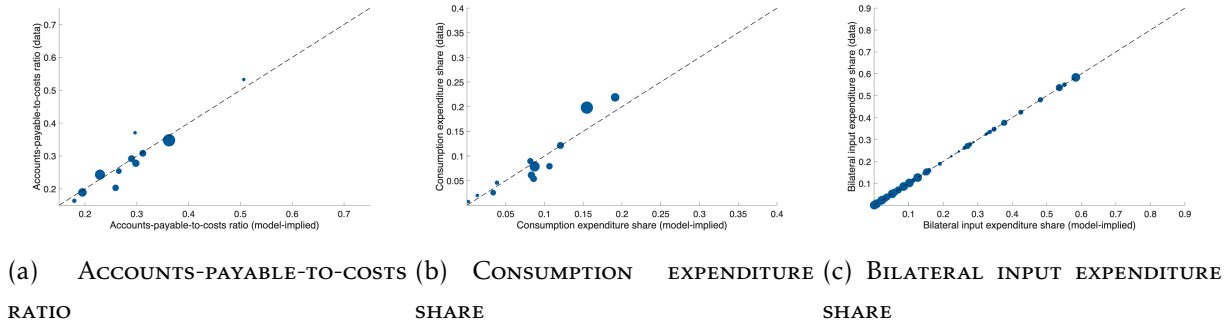
Sector	ξ_i	α_i	ψ_i	ζ_i
Mining	0.356	0.166	0.540	0.001
Utilities	0.269	0.172	0.554	0.008
Construction	0.435	0.330	0.539	0.089
Manufacturing	0.554	0.186	0.540	0.091
Wholesale trade	0.321	0.259	0.540	0.059
Retail trade	0.244	0.329	0.767	0.105
Transportation and warehousing	0.357	0.287	0.541	0.023
Information	0.367	0.198	0.541	0.064
Professional and business services	0.283	0.458	0.540	0.061
Educational services and health care	0.273	0.503	0.541	0.414
Arts and recreation services	0.324	0.340	0.539	0.061
Other services, except government	0.253	0.431	0.542	0.025

Note: The table reports sectoral parameters used in the COVID-19 recalibration. The input shares $\{\xi_i\}$ and labor shares $\{\alpha_i\}$ are constructed from the 2017 input-output table. The final-demand weights $\{\zeta_i\}$ and relationship-maintenance cost parameters $\{\psi_i\}$ are estimated jointly by matching the stationary equilibrium to pre-COVID moments.

The re-estimated model fits the targeted moments closely. Figure IV.18 compares model-implied moments with their empirical counterparts. The model matches bilateral input-expenditure shares essentially exactly and closely reproduces the sectoral accounts-payable-to-costs ratios and consumption expenditure shares. The correlation between model-implied and empirical accounts-payable-to-costs ratios is 0.94, the corresponding correlation for consumption expenditure shares is 0.95, and the correlation for bilateral input-expenditure shares is essentially one.

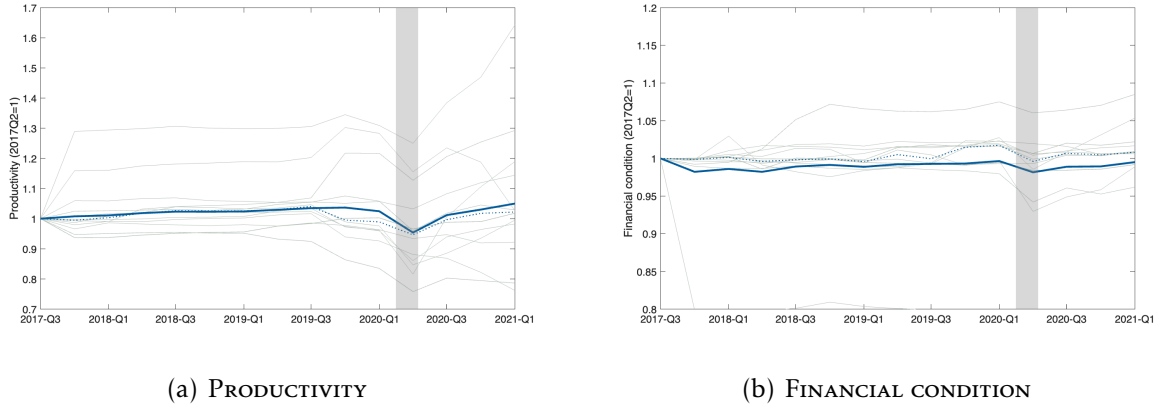
Figure IV.19 plots the recovered sectoral productivity and financial conditions, normalized to one in 2017Q2. The COVID period is associated with substantial heterogeneity in recovered productivity across sectors. The cross-sector average declines around 2020Q2 and then recovers, while some sectors experience much larger movements. In contrast, the recovered financial-condition series display less movement in the cross-sector average, although sector-specific paths remain heterogeneous. This pattern differs from the Great Recession, where deterioration in financial conditions was more central to the propagation mechanism.

Figure IV.20 reports the kernel densities of pairwise correlations of recovered productivity and financial-condition changes. Productivity changes become substantially more correlated during the COVID period: the average pairwise correlation rises from -0.03



Note: The figure compares model-implied moments on the horizontal axis with empirical moments on the vertical axis. Panel (a) reports sectoral accounts-payable-to-costs ratios. Panel (b) reports sectoral consumption expenditure shares. Panel (c) reports bilateral input-expenditure shares. The dashed line is the 45-degree line.

Figure IV.18
MODEL FIT FOR THE COVID-19 RECALIBRATION

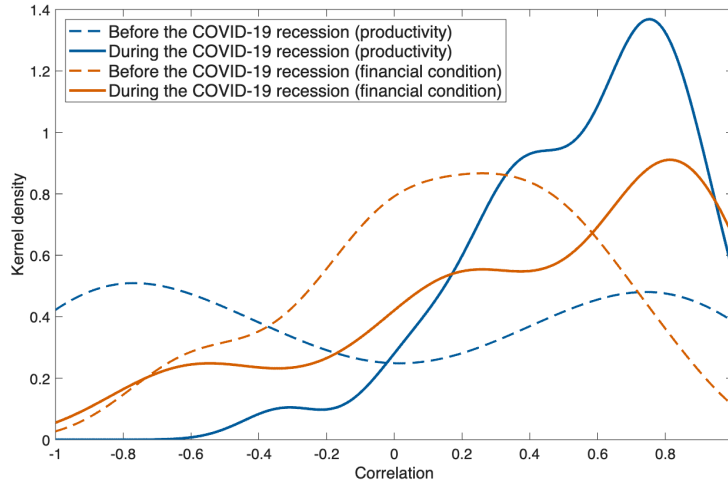


Note: The figure plots recovered sectoral productivity and financial conditions, normalized to one in 2017Q2. Thin gray lines are sector-specific series, the solid blue line is the output-share-weighted average, and the dotted blue line is the cross-sector median. The shaded area denotes the COVID-19 recession.

Figure IV.19
RECOVERED PRODUCTIVITY AND FINANCIAL CONDITIONS DURING COVID-19

before COVID-19 to 0.55 during the COVID period. Financial-condition changes also become more correlated, with the average rising from 0.15 to 0.39, although the distribution remains dispersed. These patterns suggest that the COVID-19 recession involved a broad change in the underlying shock environment. This helps explain why sectoral comovement increases sharply even when trade credit is fixed.

As shown in Panel (c)-(d) of Table IV.11, for the COVID-19 recession, pairwise output-growth correlations increase sharply under both regimes, consistent with the broad nature of the pandemic shock. Under endogenous trade credit, the average correlation rises from 0.28 before the recession to 0.84 during the recession; under fixed trade credit, it



Note: The figure plots kernel densities of pairwise correlations of recovered productivity changes and financial-condition changes before and during the COVID-19 recession. The pre-COVID window is 2017Q2–2019Q1, and the COVID window is 2019Q2–2021Q1.

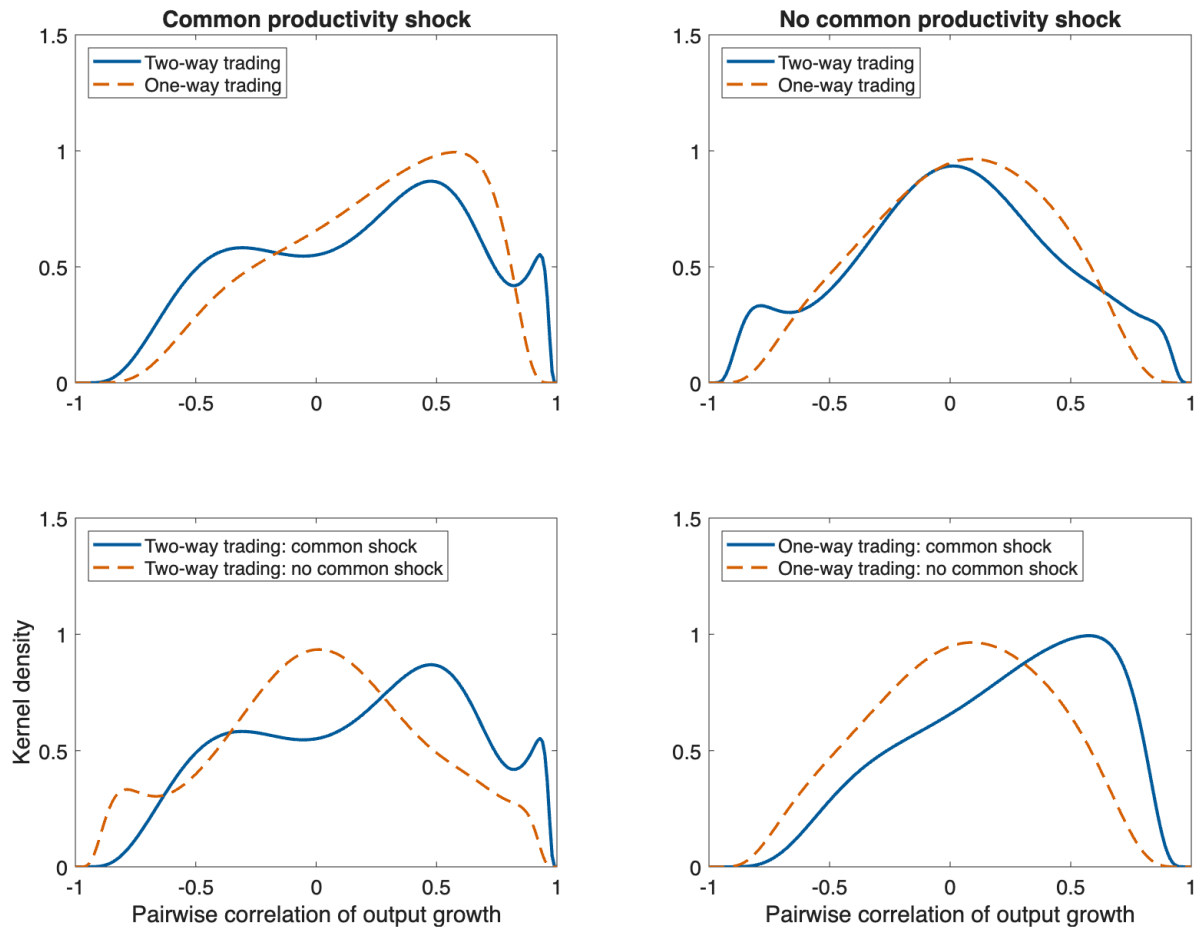
Figure IV.20

PAIRWISE CORRELATIONS OF RECOVERED CHANGES DURING COVID-19

rises from 0.20 to 0.83. The fixed-trade-credit case has a higher recession-period median and upper quartile, 0.96 and 0.98, compared with 0.89 and 0.95 under endogenous trade credit. Because correlations are bounded above by one, this indicates that many sector pairs in the fixed-trade-credit economy move close to the maximum possible comovement. Endogenous trade credit therefore appears to moderate the most extreme increases in sectoral comovement during COVID-19, even though comovement rises strongly in both economies.

IV.9 Role of common shocks

We use the calibrated model to illustrate a key implication of a common-shock environment. Specifically, we feed the model with the recovered productivity and financial shocks over 2005Q3–2007Q2 and then introduce a uniform 1.5% decline in productivity across all sectors in 2006Q1. This experiment is intended solely to isolate the effect of a broad-based common shock. The top-left panel of Figure IV.21 plots the kernel densities of pairwise output-growth correlations for the two-way and one-way trading groups, using the same classification as in Section 2.2. Relative to the baseline simulation without the common productivity shock (top-right panel), the distributions for both groups shift markedly to the right, indicating that a common shock raises sectoral comovement broadly across the economy. Although the increase is somewhat larger for the



Note: The model is fed with the productivity and financial shocks between 2005Q3 and 2007Q2, and then a 1.5% decline in productivity for all sectors is imposed in 2006Q1.

Figure IV.21
SECTORAL COMOVEMENT UNDER COMMON SHOCKS

two-way trading group, this group also exhibits stronger comovement in normal time. Thus, the model predicts that a common shock generates a broad-based increase in sectoral comovement rather than one concentrated among sectors with stronger production-network linkages.

Table IV.13
LIST OF SECTORS AND CHARACTERISTICS

Sector	# two-way	# one-way	$c\acute{o}rr_{before}$	$c\acute{o}rr_{in}$	$c\acute{o}rr_{after}$	cmp share	$\frac{AR_{before}}{sales_{before}}$	$\frac{AP_{before}}{OC_{before}}$	$\Delta \frac{AR}{sales}$	$\Delta \frac{AP}{OC}$
Oil and gas extraction	5	15	0.034	-0.218	-0.048	0.0%	55.4%	163.7%	-2.8%	-57.9%
Mining, except oil and gas	15	22	0.138	0.245	0.001	0.9%	N.A.	N.A.	N.A.	N.A.
Support activities for mining	1	27	0.138	0.206	0.085	0.0%	85.4%	39.5%	-3.5%	-3.3%
Utilities	21	35	-0.169	0.425	0.066	44.8%	45.9%	42.1%	-3.1%	-0.1%
Construction	24	31	0.134	0.481	0.002	0.0%	73.8%	33.9%	1.9%	-3.1%
Wood products	21	26	0.122	0.419	-0.017	5.0%	23.0%	18.3%	4.5%	-1.0%
Nonmetallic mineral products	26	21	0.192	0.516	0.092	11.5%	N.A.	N.A.	N.A.	N.A.
Primary metals	18	22	0.103	0.505	0.046	0.6%	52.6%	31.8%	-3.8%	-0.5%
Fabricated metal products	27	26	0.199	0.519	0.020	7.4%	68.7%	44.5%	-2.5%	0.8%
Machinery	28	25	-0.030	0.380	0.006	10.1%	70.3%	47.9%	-3.8%	-1.2%
Computer and electronic products	25	27	0.127	0.496	0.032	34.2%	66.9%	52.6%	-2.6%	-1.1%
Electrical equipment and appliances	22	25	0.038	0.486	0.066	35.9%	61.8%	45.9%	-2.5%	-1.0%
Motor vehicles, bodies and trailers	19	33	-0.023	0.424	0.002	51.6%	N.A.	N.A.	N.A.	N.A.
Other transportation equipment	2	31	-0.069	0.242	-0.012	33.8%	N.A.	N.A.	N.A.	N.A.
Furniture and related products	8	30	0.192	0.482	0.023	69.2%	61.2%	46.5%	-4.8%	-3.4%
Miscellaneous manufacturing	20	24	-0.143	0.485	-0.019	72.5%	64.5%	48.8%	-4.4%	-2.4%
Food and beverage and tobacco products	11	31	0.084	-0.076	0.067	70.4%	39.3%	32.5%	-1.2%	-1.5%
Textile mills and textile product mills	15	32	0.109	0.395	-0.039	49.6%	67.8%	31.7%	-0.3%	-1.6%
Apparel and leather and allied products	9	26	0.146	0.086	-0.157	93.4%	68.6%	39.9%	-4.0%	8.7%
Paper products	27	27	0.105	0.431	-0.057	18.7%	63.1%	40.3%	-4.5%	-2.9%
Printing and related support activities	19	27	-0.122	0.452	0.052	6.4%	46.1%	26.1%	1.3%	0.7%
Petroleum and coal products	9	47	-0.064	-0.210	0.003	40.4%	35.4%	29.4%	1.8%	1.8%
Chemical products	27	27	-0.032	0.345	0.108	40.3%	60.9%	46.8%	-0.9%	0.2%
Plastics and rubber products	30	22	0.061	0.464	0.028	18.8%	58.4%	40.0%	-5.2%	0.3%
Wholesale trade	21	25	0.016	0.542	0.074	0.0%	49.3%	38.3%	0.4%	-0.7%
Motor vehicle and parts dealers	0	36	0.004	0.405	0.130	N.A.	37.7%	51.9%	2.0%	0.5%
Food and beverage stores	0	37	0.011	0.496	0.005	N.A.	5.8%	37.4%	0.2%	-0.1%
General merchandise stores	0	31	-0.012	-0.184	-0.132	N.A.	14.5%	33.6%	0.4%	-0.8%
Other retail	0	41	0.153	0.256	0.017	59.7%	11.2%	29.8%	0.3%	0.2%
Air transportation	16	29	-0.085	0.357	-0.058	60.8%	17.9%	18.9%	-1.7%	-2.0%
Rail transportation	2	26	0.198	0.461	0.060	17.6%	47.5%	66.6%	-6.9%	-1.5%
Water transportation	0	20	0.115	-0.047	0.099	71.8%	N.A.	N.A.	N.A.	N.A.
Truck transportation	9	26	0.208	0.320	0.128	24.8%	46.9%	16.7%	-4.4%	-2.3%
Transit and ground passenger transportation	12	30	0.011	0.240	-0.097	53.8%	41.1%	61.0%	-1.7%	-3.2%
Pipeline transportation	1	23	-0.081	0.288	-0.022	0.0%	44.9%	58.7%	-6.1%	4.0%
Other transportation and support activities	22	29	0.045	0.176	0.101	6.4%	65.4%	29.8%	-0.8%	-2.0%
Warehousing and storage	19	30	0.076	0.349	0.029	0.6%	N.A.	N.A.	N.A.	N.A.
Publishing industries, except internet	13	28	0.062	0.425	-0.062	70.8%	74.4%	49.1%	-3.0%	-5.1%
Motion picture and sound recording industries	7	28	-0.075	-0.012	-0.024	43.7%	66.9%	38.2%	-1.1%	-1.6%
Broadcasting and telecommunications	30	26	0.048	0.426	-0.060	44.1%	48.6%	38.5%	-2.0%	-2.5%
Data processing and internet publishing	32	19	-0.063	0.424	0.031	11.4%	65.4%	32.5%	-1.2%	-1.7%
Legal services	25	31	-0.036	0.318	-0.016	37.8%	70.4%	21.0%	-0.7%	-1.7%
Computer systems design and related services	27	27	-0.003	0.418	0.024	0.0%	81.8%	32.4%	-4.8%	-4.4%
Professional, scientific, and technical services	37	19	0.077	0.435	0.009	5.3%	71.4%	31.3%	-2.6%	-1.1%
Management of companies and enterprises	28	26	0.065	0.310	-0.057	0.0%	N.A.	N.A.	N.A.	N.A.
Administrative and support services	36	20	0.084	0.491	-0.062	7.6%	69.2%	15.6%	-1.3%	-0.4%
Waste management and remediation services	29	26	0.040	0.262	0.065	17.7%	48.0%	112.2%	3.7%	-5.8%
Educational services	4	39	-0.025	-0.024	0.074	94.3%	N.A.	N.A.	N.A.	N.A.
Ambulatory health care services	1	35	-0.043	0.354	0.066	96.6%	60.9%	25.3%	-5.0%	-2.1%
Hospitals	0	33	-0.019	-0.120	0.021	99.6%	59.8%	35.1%	-5.0%	-1.8%
Nursing and residential care facilities	0	35	0.137	0.276	0.003	98.0%	41.5%	9.2%	-0.7%	-0.6%
Social assistance	0	41	0.037	-0.076	0.074	99.1%	N.A.	N.A.	N.A.	N.A.
Performing arts, spectator sports, and museums	18	22	0.156	0.258	-0.050	48.3%	9.9%	18.6%	0.5%	-1.5%
Amusements, gambling, and recreation industries	13	32	0.099	0.385	-0.054	92.9%	N.A.	N.A.	N.A.	N.A.
Accommodation	28	19	0.229	0.464	0.032	66.7%	34.6%	36.1%	-2.9%	-1.2%
Food services and drinking places	35	18	0.097	0.402	0.030	78.4%	6.0%	17.7%	0.7%	-1.4%
Other services, except government	38	18	-0.015	0.488	0.034	68.7%	59.1%	34.5%	-1.1%	-2.0%

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